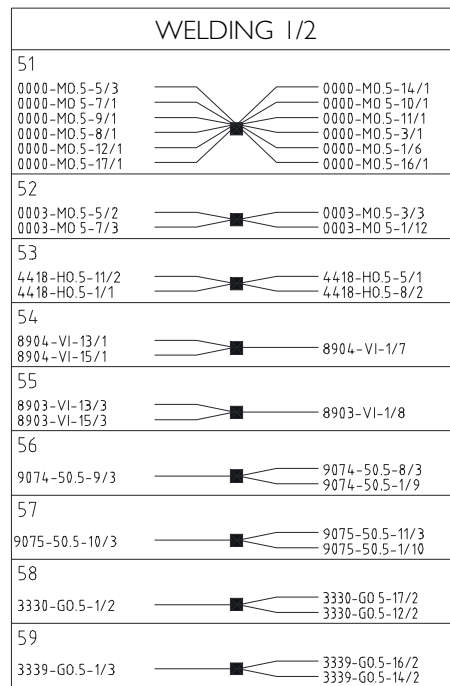


Figure 49



Component code	Description
C12	Cab-hood harness to dome lamp harness connector
C15	Dome lamp connectors to slide door bridle
C16	Dome lamp connectors to electrical lock contacts
37001	Front clearance lamp
39022	Interior door lamp with adjustable spot
39026	Side door lighting dome lamp
42102	Parking brake on indicator switch
53509	Interior lighting switch

VAN INTERIOR DOME LAMP CABLES 2/2

Figure 50

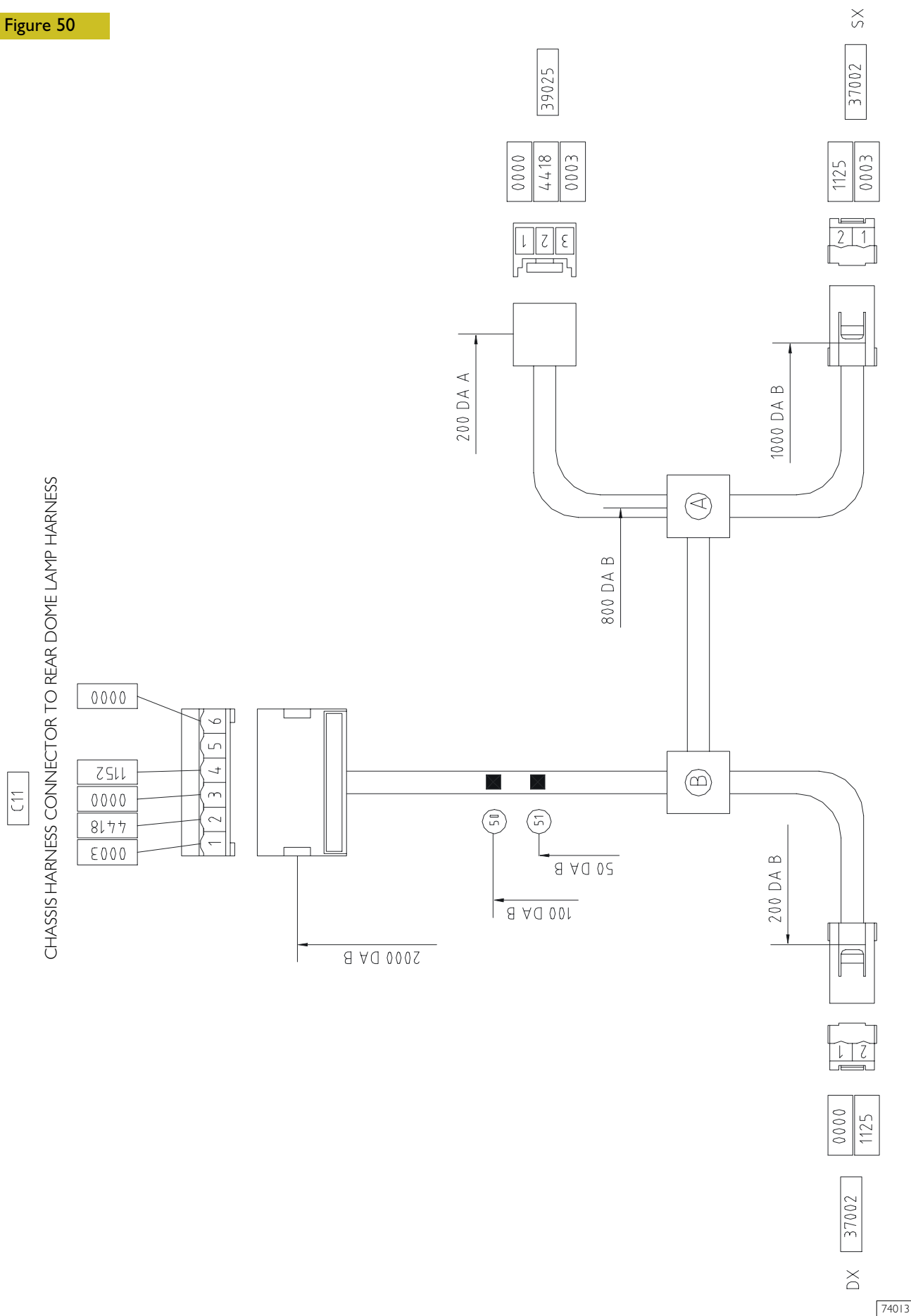
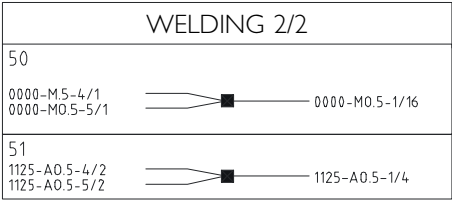


Figure 51



74014

Component code	Description
37002	Rear clearance lamp
39025	Liftgate lighting dome lamp
C I I	Chassis harness connector to rear dome lamp harness

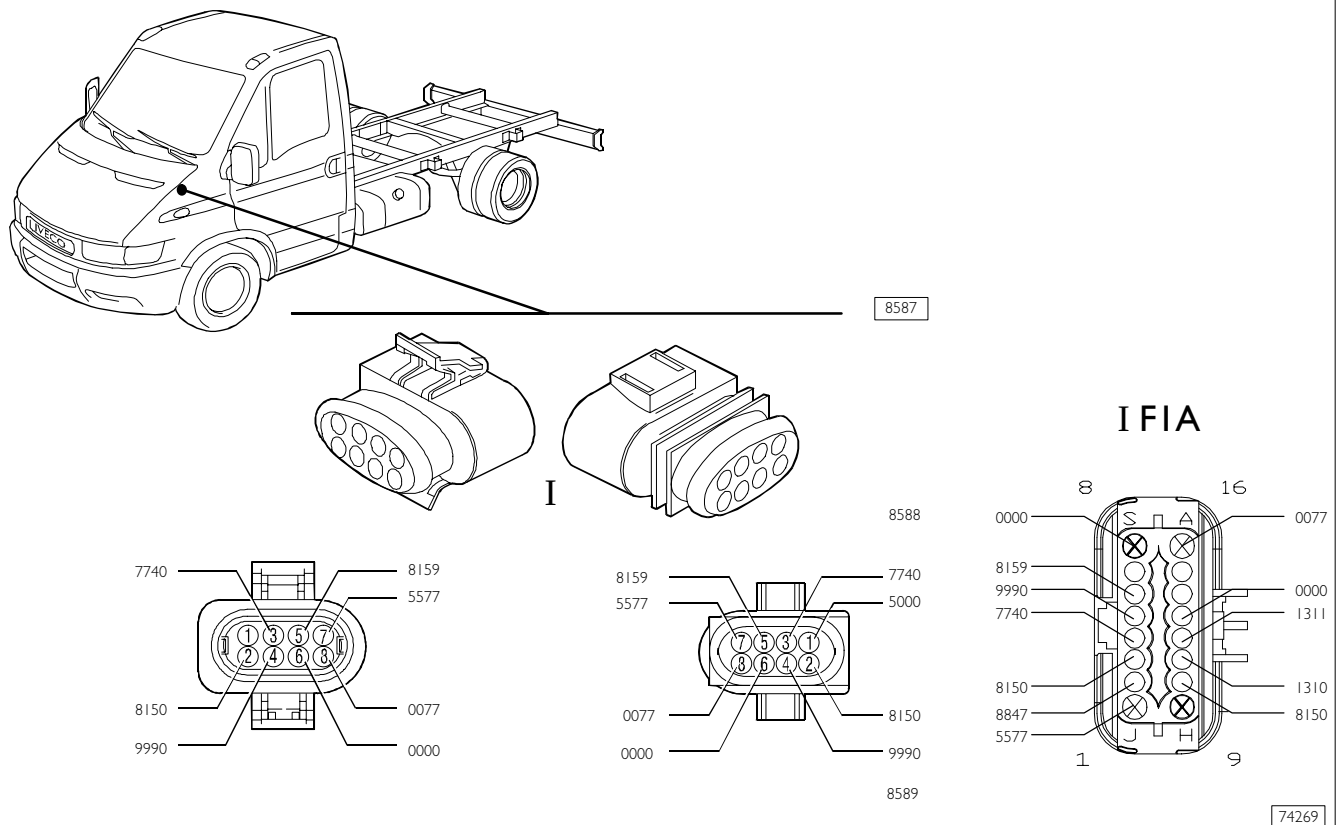
Figure 52



JUNCTION CONNECTORS

Connection between cab/bonnet cable and injection cable (Unijet)

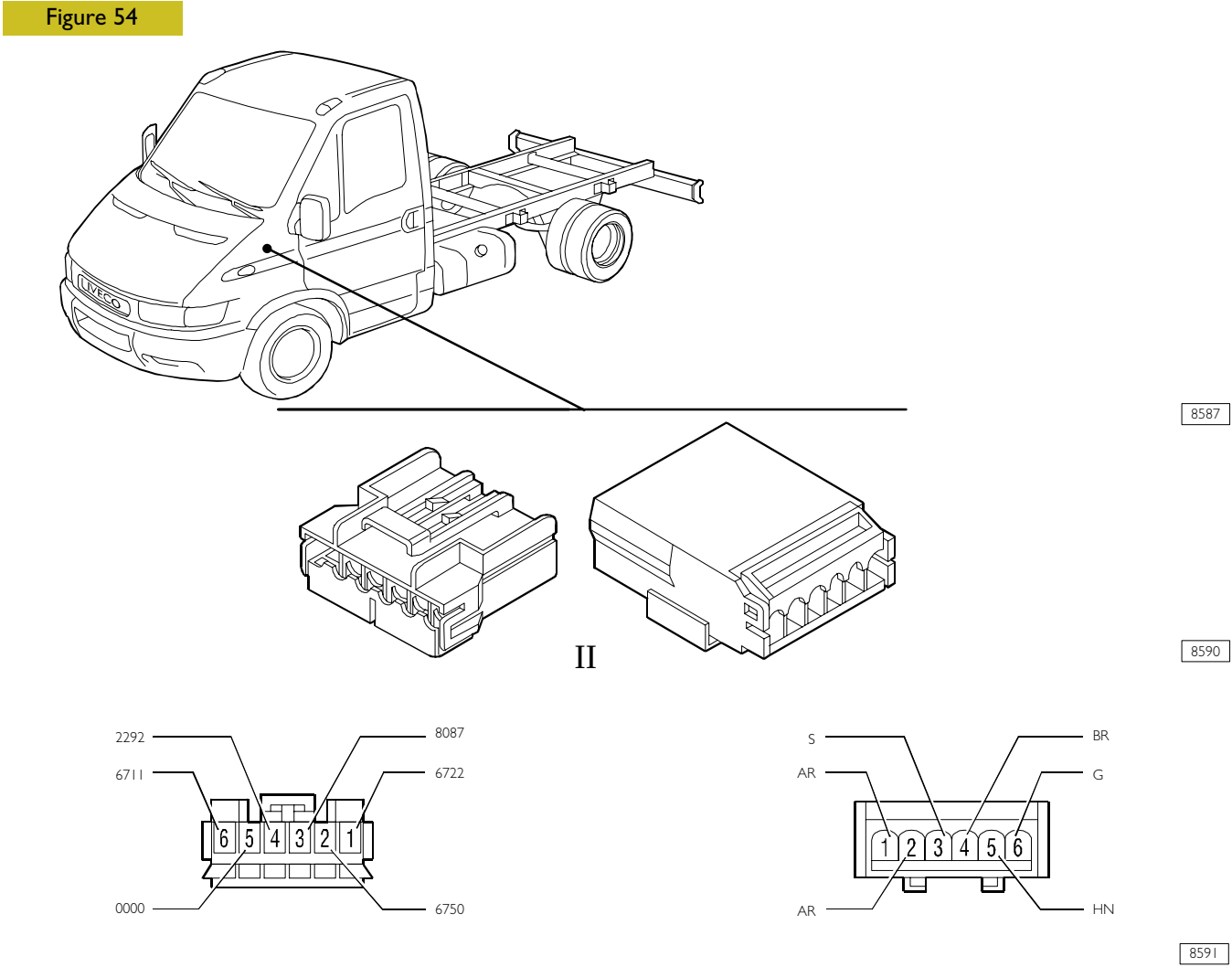
Figure 53



Ref.	Ref. (FIA)	Cable colour code	Function
I	1	5000	To pin 36 of EDC electronic control unit
	2	8150	Fuel pump engagement relay
	3	7740	Relay engagement for engine cooling electromagnetic joint control
	4	9990	Air conditioning compressor switching off
	5	8159	Fuel oil heating circuit switching on
	6	0000	Earth
	7	5577	Fuel oil heating resistance supply
	8	0077	Fuel oil heating resistance earth
	10	8150	Preheat plug centre positive (pin-86)
	11	1310	Preheat plug centre control (pin-D1), (EDC pin-B37)
	12	1311	Preheat plug centre control (pin-ST), (EDC pin-B42)
	13	0000	Preheat plug centre mass (pin-31)
	2	8847	Speed adjustment sensor positive
	7	—	Free
	9	—	Free
	14	—	Free
	15	—	Free

Connection between cab/bonnet cable and air bag cable

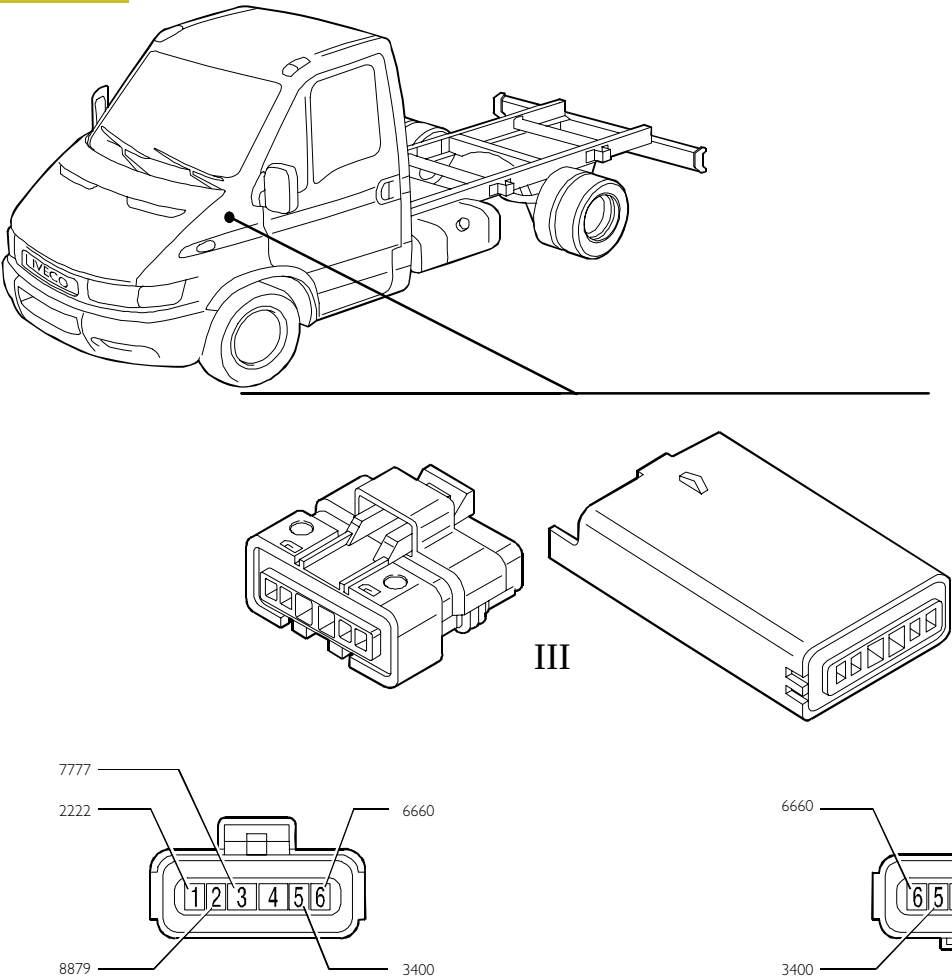
Figure 54



Ref.	Cable colour code	Function
II	1 6722-AR	Supply (+15)
	2 6750-AR	Supply (+15)
	3 8087-S	Supply (+15) for air bag electronic control unit
	4 2292-BR	To diagnostic connector
	5 0000-HN	Earth
	6 6711-G	To air bag failure warning light

Connection between cab/bonnet cable and rear differential lock cable

Figure 55



8587

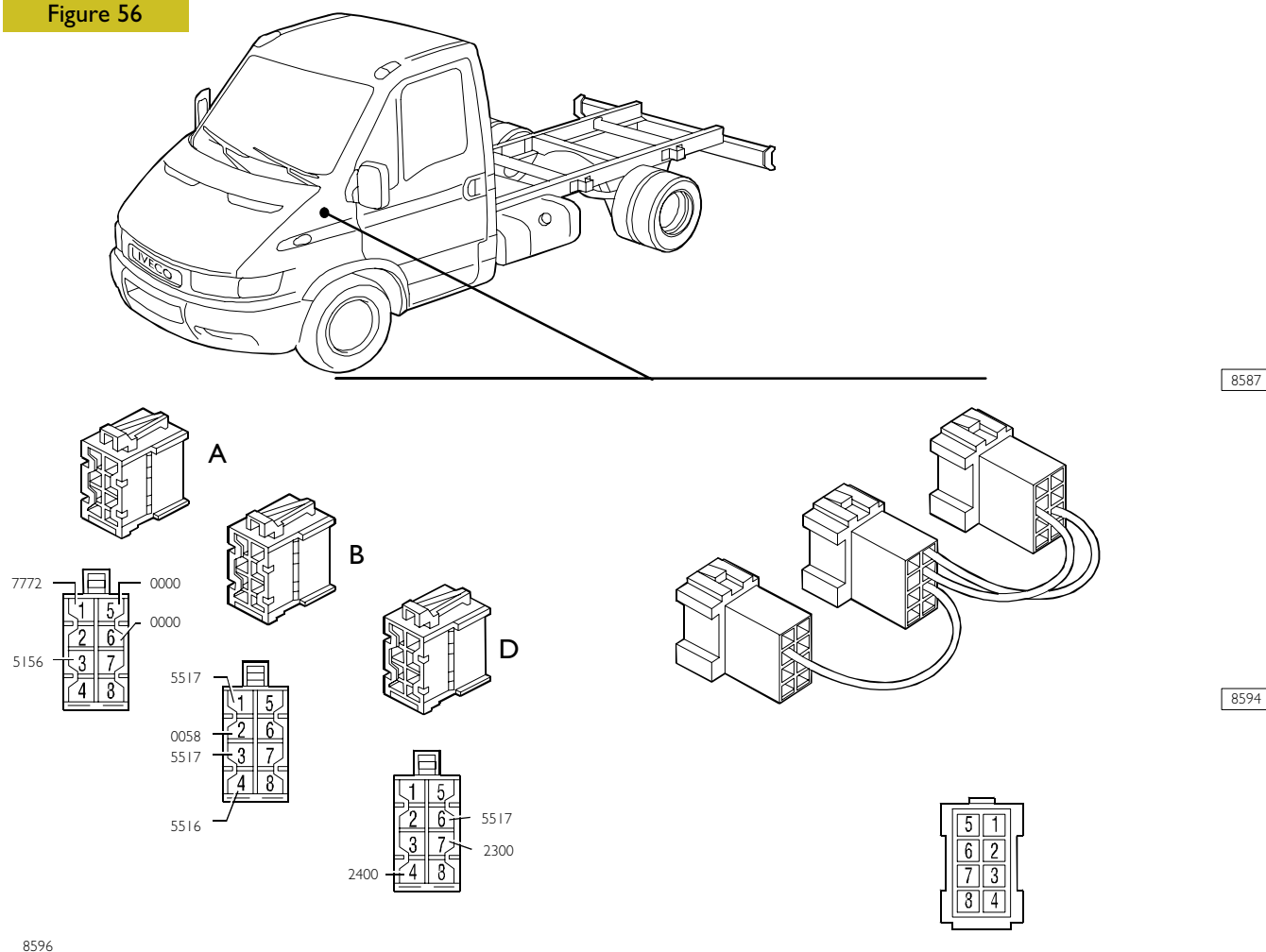
8592

8593

Ref.	Cable colour code	Function
III	1 2222	Tachometric signal
	2 8879	Supply (+15)
	3 7777	Supply (+30)
	4 –	Spare
	5 3400	Rear differential lock engagement
	6 6660	To rear differential lock on warning light

Connection between cab/bonnet cable and cable for tachometer

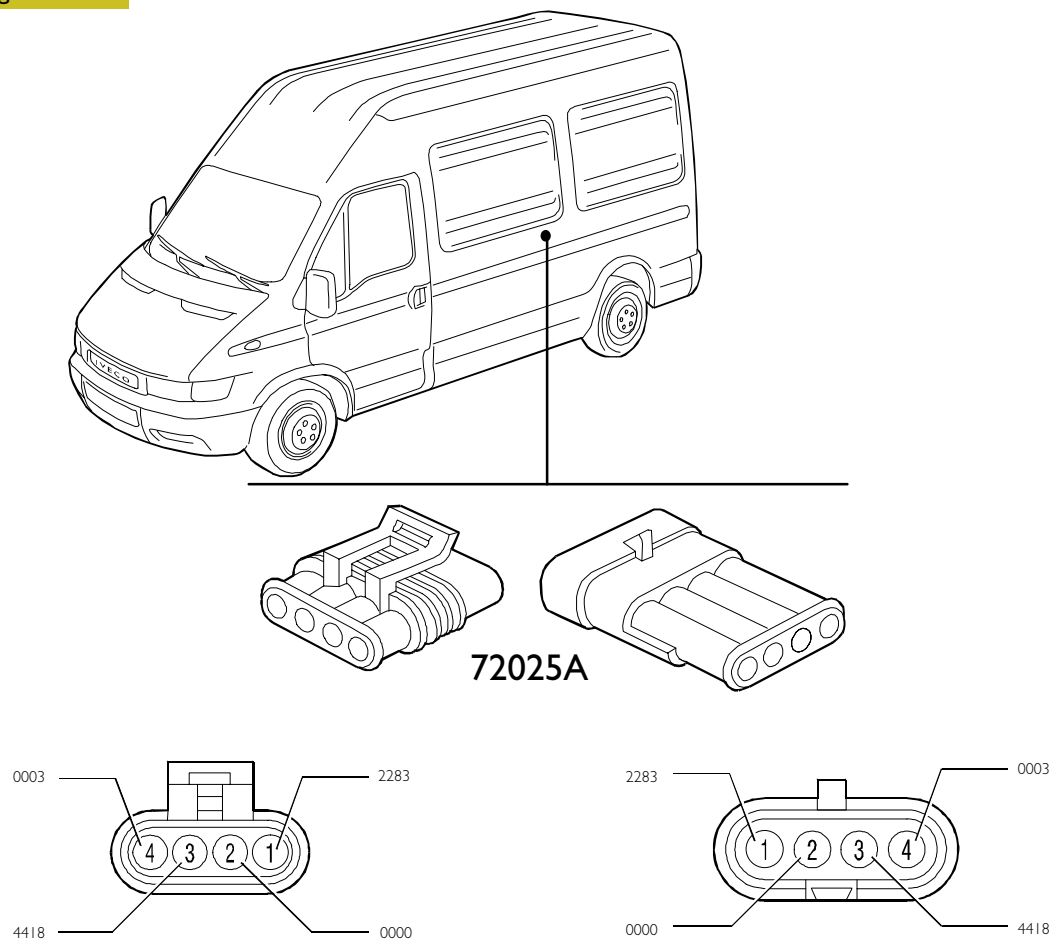
Figure 56



Ref.	Cable colour code	Function
4001IA (white)	1 7772	Supply (+30)
	2 —	Spare
	3 5156	Supply (+15)
	4 —	Spare
	5 0000	Earth
	6 0000	Earth
	7 —	Spare
	8 —	Spare
4001IB (yellow)	1 5517	Supply (+15)
	2 0058	Earth
	3 5517	Tachometric signal
	4 5516	To transmitter for tachometer
	5 —	Spare
	6 —	Spare
	7 —	Spare
	8 —	Spare
4001ID (brown)	1 —	Spare
	2 —	Spare
	3 —	Spare
	4 2400	To instrument cluster with warning lights
	5 —	Spare
	6 5517	Tachometric signal
	7 2300	To instrument cluster with warning lights
	8 —	Spare

Connection between frame cable and right tail lamp cable

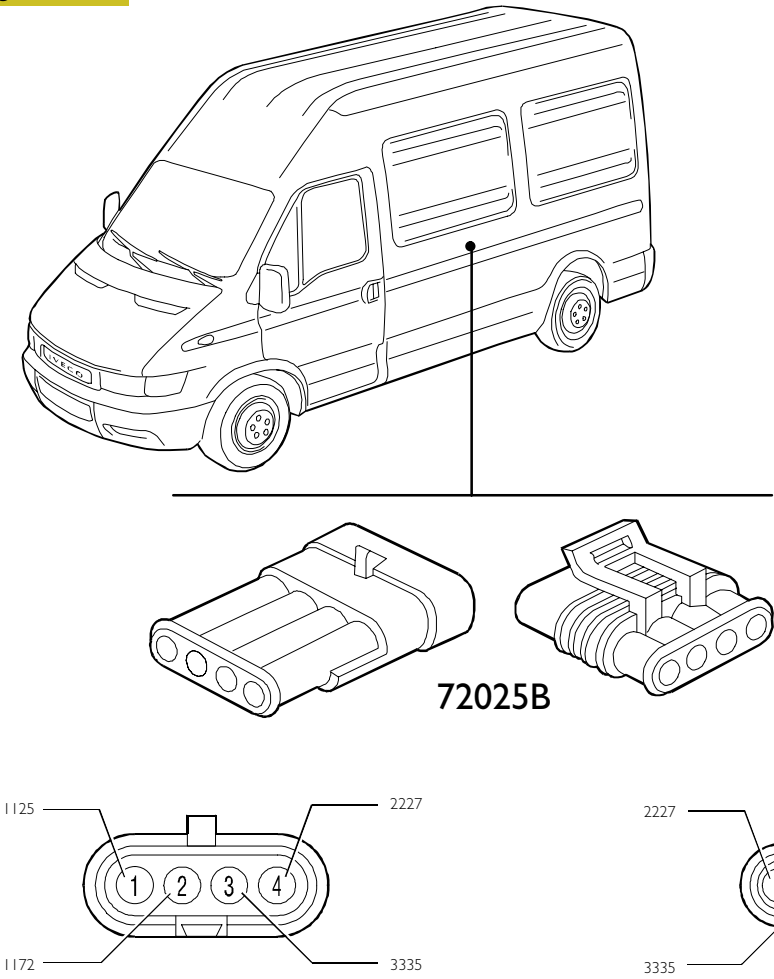
Figure 57



Ref.	Cable colour code	Function
72025A	1 2283	Rear fog guard
	2 0000	Earth
	3 4418	Rear roof lamp supply
	4 0003	Switching on interior lights

Connection between frame cable and right tail lamp cable

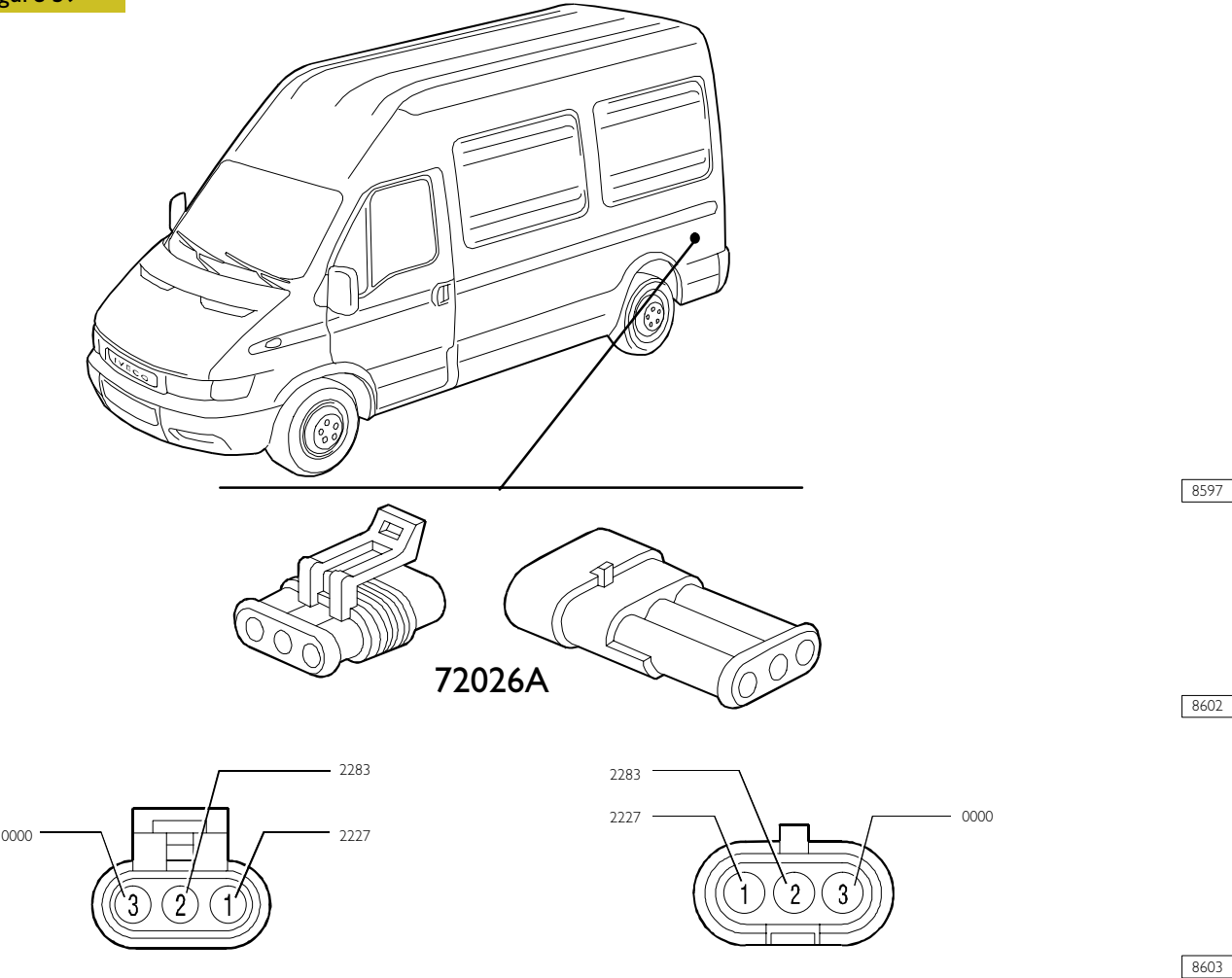
Figure 58



Ref.	Cable colour code	Function
72025B	1 1125	Direction indicator
	2 1172	Stop light
	3 3335	Side light
	4 2227	Reversing light

Connection between frame cable and left tail lamp cable

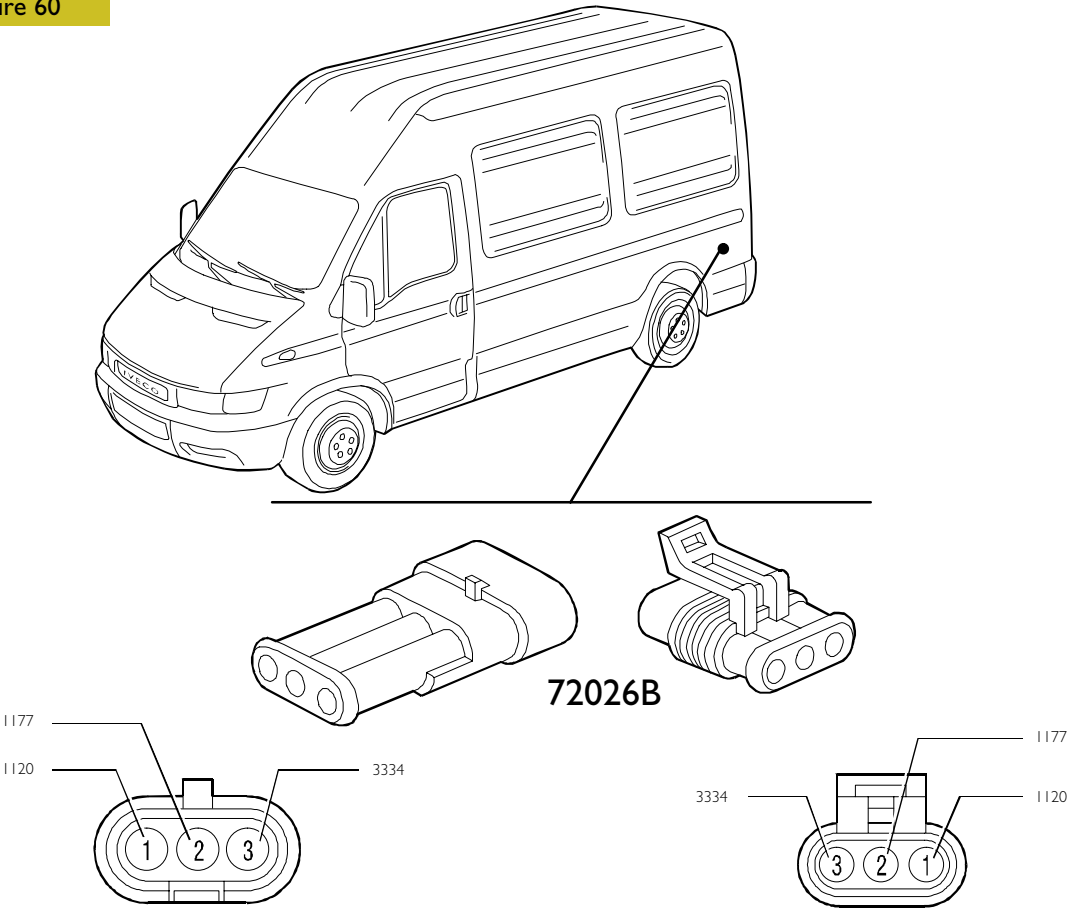
Figure 59



Ref.	Cable colour code	Function
72026A	1 2227	Reversing light
	2 2283	Rear fog guard
	3 0000	Earth

Connection between frame cable and left tail lamp cable

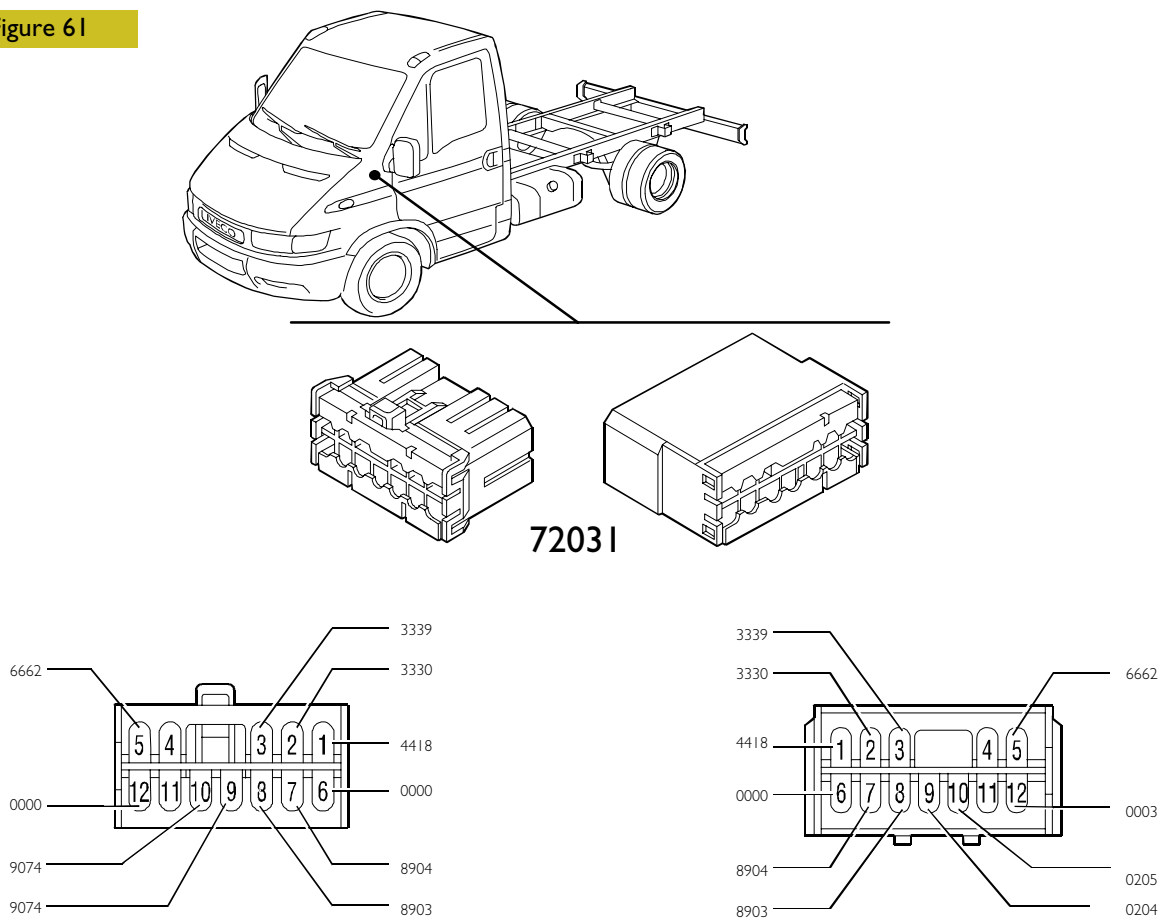
Figure 60



Ref.	Cable colour code	Function
1	1120	Direction indicator
72026B 2	1177	Stop light
3	3334	Side light

Connection between cab/bonnet cable and roof lamp cable inside cab

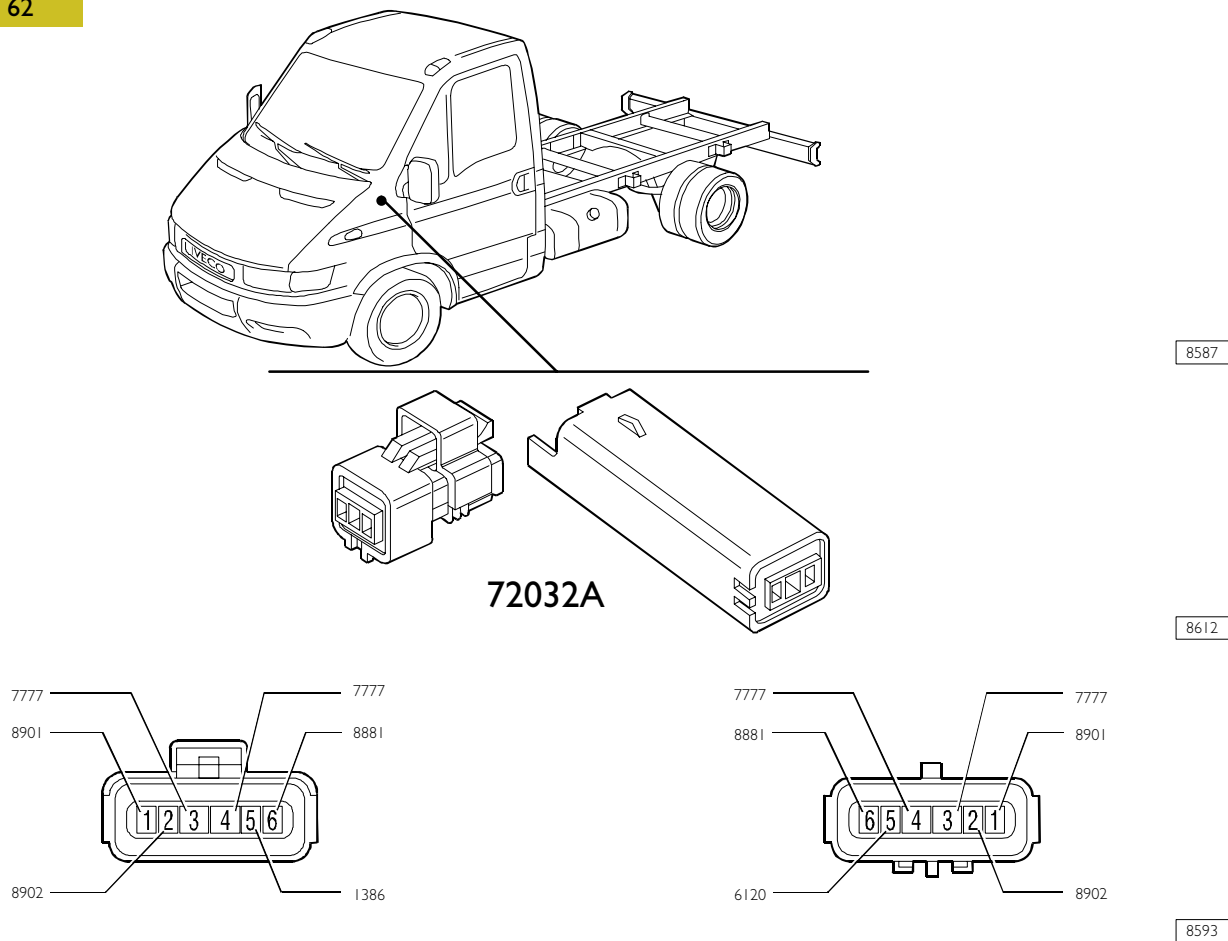
Figure 61



Ref.	Cable colour code	Function
72031	1 4418	Supply (+30)
	2 3330	Left front clearance light
	3 3339	Right front clearance light
	4 –	Spare
	5 6662	Handbrake on signal
	6 0000	Earth
	7 8904	To motor for side door closing control (van)
	8 8903	To motor for side door closing control (van)
	9 9074–0204	To switch for turning on interior lights (van)
	10 9074–0205	To switch for turning on interior lights (van)
	11 –	Spare
	12 0000–0003	Turning on roof lamp

Connection between cab/bonnet cable and 13 pin current socket cable or rear door opening/closing cable (van)

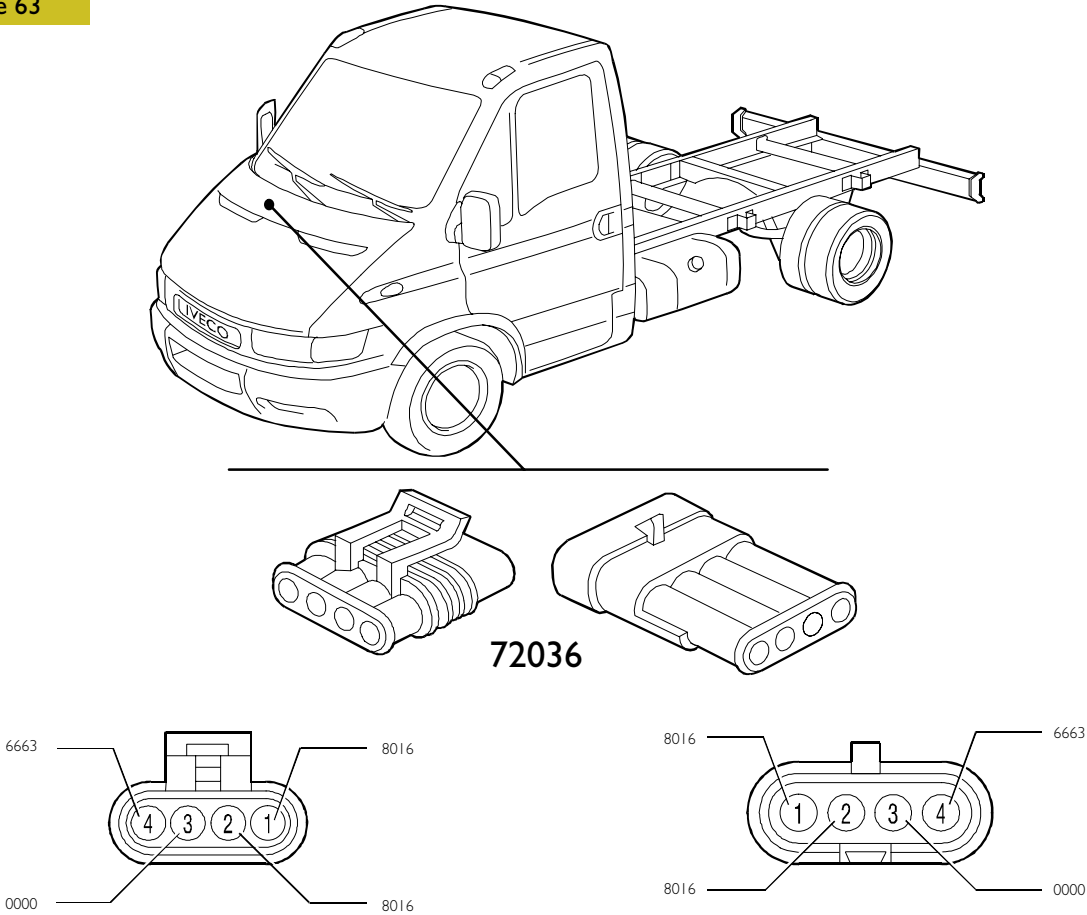
Figure 62



Ref.	Cable colour code	Function
72032A	1 8901	To pin 8 connector A of electronic control unit for central door locking
	2 8902	To pin 7 connector A of electronic control unit for central door locking
	3 7777	Supply (+30)
	4 7777	Supply (+30)
	5 1386-6120	Trailer direction indicator on signal
	6 8881	Supply (+15)

Connection between cab/bonnet cable and brake wear/air cleaner clogged cable

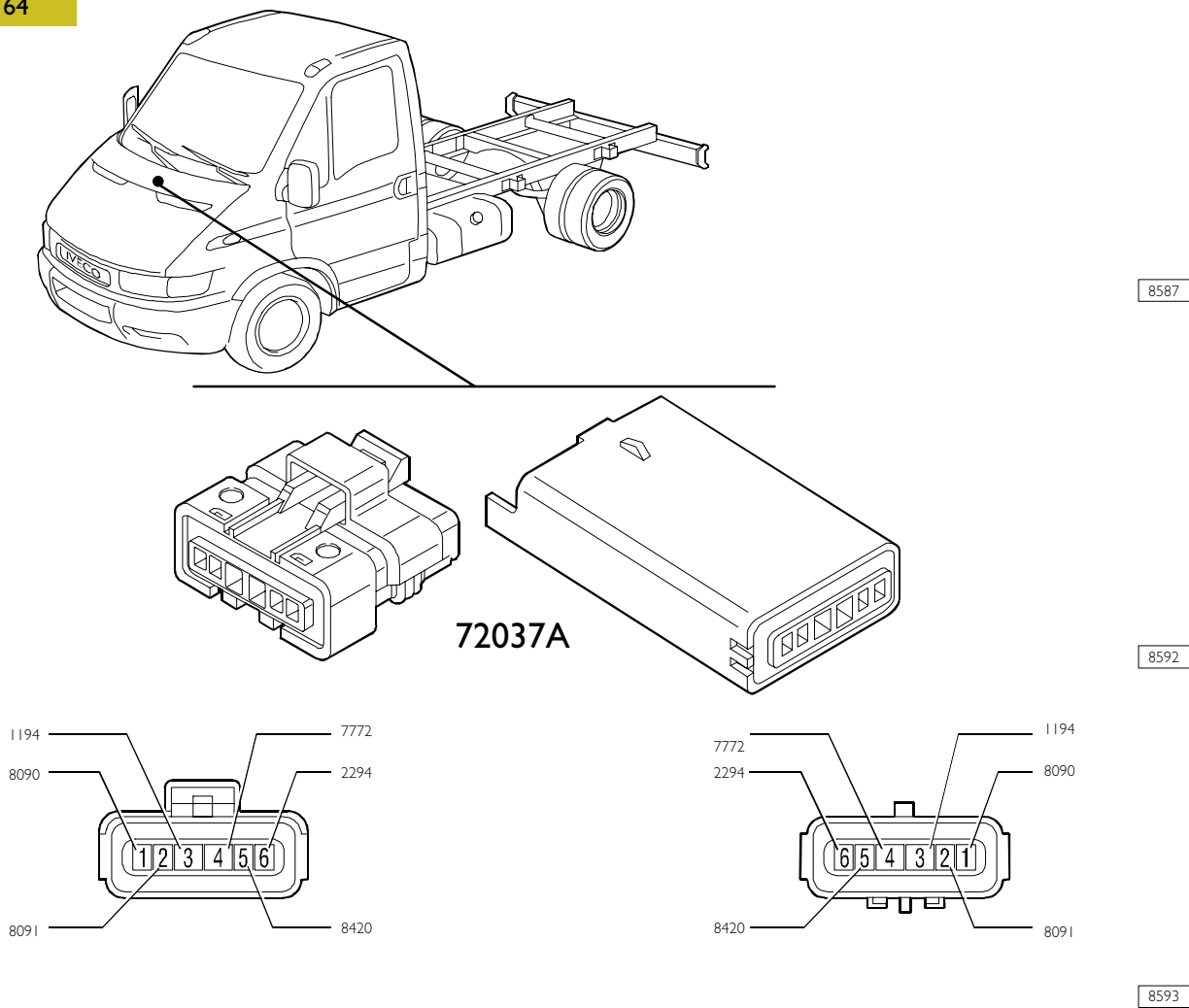
Figure 63



Ref.	Cable colour code	Function
72036	1 8016	Brake wear signal
	2 8016	Brake wear signal
	3 0000	Earth
	4 6663	Air cleaner clogged signal

Connection between cab/bonnet cable and self-levelling suspension cable

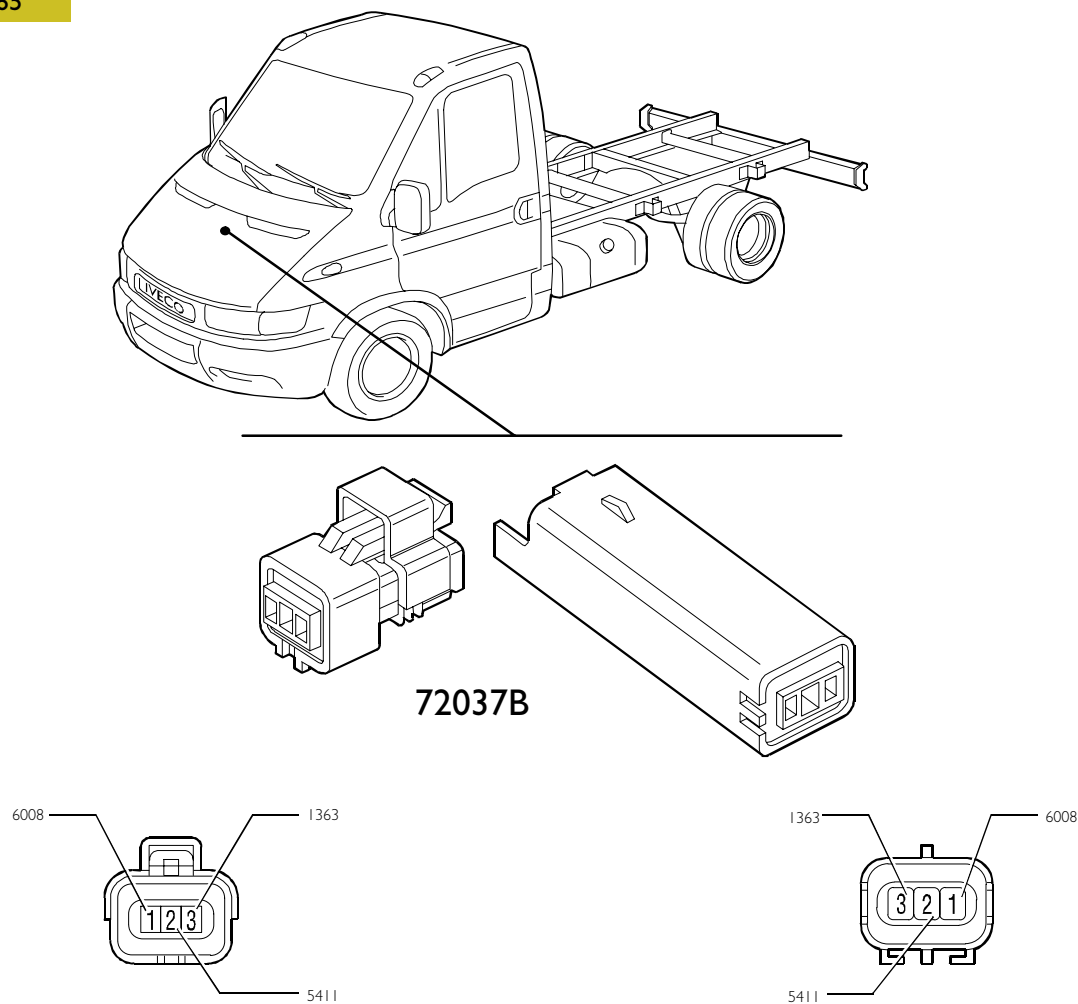
Figure 64



Ref.	Cable colour code	Function
72037A	1 8090	Supply (+15)
	2 8091	To pin 26 of electronic control unit for vehicle raising/lowering control
	3 1194	To 38-pin diagnostic connector (cell 15)
	4 7772	Supply (+30)
	5 8420	Suspension levelling control
	6 2294	To 38-pin diagnostic connector (cell 16)

Connection between cab/bonnet cable and self-levelling suspension cable

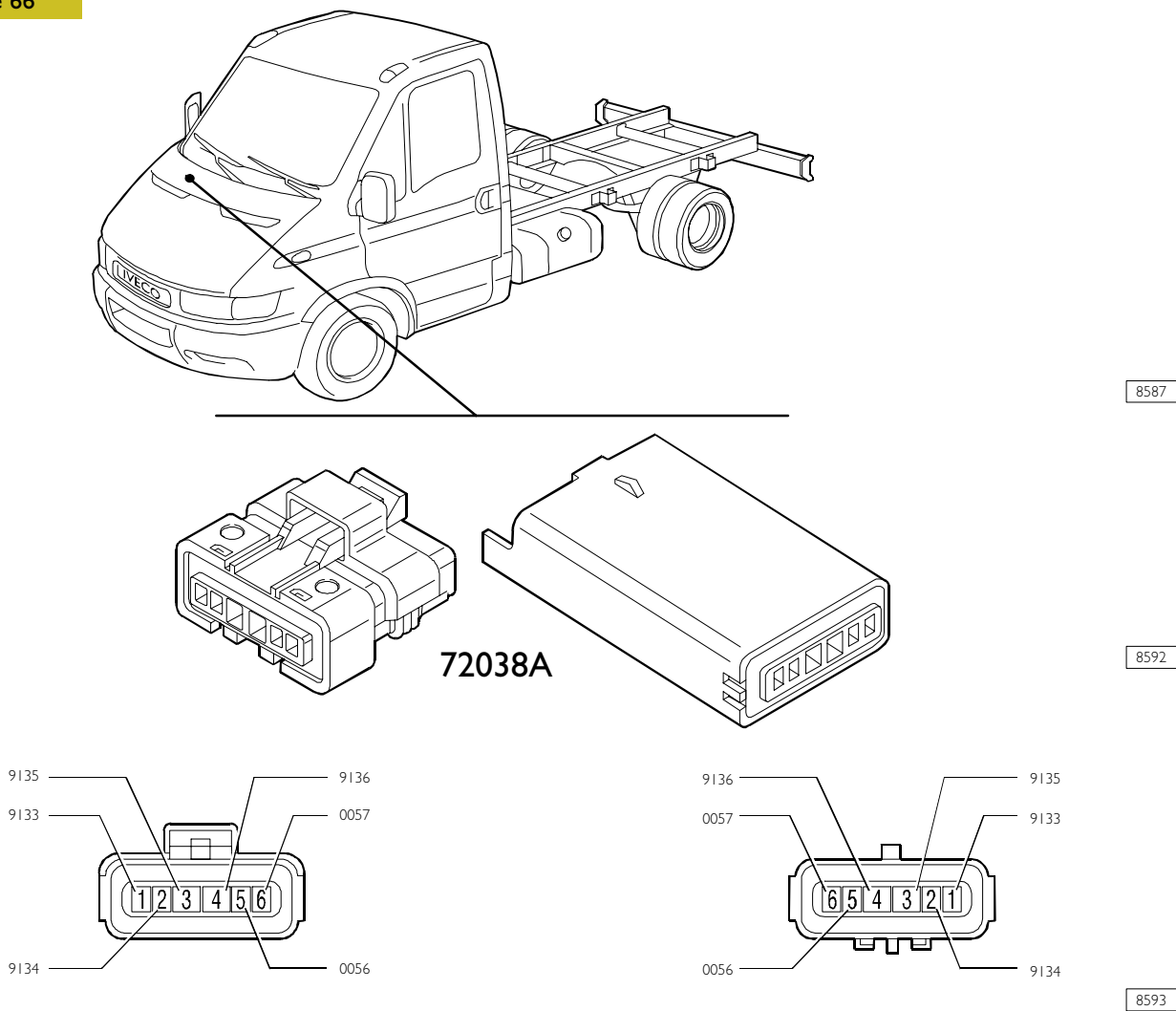
Figure 65



Ref.	Cable colour code	Function
1	6008	Self-levelling suspension system failure signal
72037B 2	5411	Tachometric signal
3	1363	Positive with vehicle stop signal switch engaged

Connection between cab/bonnet cable and total power takeoff cable

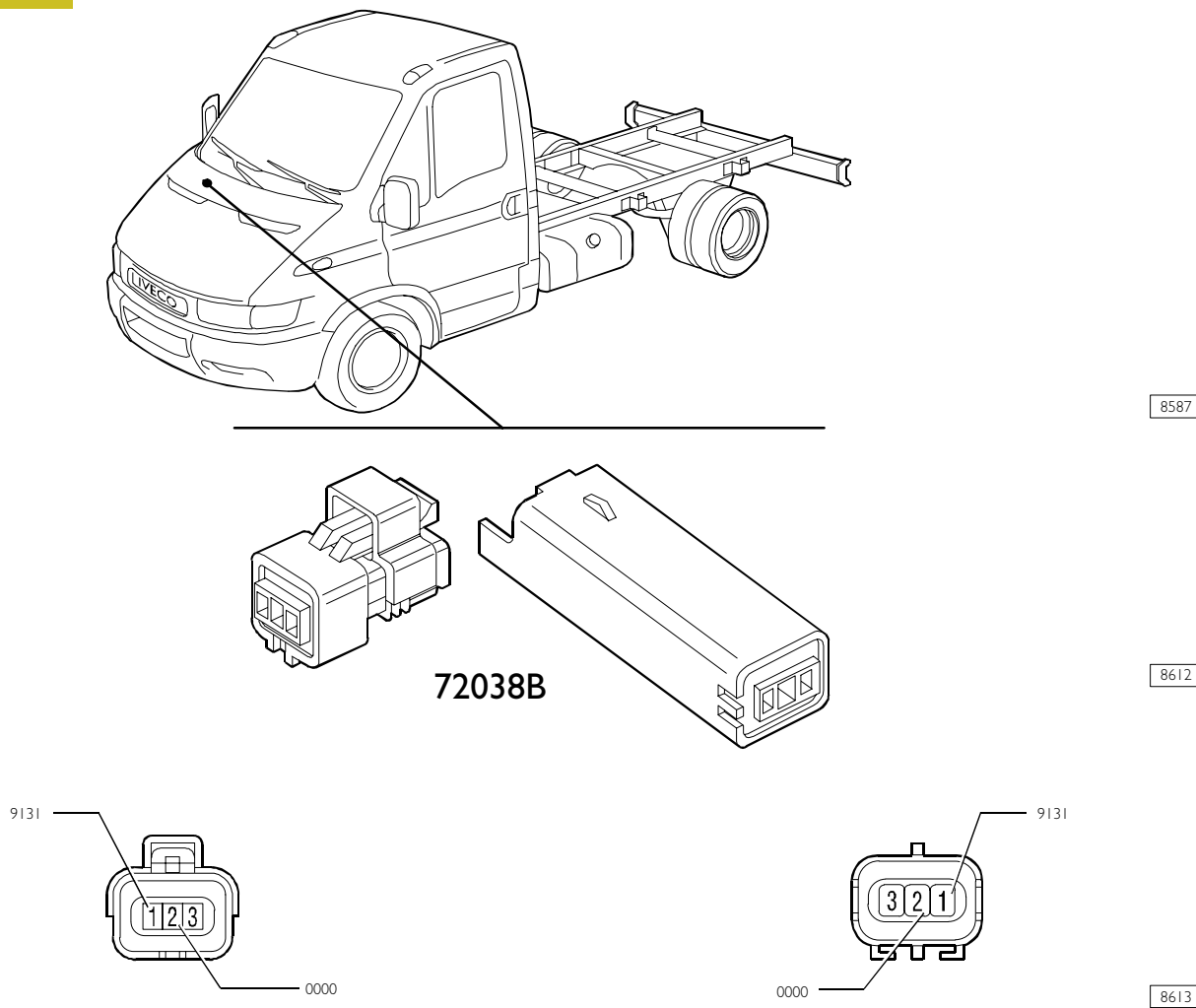
Figure 66



Ref.	Cable colour code	Function
72038A	1 9133	To electronic control unit for switching on total power takeoff (connector P2 cell 3)
	2 9134	To electronic control unit for switching on total power takeoff (connector P2 cell 4)
	3 9135	To electronic control unit for switching on total power takeoff (connector P2 cell 5)
	4 9136	To electronic control unit for switching on total power takeoff (connector P2 cell 6)
	5 0056	To electronic control unit for switching on total power takeoff (connector P2 cell 7)
	6 0057	To electronic control unit for switching on total power takeoff (connector P2 cell 8)

Connection between cab/bonnet cable and total power takeoff cable

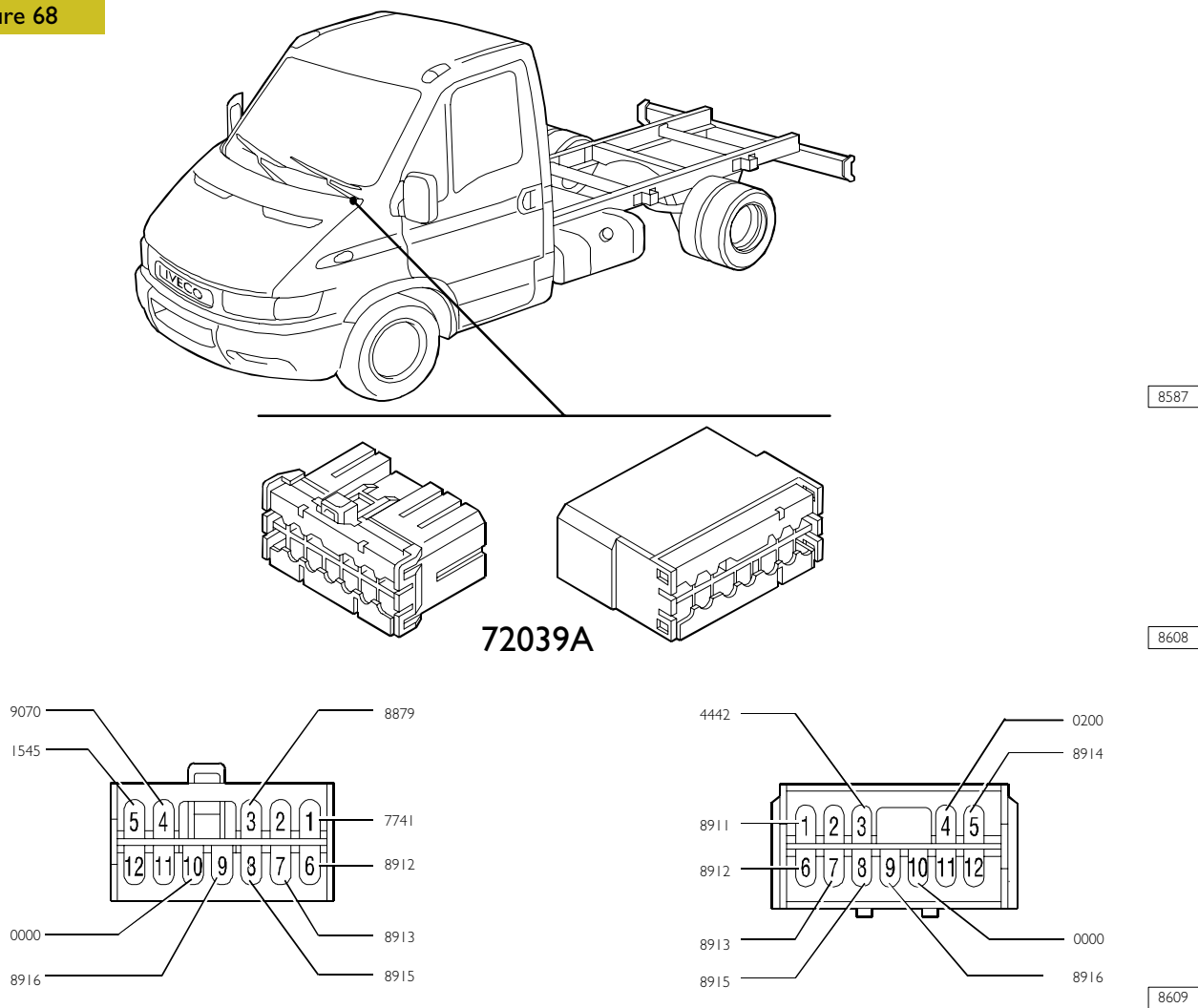
Figure 67



Ref.	Coble colour code	Function
1	9131	To switch for total power takeoff engaged signal
72038B 2	0000	Earth
3	—	Spare

Connection between cab/bonnet cable and antitheft cable

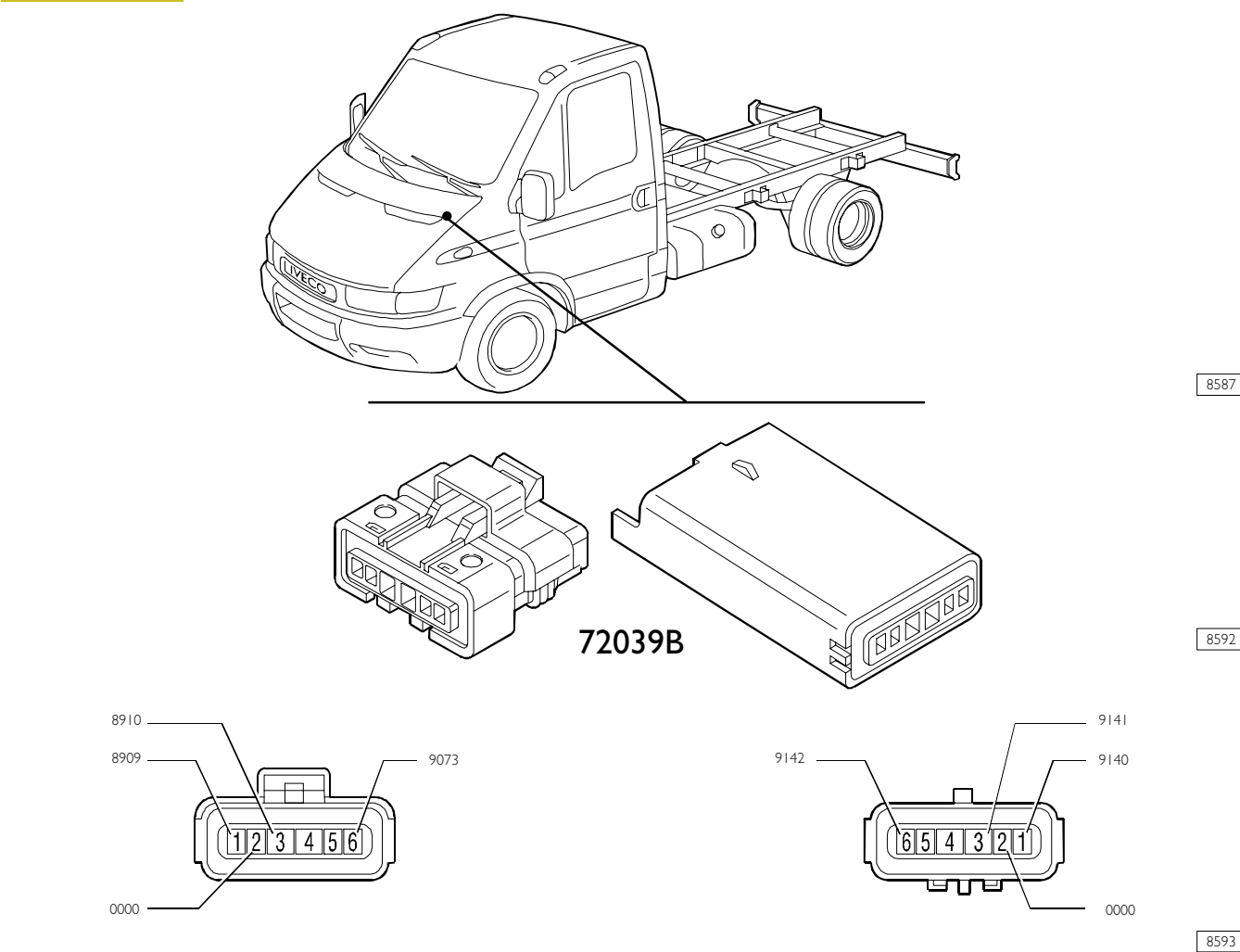
Figure 68



Ref.	Cable colour code	Function
72039A	1 7741-8911	Supply (+30)
	2 -	Spare
	3 8879-4442	Supply (+15)
	4 9070-0200	To switch for locking rear door (van)
	5 1545-8914	To volumetric sensor
	6 8912	To volumetric sensor
	7 8913	To volumetric sensor
	8 8915	To volumetric sensor
	9 8916	To volumetric sensor
	10 0000	Earth
	11 -	Spare
	12 -	Spare

Connection between cab/bonnet cable and antitheft cable with central door locking

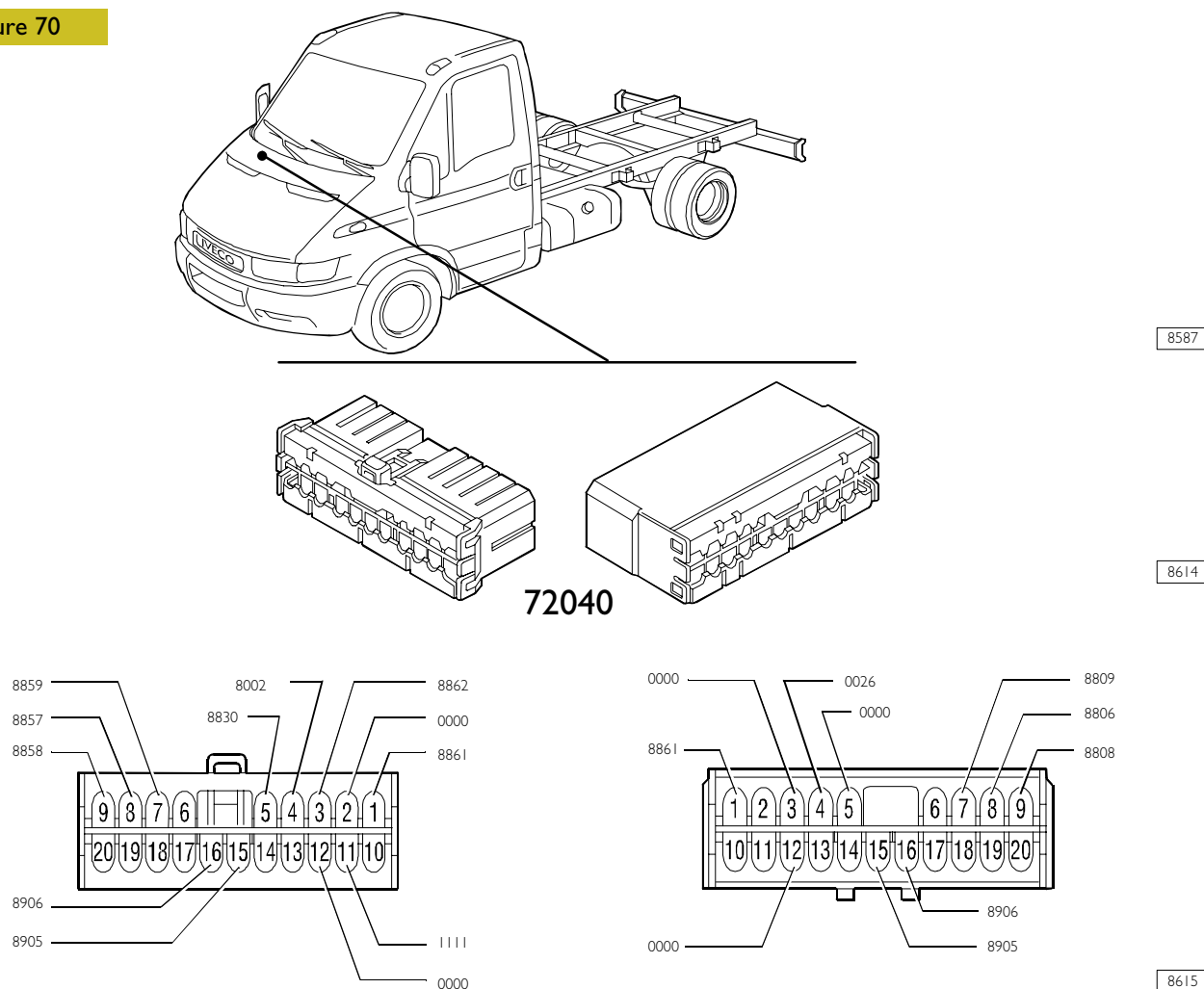
Figure 69



Ref.	Cable colour code	Function
72039B	1 8909-9140	Supply (+15)
	2 0000	Earth
	3 8910-9141	Supply (+30)
	4 -	Spare
	5 -	Spare
	6 9073-9142	To self-powered siren for alarm

Connection between cab/bonnet cable and right door cable

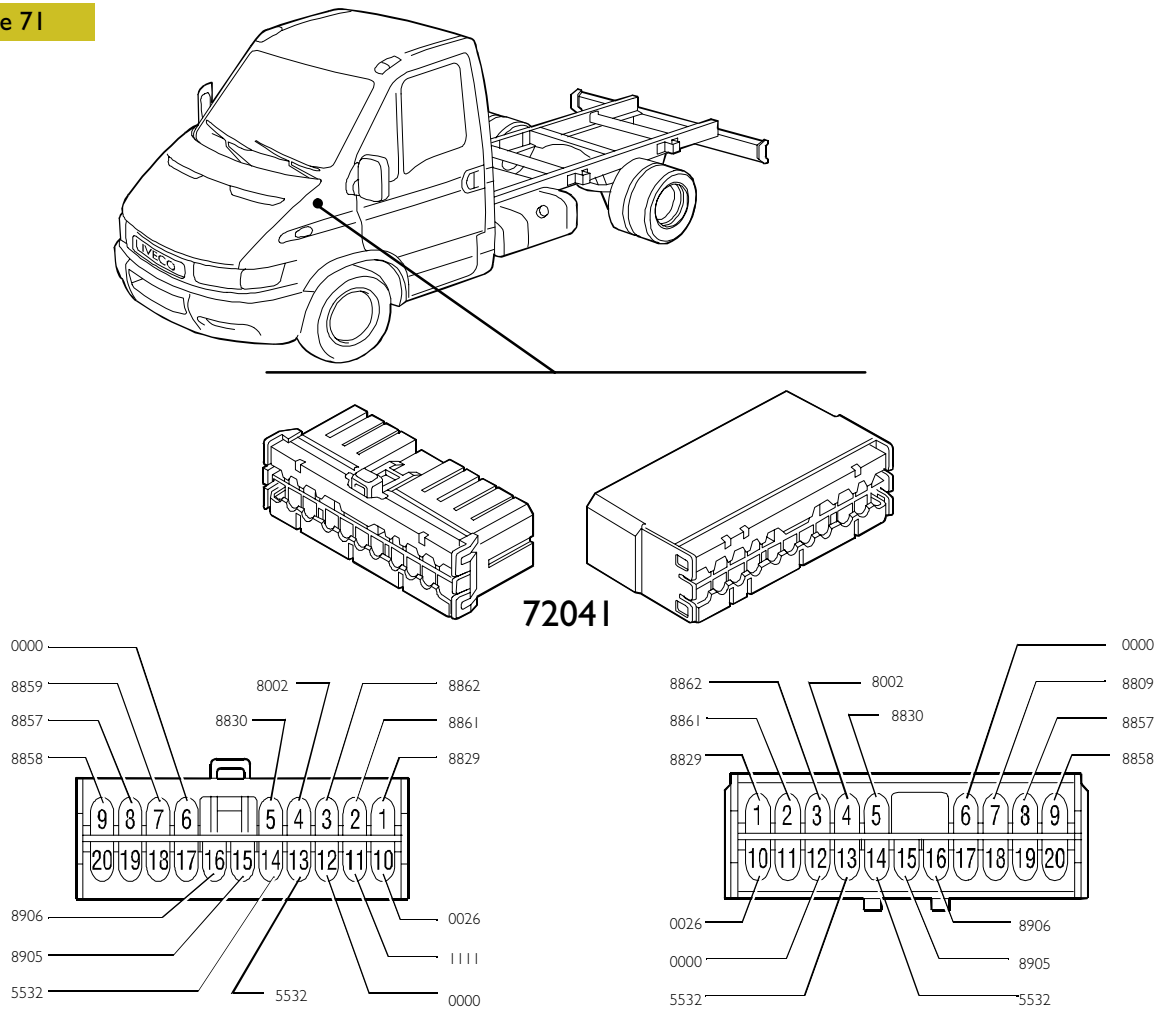
Figure 70



Ref.	Cable colour code	Function
72040	1 8861	Supply (+15/A)
	2 0000	Earth
	3 8862-0000	Connection between switches for passenger's door power window
	4 8002-0026	Connection between switches for passenger's door power window
	5 8830	Heated rear-view mirror
	6 -	Spare
	7 8859-8809	Electrically-operated aimable mirror
	8 8857-8806	Electrically-operated aimable mirror
	9 8858-8808	Electrically-operated aimable mirror
	10 -	Spare
	11 1111	To connection between cab/bonnet cable and left door cable
	12 0000	Earth
	13 -	Spare
	14 -	Spare
	15 8905	To door closing control motor
	16 8906	To door closing control motor
	17 -	Spare
	18 -	Spare
	19 -	Spare
	20 -	Spare

Connection between cab/bonnet cable and left door cable

Figure 71



8587

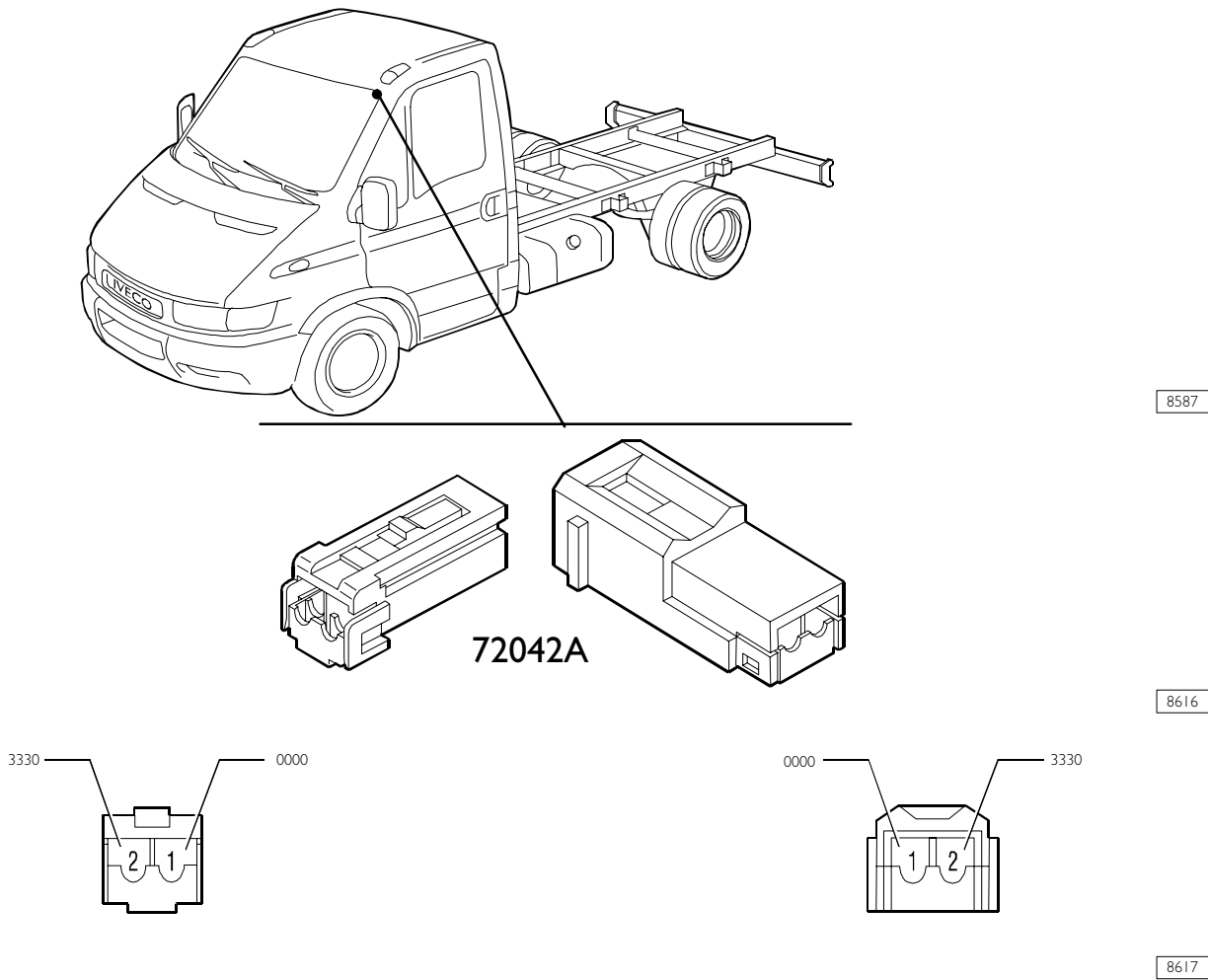
8614

8615

Ref.	Cable colour code	Function
72041	1 8829	Supply (+15)
	2 8861	Supply (+15/A)
	3 8862	Connection between switches for passenger's door power window
	4 8002	Connection between switches for passenger's door power window
	5 8830	Heated rear-view mirror
	6 0000	Earth
	7 8859-8809	Electrically-operated aimable mirror
	8 8857	Electrically-operated aimable mirror
	9 8858	Electrically-operated aimable mirror
	10 0026	Earth
	11 1111	To connection between cab/bonnet cable and left door cable
	12 0000	Earth
	13 5532	To outside temperature sensor
	14 5532	To outside temperature sensor
	15 8905	To door closing control motor
	16 8906	To door closing control motor
	17 -	Spare
	18 -	Spare
	19 -	Spare
	20 -	Spare

Connection between cab roof lamp and left front clearance light cable

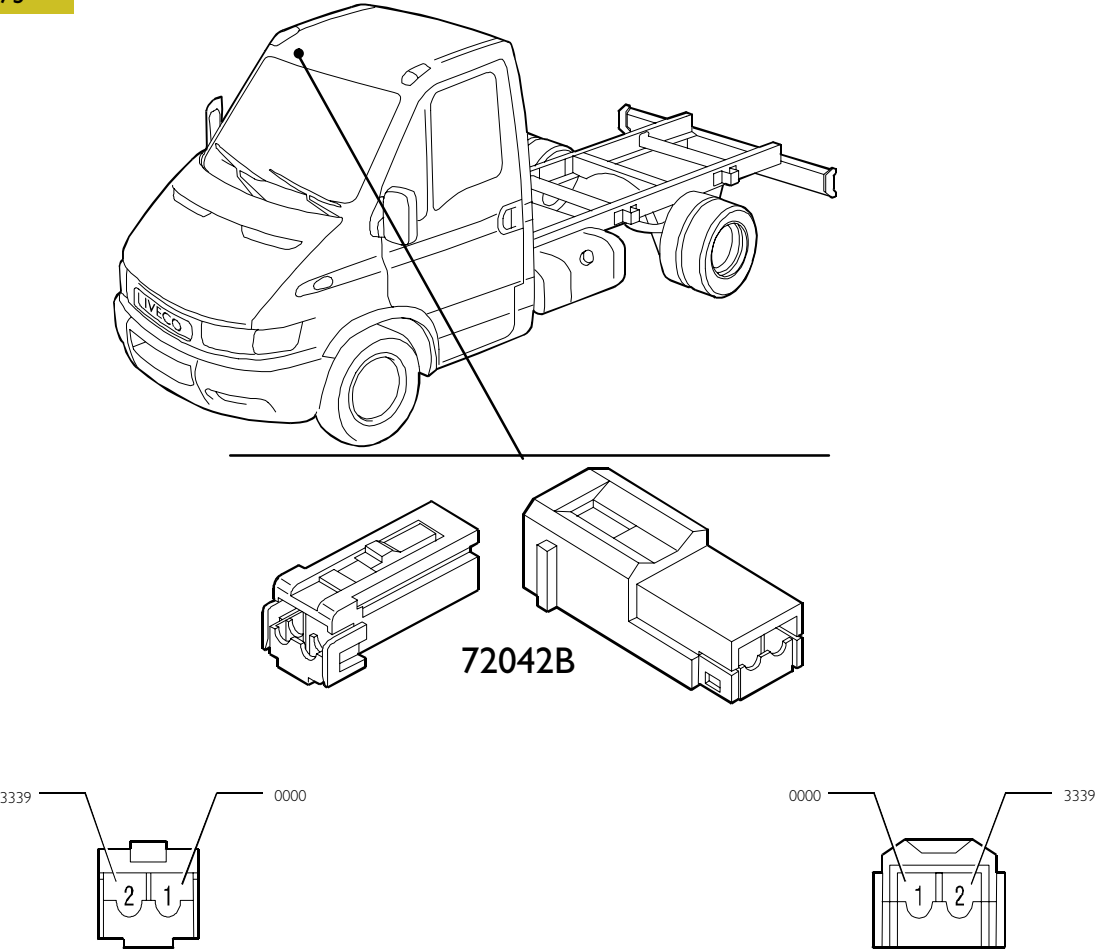
Figure 72



Ref.	Cable colour code	Function
72042A	1 0000 2 3330	Earth Front clearance light

Connection between cab roof lamp and right front clearance light cable

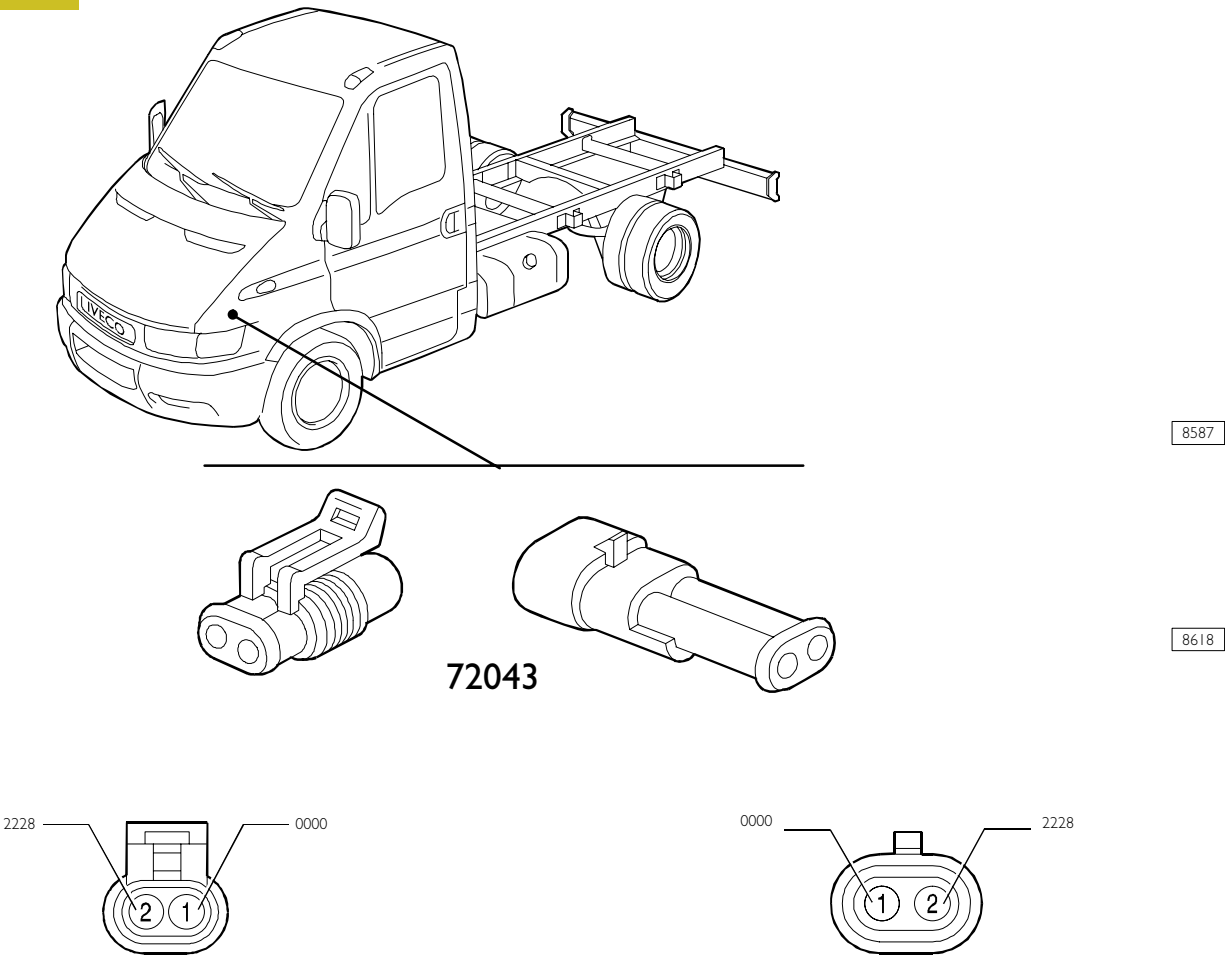
Figure 73



Ref.	Cable colour code	Function
72042B	1 0000	Earth
	2 3330	Front clearance light

Connection between cab/bonnet cable and fog lamp cable

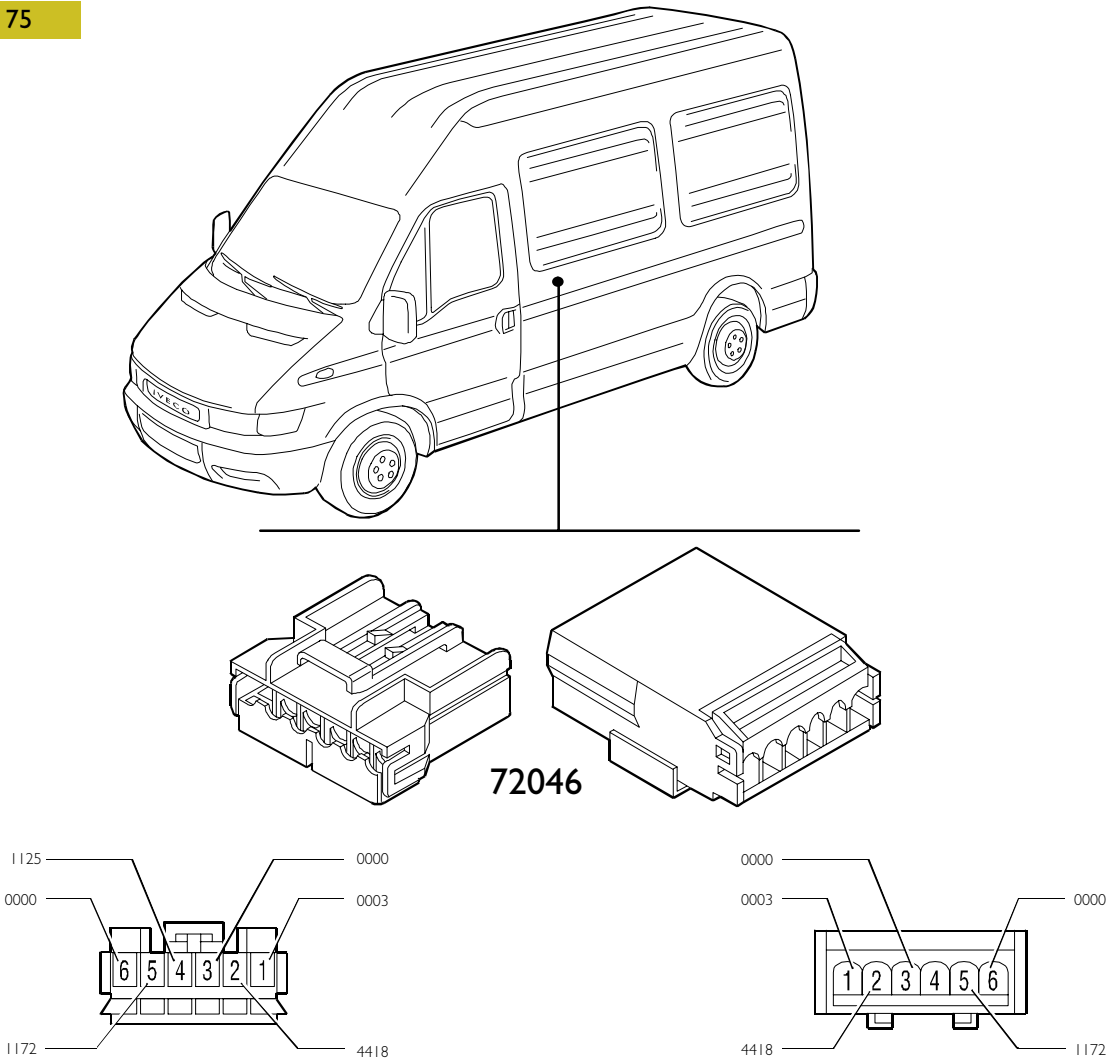
Figure 74



Ref.	Cable colour code	Function
72043	1 0000 2 2228	Earth Fog lamps

Connection between right tail light cable and rear roof lamp cable

Figure 75



8587

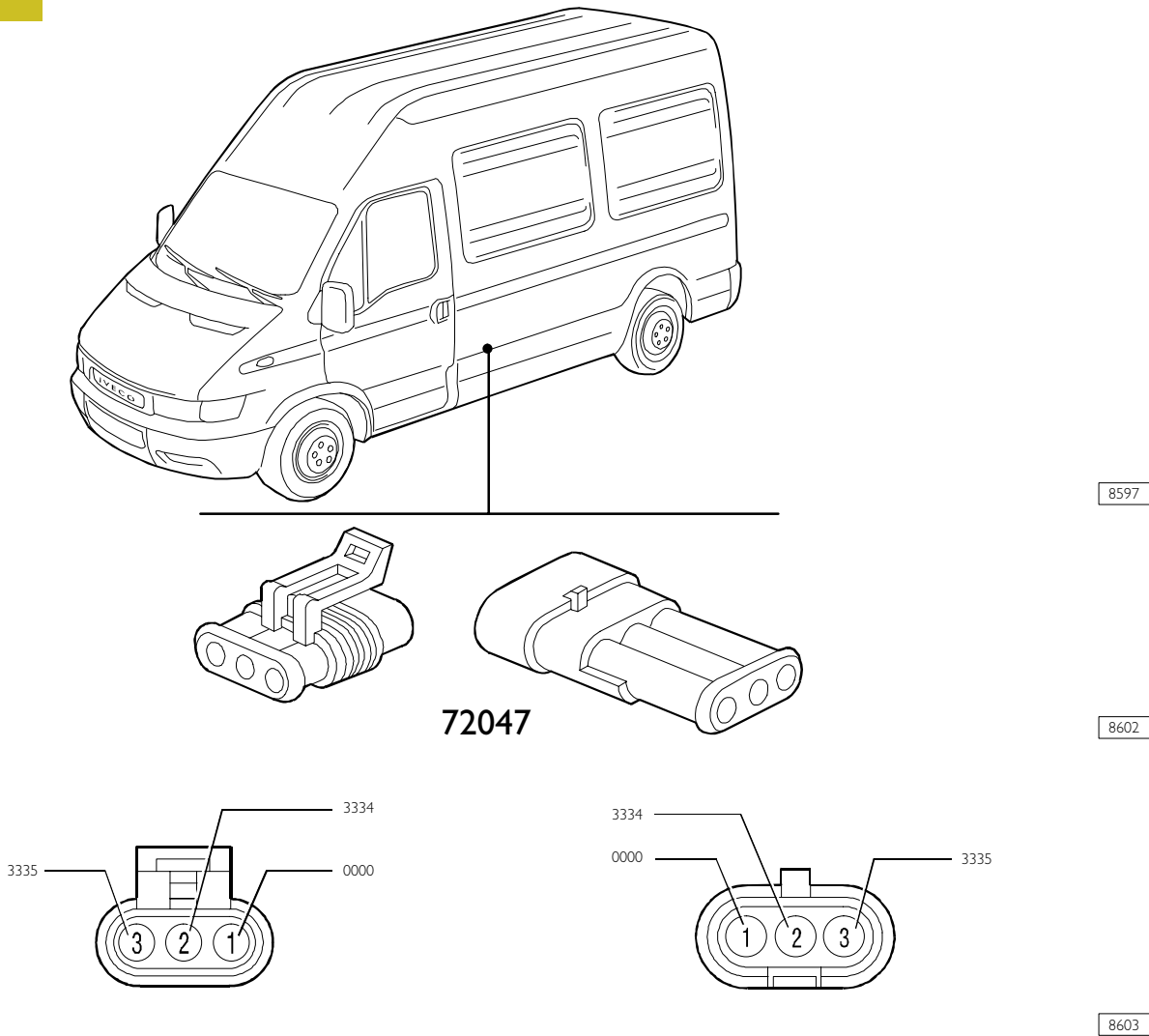
8590

8591

Ref.	Cable colour code	Function
72046	1 0003	Switching on roof lamp
	2 4418	Supply (+30)
	3 0000	Earth
	4 1125	Positive with direction indicators/hazard warning lights on
	5 1172	Additional stop light
	6 0000	Earth

Connection between frame cable and side clearance lights cable

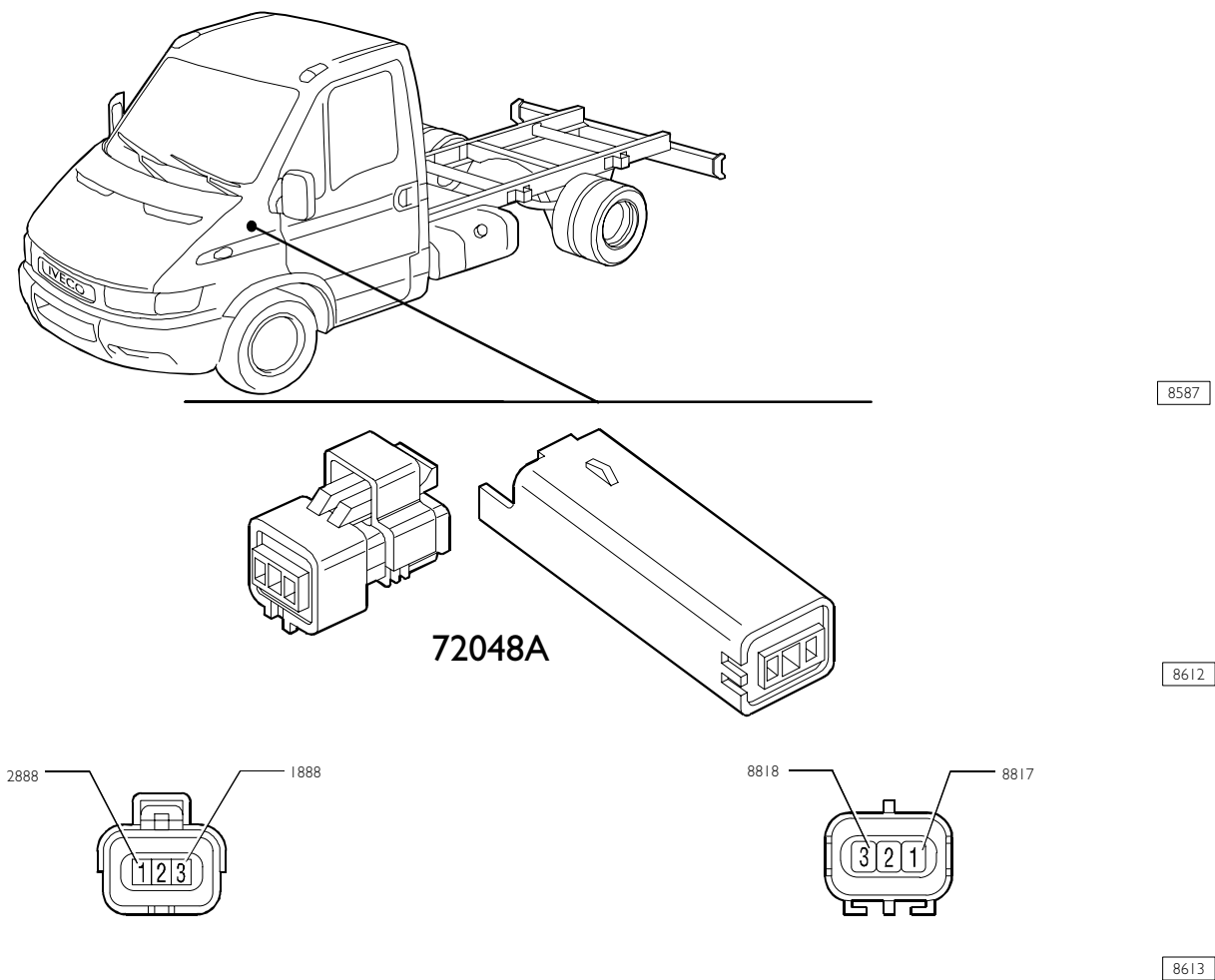
Figure 76



Ref.	Cable colour code	Function
72047	1 0000	Earth
	2 3334	Vehicle right side clearance lights
	3 3335	Vehicle left side clearance lights

Connection between cab/bonnet cable and ABS cable

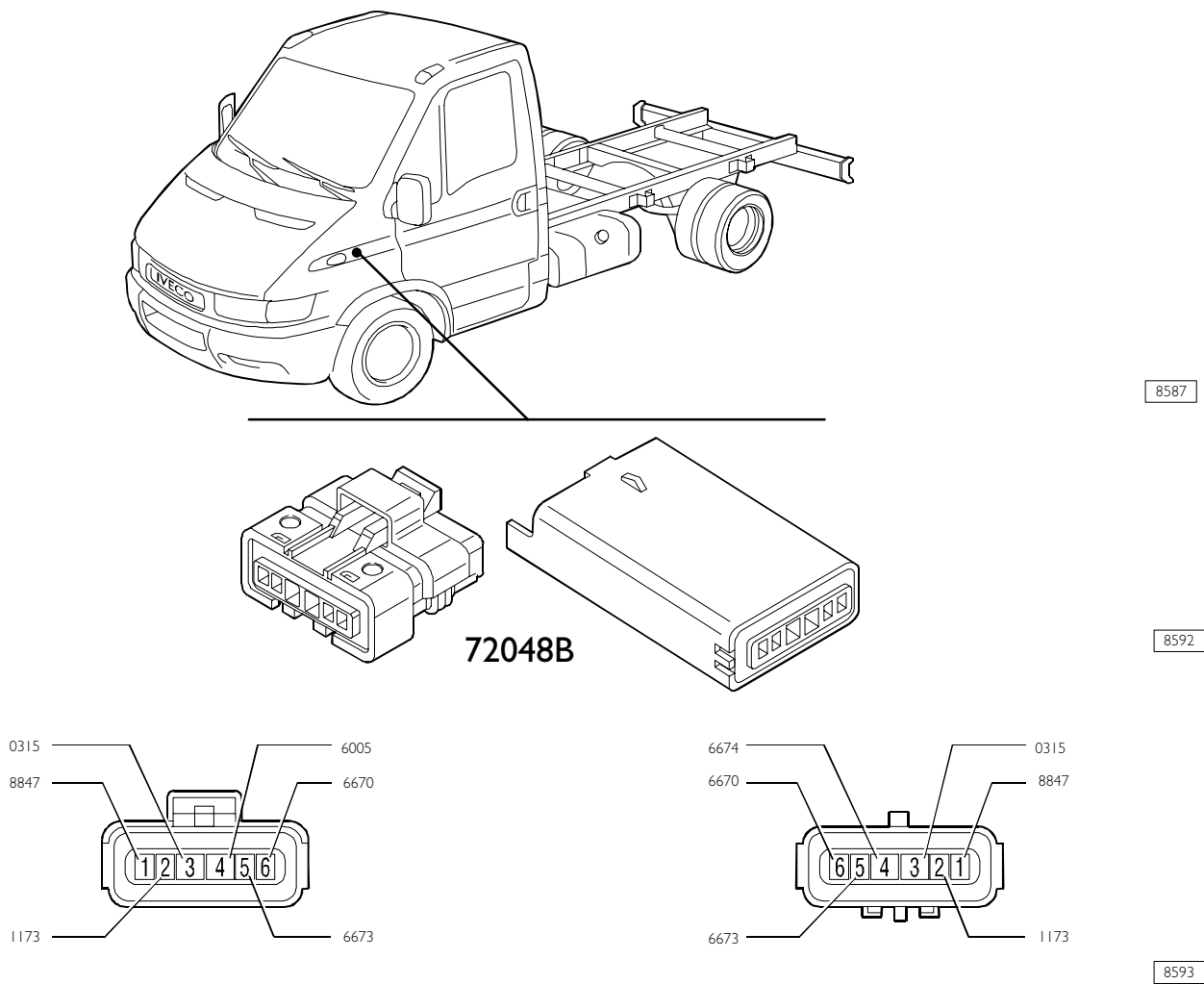
Figure 77



Ref.	Cable colour code	Function
72048A	1 2888-8817	To 38-pin diagnostic connector (cell 4)
	2 -	Spare
	3 1888-8818	To 38-pin diagnostic connector (cell 3)

Connection between cab/bonnet cable and ABS cable

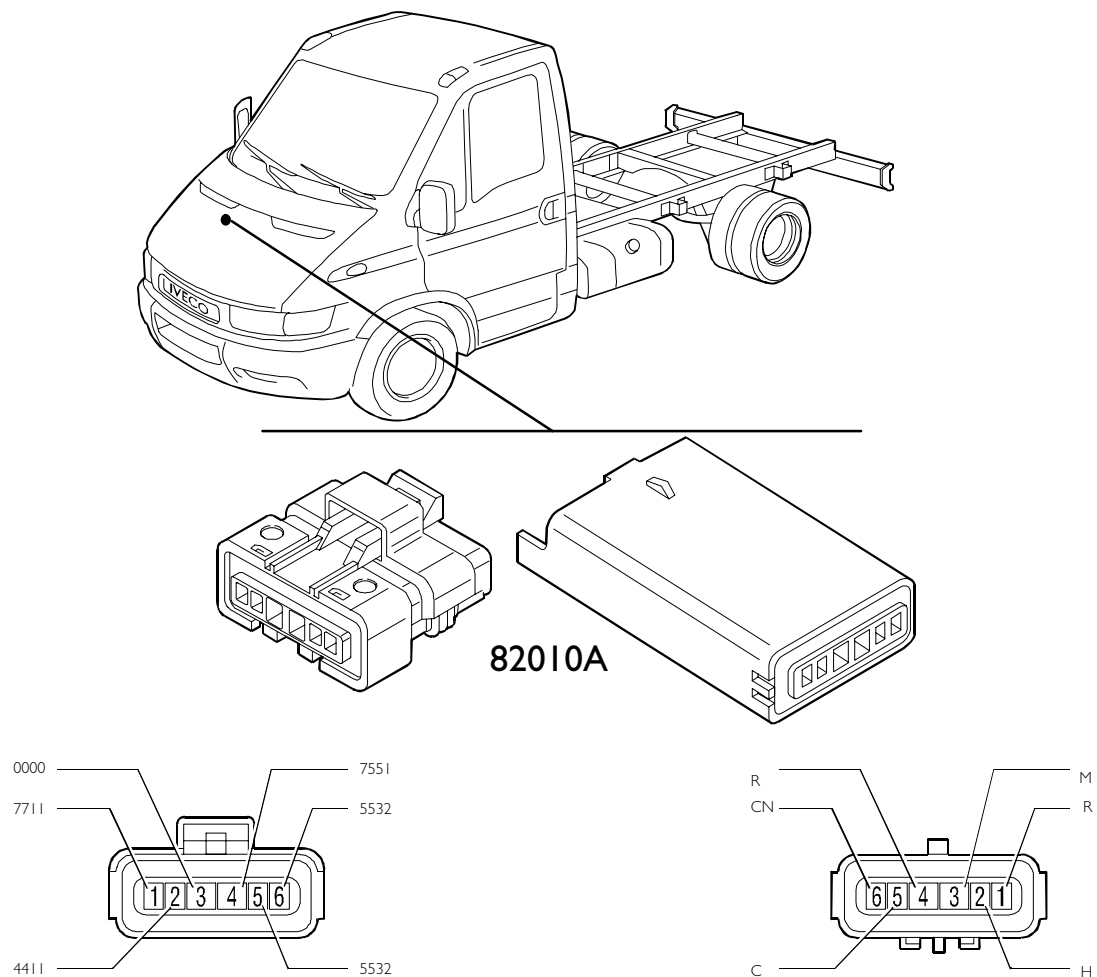
Figure 78



Ref.	Cable colour code	Function
72048B	1 8847	Supply (+15)
	2 1173	Positive with switch for vehicle stop engaged signal
	3 0315	Switching off retarder
	4 6005–6674	ABD device on signal
	5 6673	EBD device failure signal
	6 6670	ABS device failure signal

Connection between cab/bonnet cable and climate control system cable

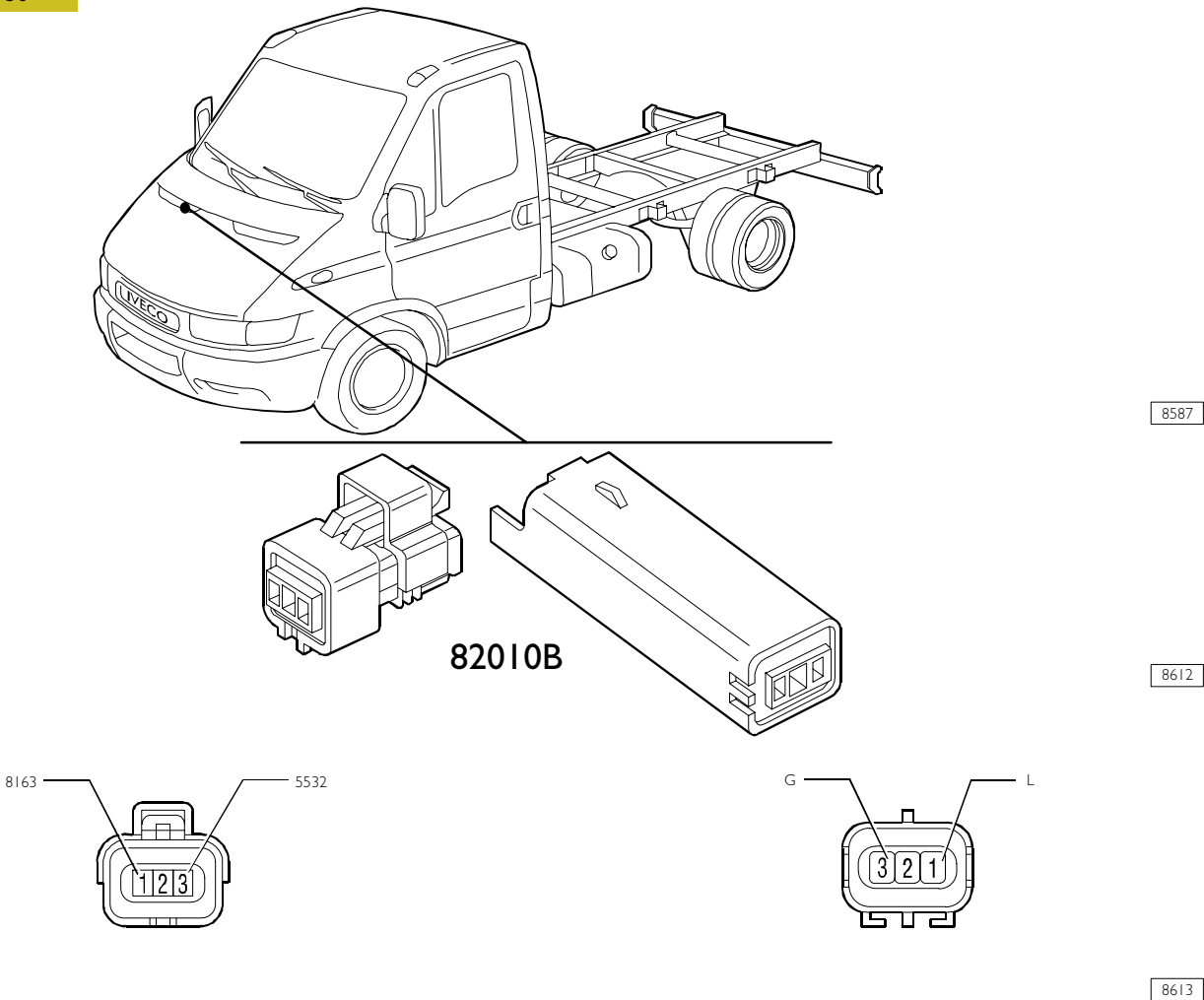
Figure 79



Ref.	Cable colour code	Function
82010A	1 7551-R	Supply (+30)
	2 4411-H	Positive with side lights on
	3 0000-M	Earth
	4 7711-R	Supply (+15/A)
	5 5532-C	Outside temperature detection
	6 5532-CN	Outside temperature detection

Connection between cab/bonnet cable and climate control system cable

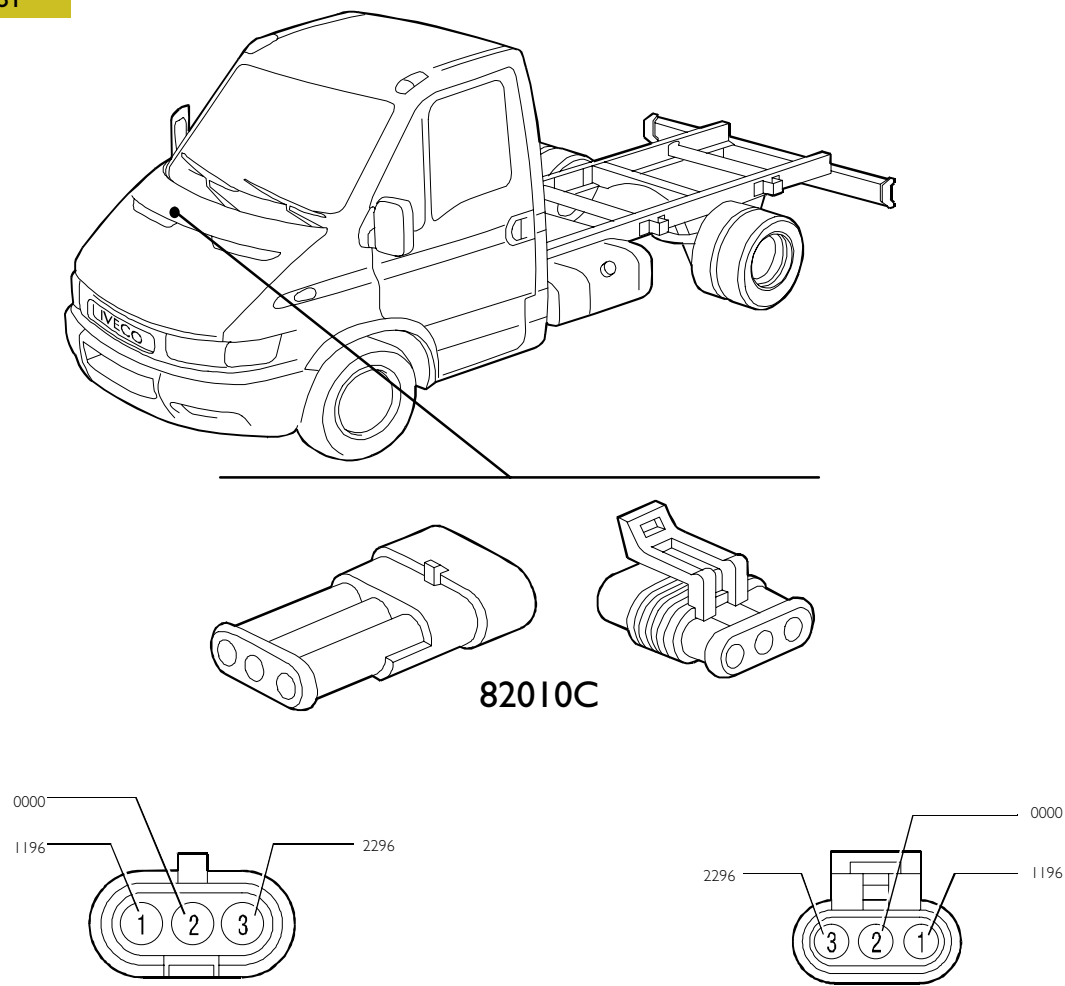
Figure 80



Ref.	Cable colour code	Function
82010B	1 8163-L	To set of 3 switches for coolant fluid pressure signal
	2 -	Spare
	3 5532-G	To instrument cluster with warning lights (connector B pin 11)

Connection between cab/bonnet cable and climate control system cable

Figure 81



Ref.	Cable colour code	Function
82010C	1 1196	To 38-pin diagnostic connector (cell 13)
	2 —	Spare
	3 2296	To 38-pin diagnostic connector (cell 14)

VENDOR-DERIVED BUS VERSION

General Information

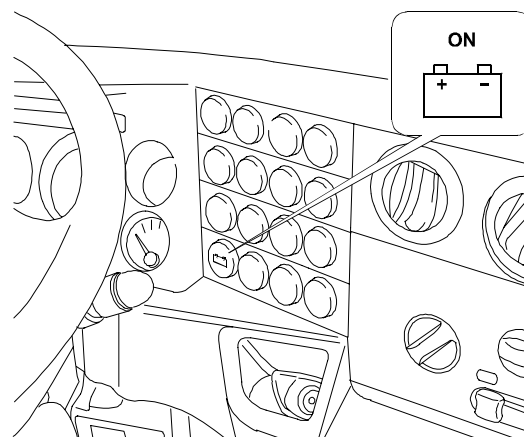
Within this handbook, school-buses are included in the Bus version. If compared to other vehicles, Buses are provided with different interior lighting harness.

The TGC switch position on dashboard panel is shown in the Push-button location Figure 82 while the main switch is located in the engine compartment near the oil filter (Figure 83). To turn the TGC off, turn the ignition key-switch to STOP and turn off the general safety switch 52029 Figure 84 ref. 1, on the left of the steering wheel.

A perspective view of the interior lighting harness is provided in Figure 85.

Figure 84 ref. 2 shows the fan control switch.

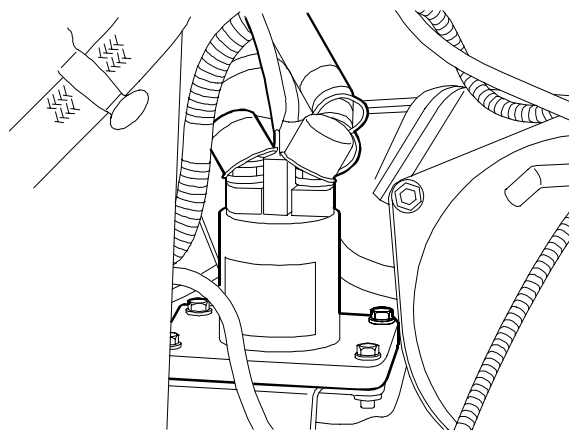
Figure 82



73728

TGC CONTROL SWITCH

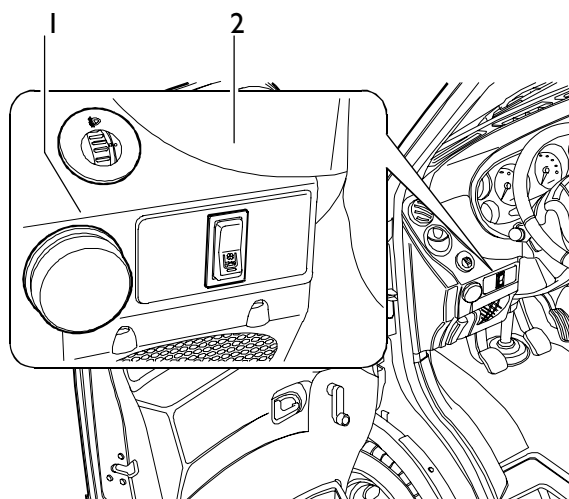
Figure 83



73717

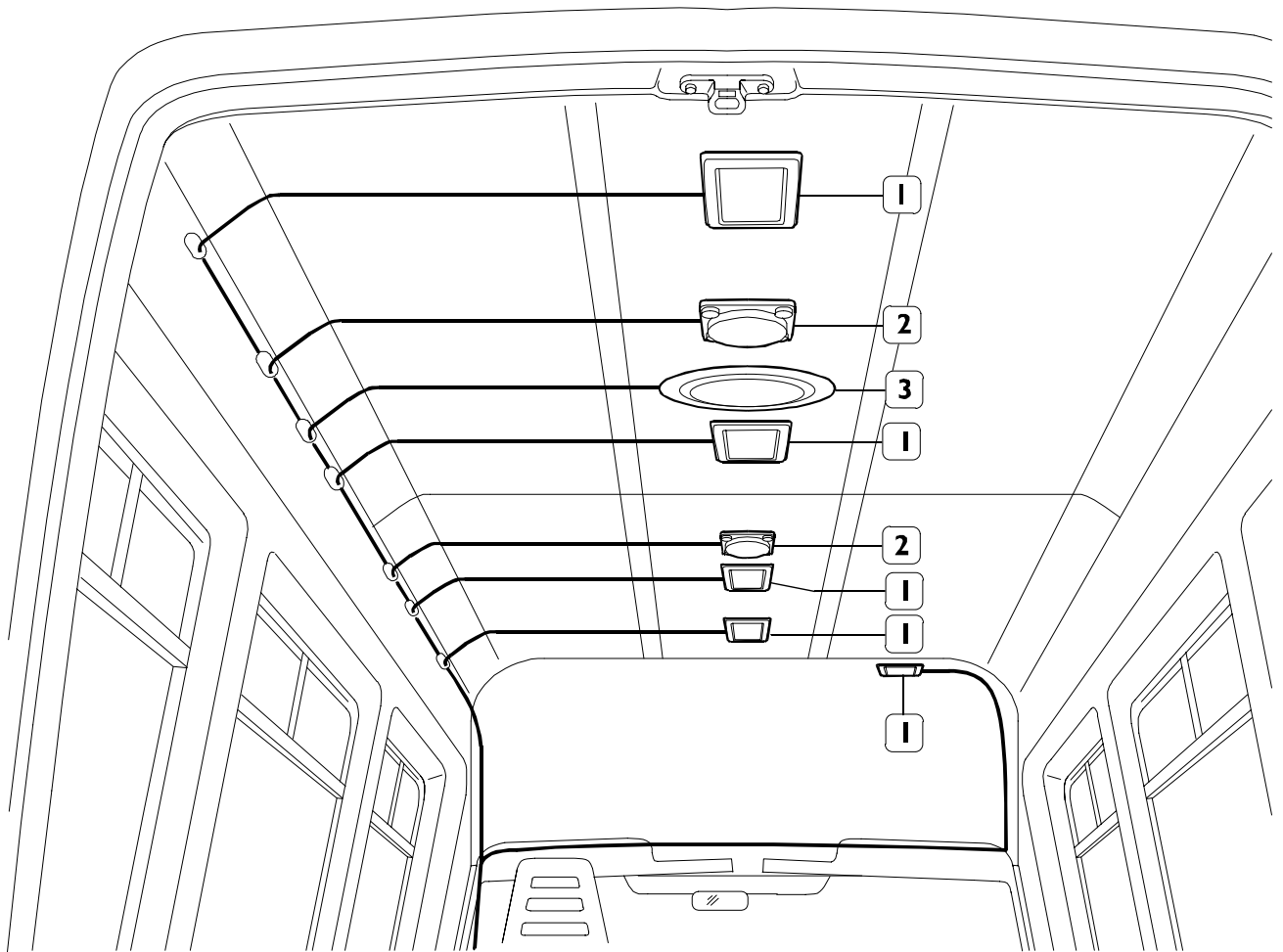
VOLTAGE MAINS REMOTE SWITCH

Figure 84



73712

1. General safety switch – 2. Fan control switch

Perspective view of Vendor-derived Bus version interior lighting harness**Figure 85**

73716

1. Interior lighting – 2. Driver loudspeaker provisions – 3. Fan

Diagnostic connector

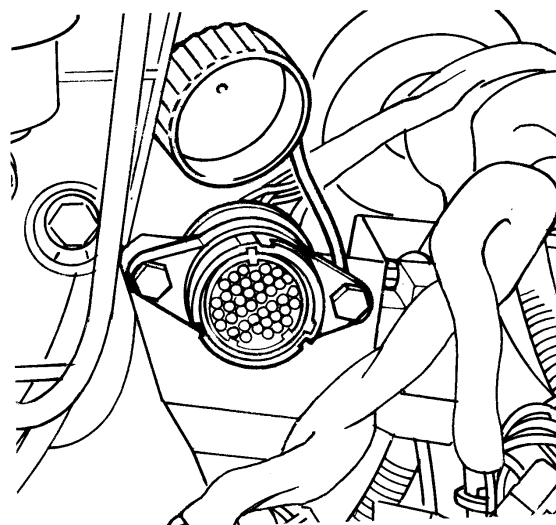
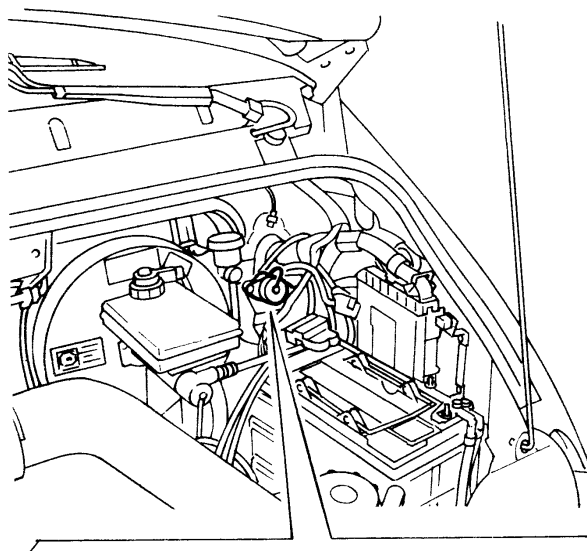
Near the positive terminal of the battery there is a diagnostic connector. It is fastened by 2 screws to a special bracket fitted on the body and access to it is gained by unscrewing the protective cap; suitably connected to the diagnostic system (IWT, MODUS), it allows quick identification of the cause of faults in the various electronic devices.

The diagnosis connector on FI engine vehicles is located on the conveyor unit lower part on the passenger side.

The table on the following page shows the correspondence between the various electronic systems and the connector pins referring to them.

NOTE The connector shown is seen from the pin side.

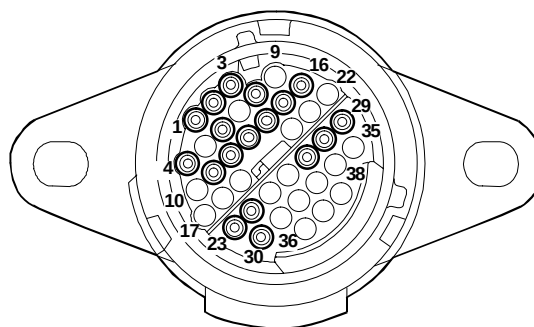
Figure 86



8620

38-PIN DIAGNOSTIC CONNECTOR

Figure 87



8621

IDENTIFICATION OF THE PINS OF THE 38-PIN DIAGNOSTIC CONNECTOR

System	Pin	Function
EDC/EGR/Methane	1	L
	2	K
ABS/EBD/ABD	3	L
	4	K
Air Bag/Retarder	6	K
Tachometer (connector A cell 2 instrument cluster)	8	K
Supply +15/A services	11	Input
Immobilizer/Central door locking/Alarm	12	K
Climate control system	13	L
	14	K
Self-levelling suspension	15	L
	16	K
Engine timing signal (connector B pin 28 EDC control unit) Engine timing earth (battery –)	23	–
	24	Screening
Supply +30	27	Battery +V
Engine rpm	28	RPM
Vehicle speed	29	–
Earth	30	–

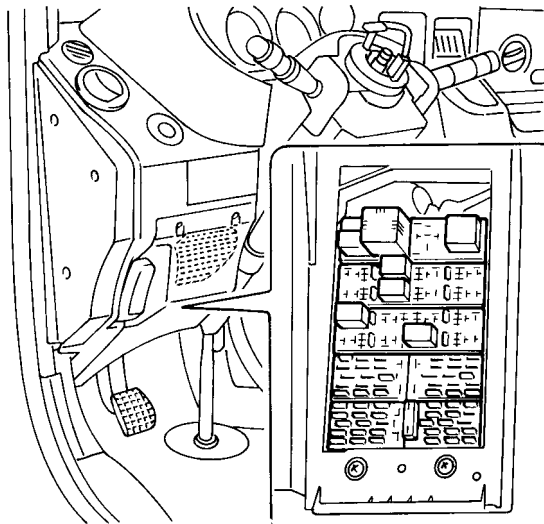
☐ With EDC16: pins 1/23 are not connected.

☐ Can Line pin 21/22.

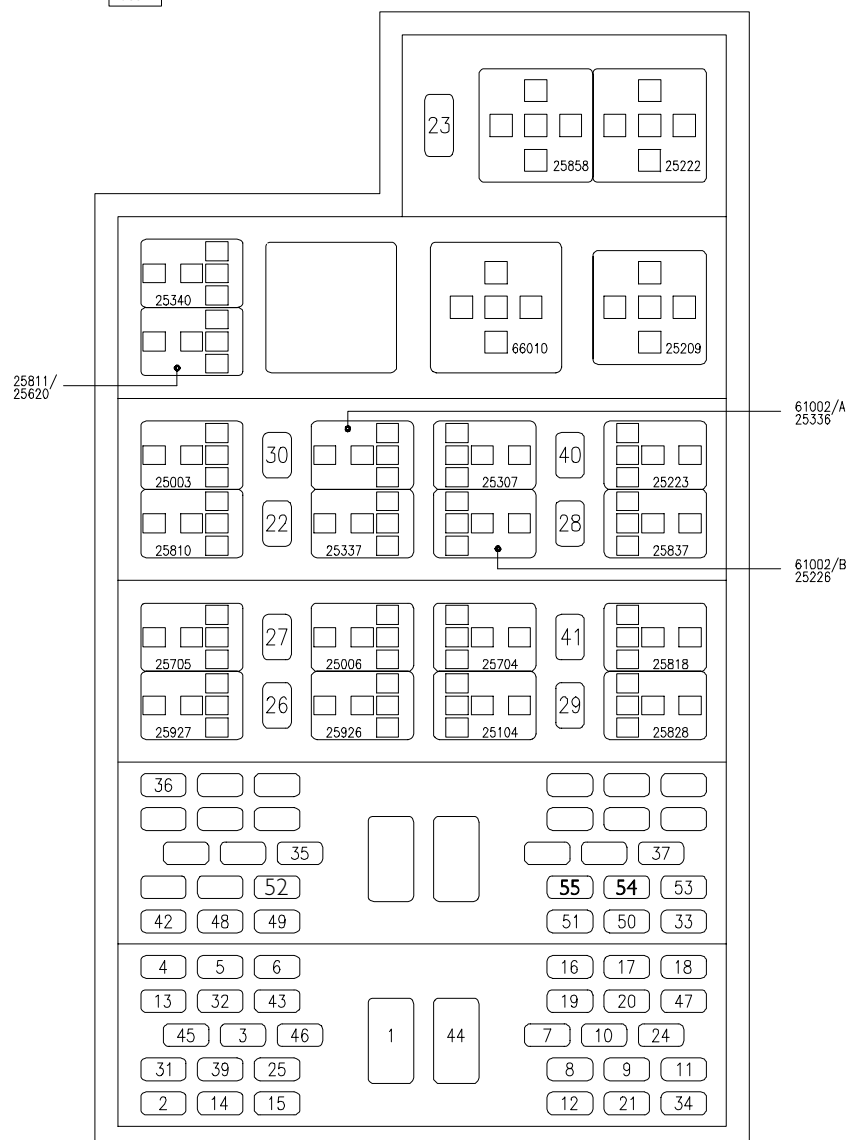
The unused pins are not indicated.

RELAY AND FUSE HOLDER SUPPORT

Figure 88



8631



Identification of fuses

N.	Functions	
1.	Additional heater	30
2.	Immobilizer/Warming/EDC	5
3.	—	—
4.	Instrument cluster (+30)	5
5.	Instrument cluster (+15)	5
6.	Stalk unit (rear fog guard, flasher)	5
7.	Stalk unit (low beams)	10
8.	Stalk unit (high beams)	15
9.	Stalk unit (side lights)	15
10.	Windscreen wiper	15
11.	Stalk unit (direction indicators/hazard warning lights)	15
12.	Stalk unit (horns)	15
13.	Reversing light	5
14.	Side light, clearance, no. plate light	5
15.	Side light cluster lighting	5
16.	Left low beam headlamp	10
17.	Right low beam headlamp	10
18.	Left high beam headlamp	10
19.	Right high beam headlamp/headlamp aiming	10
20.	Interior lighting/radio/cigar lighter	10
20.	Interior lighting/radio	10
21.	Electric windscreen defrosting unit	30
21.	Cigar lighter	15
22.	Heated fuel filter	15
23.	EDC electronic control unit	25
24.	EDC	10
25.	Stop lights	5
26.	13-pin current socket	15
27.	13-pin current socket	15
28.	13-pin current socket	10
29.	38-pin diagnostic connector	5
30.	Fog lamps	10
31.	Climate control, additional heater, engine cooling	5
31.	Compressor, timer, engine fan	10
32.	Climate control	5
33.	Additional heater	20
34.	Headlamp washer	20
35.	Rear differential lock	5
36.	ABS, EBD, ABD failure warning light	5
37.	ABS	5
39.	Heated mirrors	10
39.	Heated mirrors, heated rear-screen, heated windscreen	15
40.	Power windows	30
40.	Power windows (left)	20
41.	Heated rear-screen, heated windscreen	10/ 20
42.	Self-levelling suspensions	5
43.	Air bag failure warning light	5
44.	Self-levelling suspensions (Wabco)	40
45.	Air-bag	5
46.	—	—
46.	Power windows (right)	20
47.	Electric fuel pump	10
48.	Door locking/alarm	30
49.	Door locking/alarm	5
50.	Rear differential lock	30
51.	Total power takeoff	10
52.	Front differential lock (4x4)	30
53.	Self-levelling suspensions Technik	30
54.	Electric hatch (van and 6+1)	15
55.	Rototranslating door	15

Identification of relays/diode holders

Code	Function
25035	Switching on external lights
25003	Switching on fog lamps
25006	Switching on stop lights
25704	Total power takeoff engagement control
25104	Switching off Retarder with ABS on
25209	Loads cut-off during starting
25222	Thermal starter engagement enable
25223	Thermal starter fan control
25307	Climate control compressor engagement
25336	Engine cooling
25337	Climate control compressor disengagement
25340	Compressor on signal
25620	Fuel filter clogged signal
25705	Diagnostics enable
25810	Heated fuel filter circuit control
25811	Advance variator control (KSB)
25818	Heated windscreen switching on
25837	Electric fuel pump switching on
25858	EDC switching on
25926	Suspension raising enable
25927	Suspension lowering enable
25928	Rear-screen heating switching on
61002A	Anti-return from Trinary
61002B	Anti-return from TGC (bus)
66010	Timer for headlamp washer



Fuse 39 must be replaced with a 10A fuse if the following optionals are present on the vehicle at the same time:

- ☐ heated wing mirrors with heated driver's seat;
- ☐ heated wing mirrors with heated driver's and passenger's seat;
- ☐ heated wing mirrors with electric aiming with heated driver's seat;
- ☐ heated wing mirrors with electric aiming with heated driver's and passenger's seat.

Fuse 41 (39 for F1A) must be replaced with a 15A fuse if the following optionals are present on the vehicle at the same time:

- ☐ heated windscreen with heated rear-screen.

Fuse 55 must be replaced with a 15A fuse if both the seats are heated.

Fuse 70060 (40A) relating to the ABS, device is located in the engine compartment, near the control box.

For the F1A, plug preheat centre 25231 is located on the engine compartment to the right of the EDC centre and is powered by 60A fuse 70064.

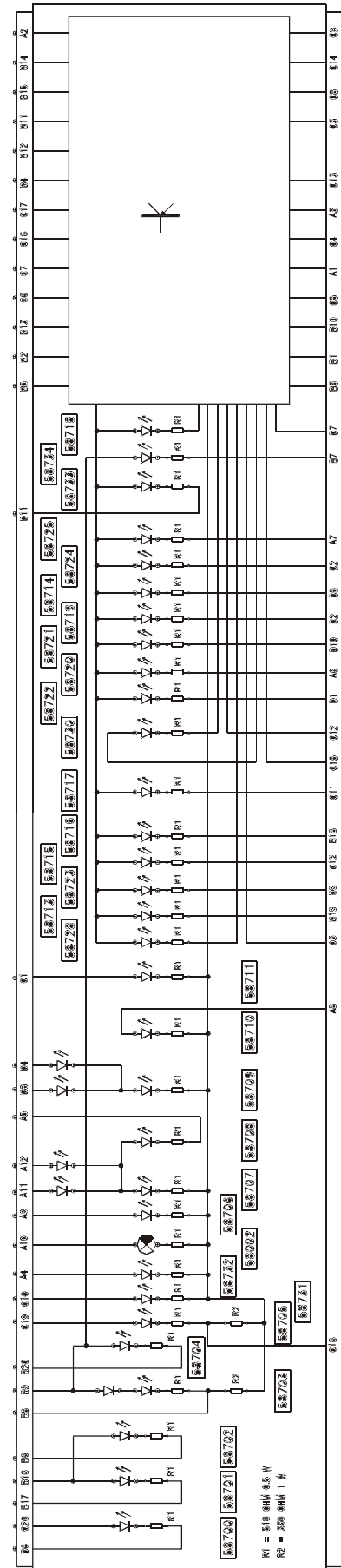
The previous indications (fuses/remote control switches) can be changed. Should it occur, make reference to the indications on the sticker label next to the fuses.

OPTICAL INDICATORS

A table is shown below with pin out 58918 in 4–connector version, as well as a 2–connector version one.

1	2	1	2	1	2
D6	B5	* B15	A2	B3	B156
C20	A5	B2	A17	B1	A22
B17	A29	B12	A18	B10	A21
B16	A14	C8	A4	C5	B26
B6	A30	* B4	A3	A1	A23
B8	A32	* B14	A19	C4	B24
B9	A16	A2	A6	A3	A7
B20	A15	C18	B14	C13	B27
C19	B13	A8	A11	C3	B25
C10	B15	D3	B29	B11	B2
A4	A8	B19	A13	C14	B1
A10	A9	D9	B30	C9	B23
A9	A10	D12	B9		
A11	A24	B18	A12		
A12	A25	C11	B11		
A5	A26	C15	B22	1	2
D8	B3	C12	B21	* B4	A19
D4	B4	D1	B6	* B14	A2
C1	B12	A6	A27	* B15	A3
D11	B10	D10	B8	* Variant for versions .11/.13/.15.	
B5	A20	D2	B7		
B13	A1	D5	B31		
C6	B18	C2	B32		
C7	B20	A7	A28		
C16	B19	B7	A31		
C17	B17	D7	B28		

Figure 89

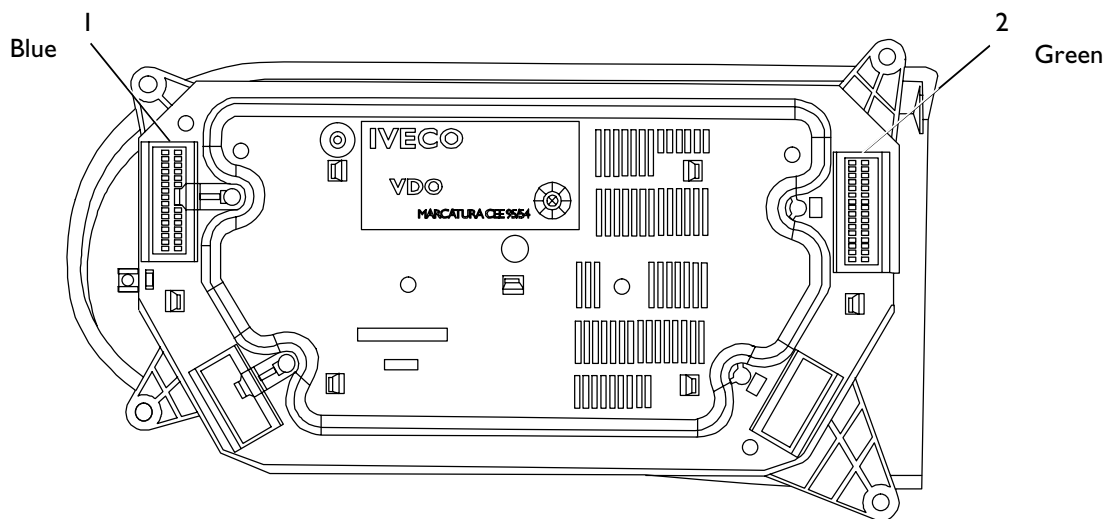
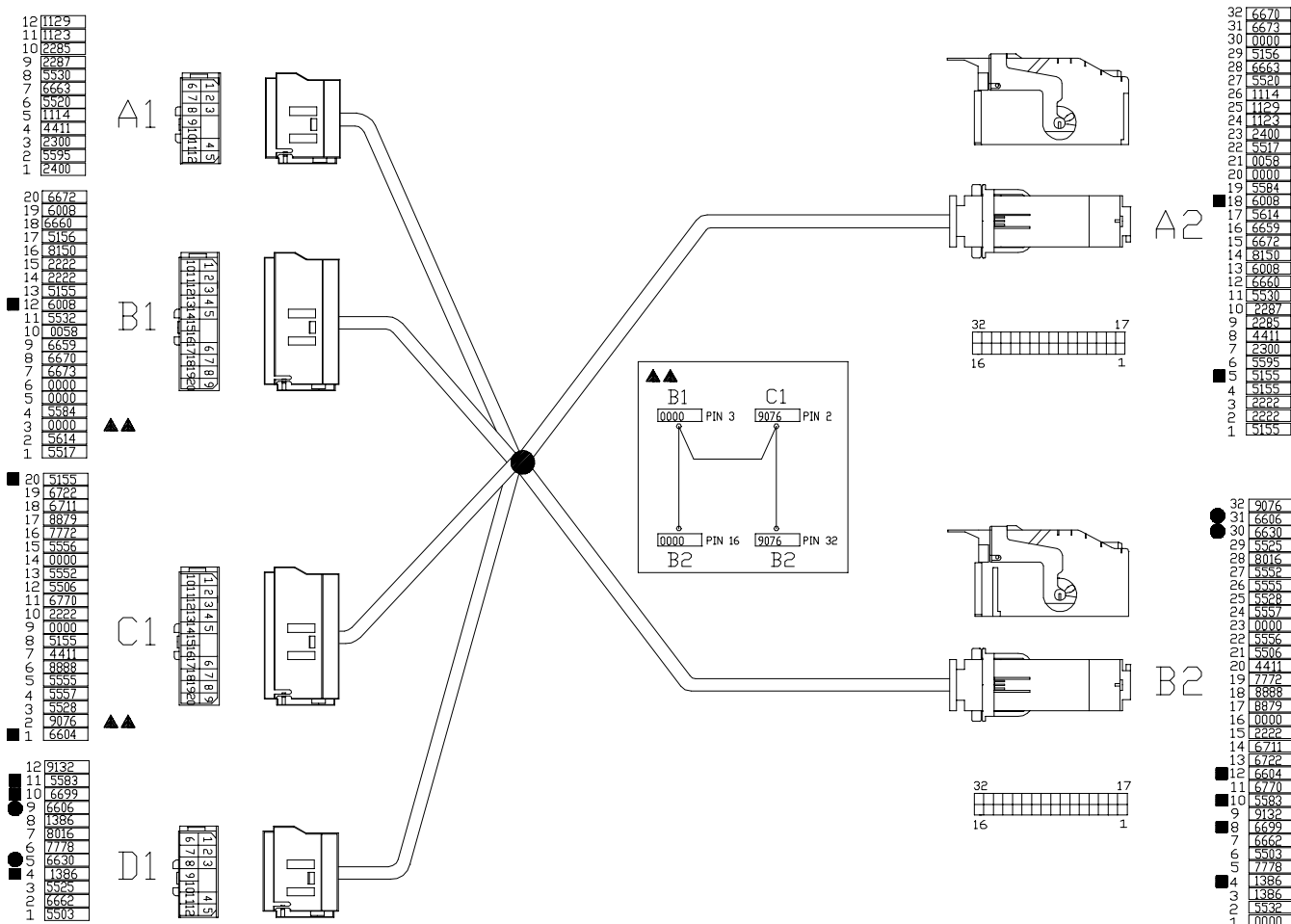


058918 PIN OUT OF THE PREVIOUS VERSION

74024

Interface bride with the new tool with 32-way connectors

Figure 90



REAR VIEW OF 32-WAY CONNECTOR UNIT

1. Blue colour seat "A" for 32-way female holder connector complete with terminals
2. Blue colour seat "B" for 32-way female holder connector complete with terminals

77025

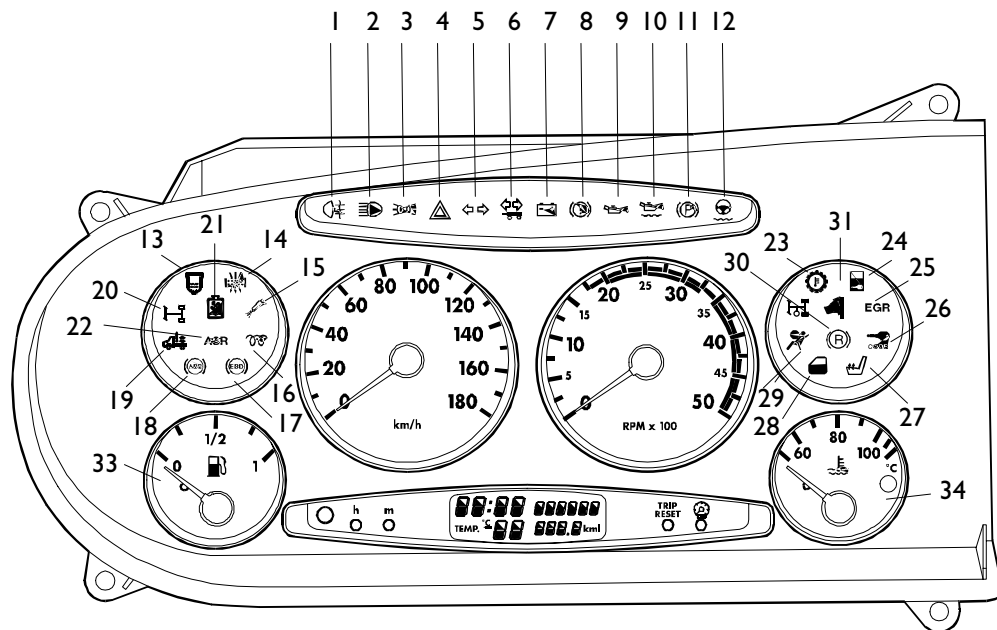
0742528t

INSTRUMENT CLUSTER

Warning lights assembly

Figure 91

58918

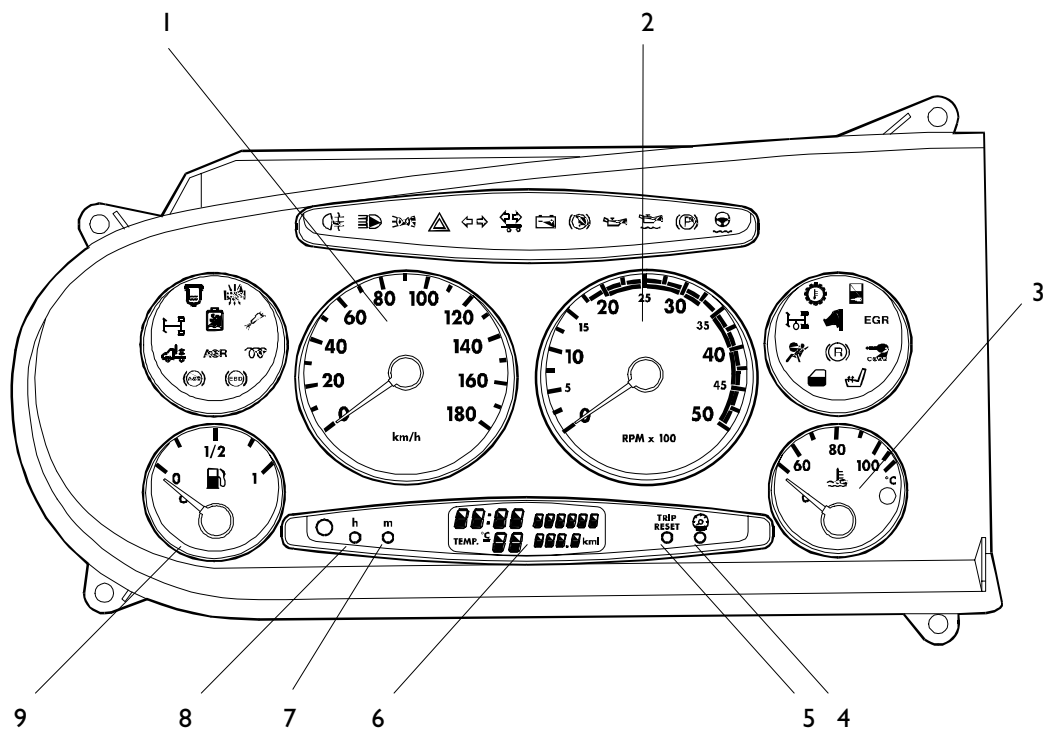


8655

Ref.	Component code	Description
1	58706	Rear fog guard
2	58002	High beam headlamps
3	58732	Side lights
4	58708	Hazard warning lights
5	58707	Tractor direction indicators
6	58709	Trailer direction indicators
7	58700	No battery charge
8	58718	Brake system failure (under TEST)
9	58722	Low engine oil pressure
10	58730	Low engine oil pressure (under TEST)
11	58719	Handbrake on (under TEST)
12	58728	Low power steering fluid level (under TEST)
13	58710	Water in fuel filter (under TEST)
14	58725	Air cleaner clogged (under TEST)
15	58701	EDC failure
16	58702	Engine warming
17	58734	EBD failure
18	58703	ABS failure
19	58713	ECAS failure
20	58735	Rear transversal differential lock
21	58720	Low engine coolant fluid level (under TEST)
22	58704	ABD on
23	58721	High gearbox oil temperature (under TEST)
24	58723	Emergency handle lock (under TEST)
25	58733	EGR failure
26	58717	Immobilizer engaged
27	58731	Seat heating
28	58724	Door open indicator
29	58705	Airbag failure
30	58715	Total power takeoff (PTO) on
31	58714	Emergency handle activated (under TEST)
32	58711	Retarder on
33	44031	Low fuel level (under TEST)
34	47207	High engine coolant fluid temperature (under TEST)

Instrument assembly

Figure 92

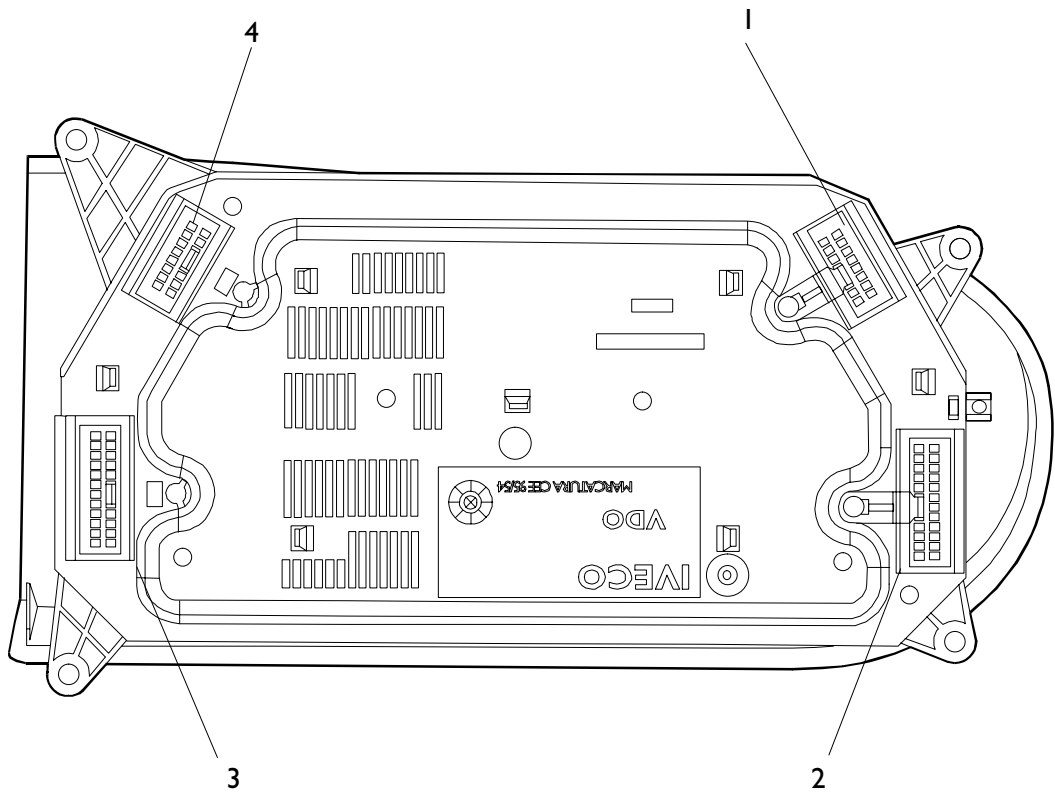


8655

Ref.	Description
1	Electronic tachometer
2	Electronic gyrometer
3	Engine coolant fluid temperature gauge with incorporated warning light
4	Instrument lighting adjustment button
5	Trip meter reset button
6	Display for: clock, outside temperature (only with climate control), total odometer (km), trip meter (km)
7	Clock adjustment button (minutes)
8	Clock adjustment button (hours)
9	Fuel level gauge with reserve warning light

Rear view

Figure 93

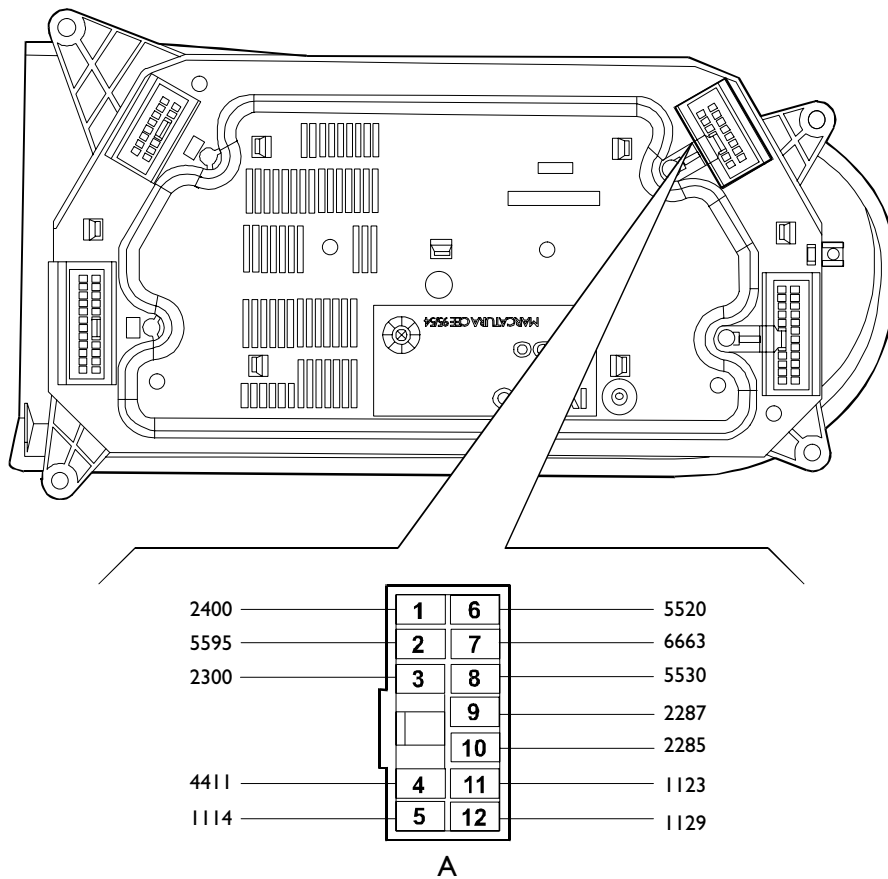


8643

Ref.	Description
1	Seat "A" for 12-way, blue, female holder connector complete with terminal
2	Seat "B" for 20-way, blue, female holder connector complete with terminal
3	Seat "C" for 20-way, black, female holder connector complete with terminal
4	Seat "D" for 12-way, black, female holder connector complete with terminal

Connector assembly (cable input side view)

Connector A

Figure 94

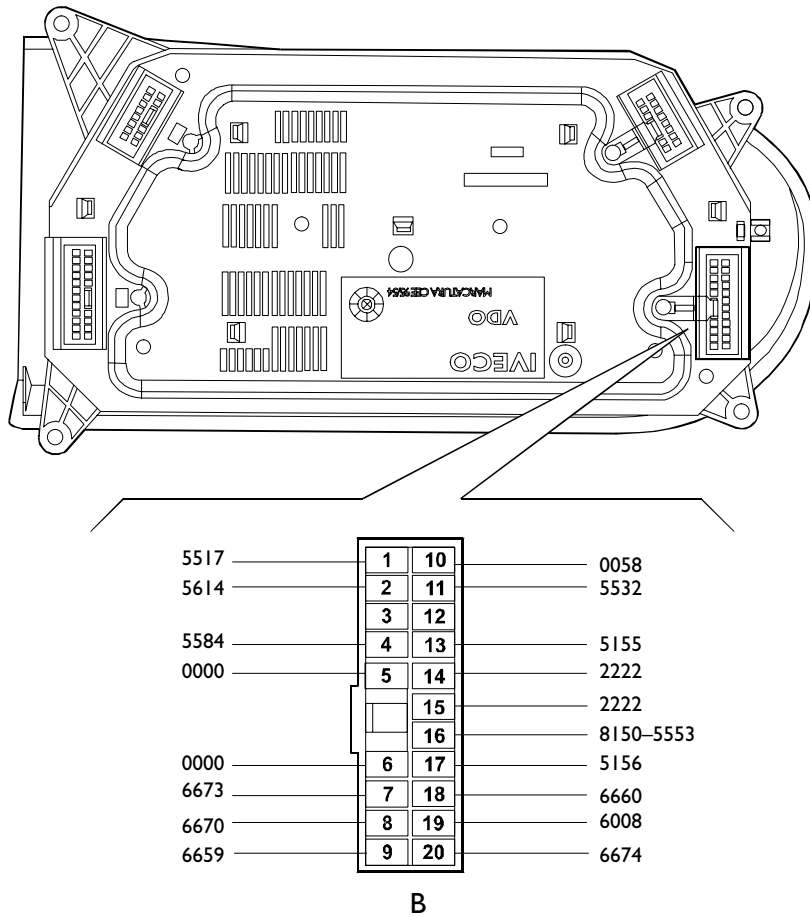
8643

8644

Ref.	Cable colour code	Function
A	1 2400	Tachograph failure warning light
	2 5595	To cell 8 of diagnostic connector (serial line K)
	3 2300	Speed indication from tachograph
	4 4411	Side lights on warning light
	5 1114	Hazard warning lights on warning light
	6 5520	Low engine coolant fluid level warning light
	7 6663	Air cleaner clogged warning light
	8 5530	Water in fuel oil filter warning light
	9 2287	Rear fog guard on warning light
	10 2285	High beam headlamps on warning light
	11 1123	Direction indicators on warning light
	12 1129	Direction indicators on warning light

Connector B

Figure 95



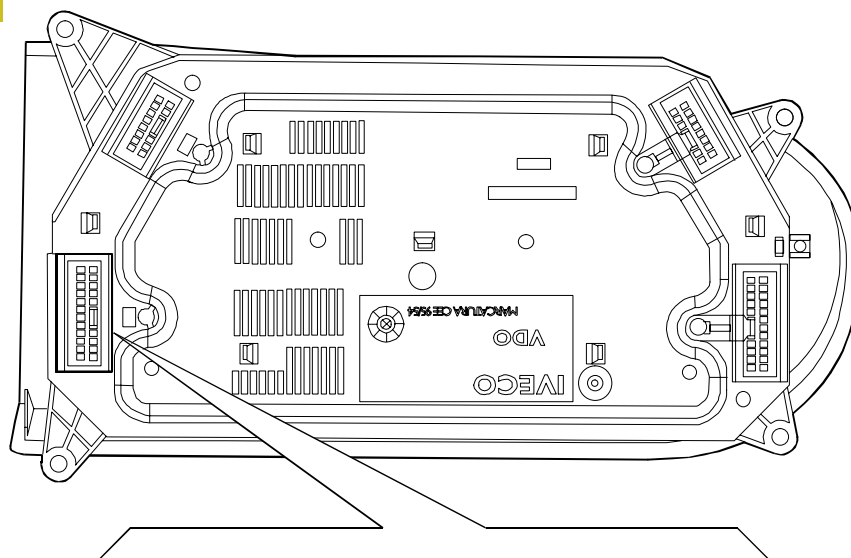
8643

8645

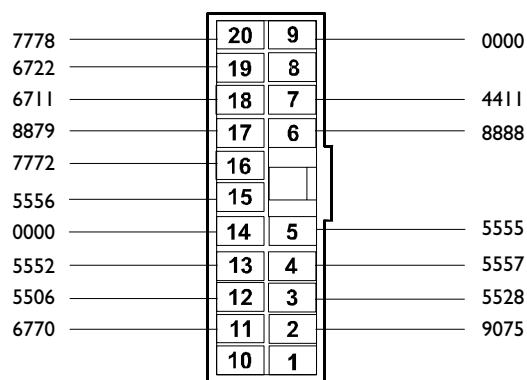
Ref.	Cable colour code	Function
B	1 5517	Tachometric signal
	2 5614	Rev counter signal
	3 -	-
	4 5584	To cell 28 of diagnostic connector (engine rpm repeater)
	5 0000	To pin 14 of connector B of electronic control unit for EDC
	6 0000	Earth for warning on warning light
	7 6673	EBD Failure warning light
	8 6670	ABS failure warning light
	9 6659	Supply (+15) for ABD, ABS and EBD failure warning lights
	10 0058	To electronic tachometer transmitter
	11 5532	To pin 2 of connector B of air conditioning system electronic control unit
	12 -	-
	13 5155	To pin 4 of connector B of EDC electronic control unit (tachometric signal repeater)
	14 2222	Tachometric signal repeater for rear differential lock
	15 2222	Tachometric signal repeater
	16 8150	Positive for EDC failure and warning on warning lights (Unijet)
	5553	Positive for warning on warning light
	17 5156	EDC failure warning light
	18 6660	Rear differential lock warning light
	19 6008	Self-levelling suspension system failure warning light
	20 6674	A.S.R. pilot lamp

Connector C

Figure 96



8643



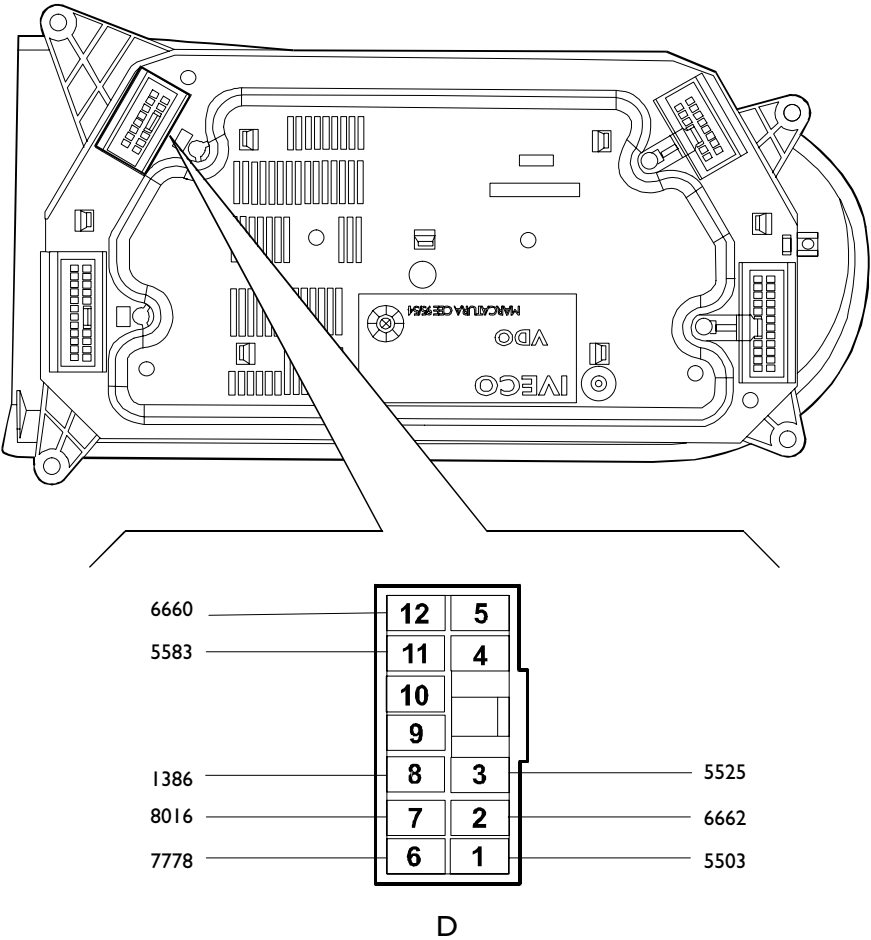
8646

C

Ref.	Cable colour code	Function
C	1	Retarder on warning light (bus)
	2	9075 Door open warning light
	3	5528 Engine coolant fluid temperature gauge
	4	5557 Fuel level gauge
	5	5555 Fuel reserve warning light
	6	8888 Supply (+50)
	7	4411 Positive with exterior light switch on
	8	— To pin A1 of radio receiver set
	9	0000 Connection to signal earth ms7
	10	— Fog lamp pilot lamp
	11	6770 Sensor for immobilizer on signal
	12	5506 Low engine oil level sensor
	13	5552 Engine coolant fluid high temperature warning light
	14	0000 Earth
	15	5556 Low engine oil level sensor
	16	7772 Supply (+30)
	17	8879 Supply (+15)
	18	6711 Air bag failure warning light
	19	6722 Supply (+15) for air bag failure signal
	20	7778 Supply (+15) for battery charge failure warning light

Connector D

Figure 97



8643

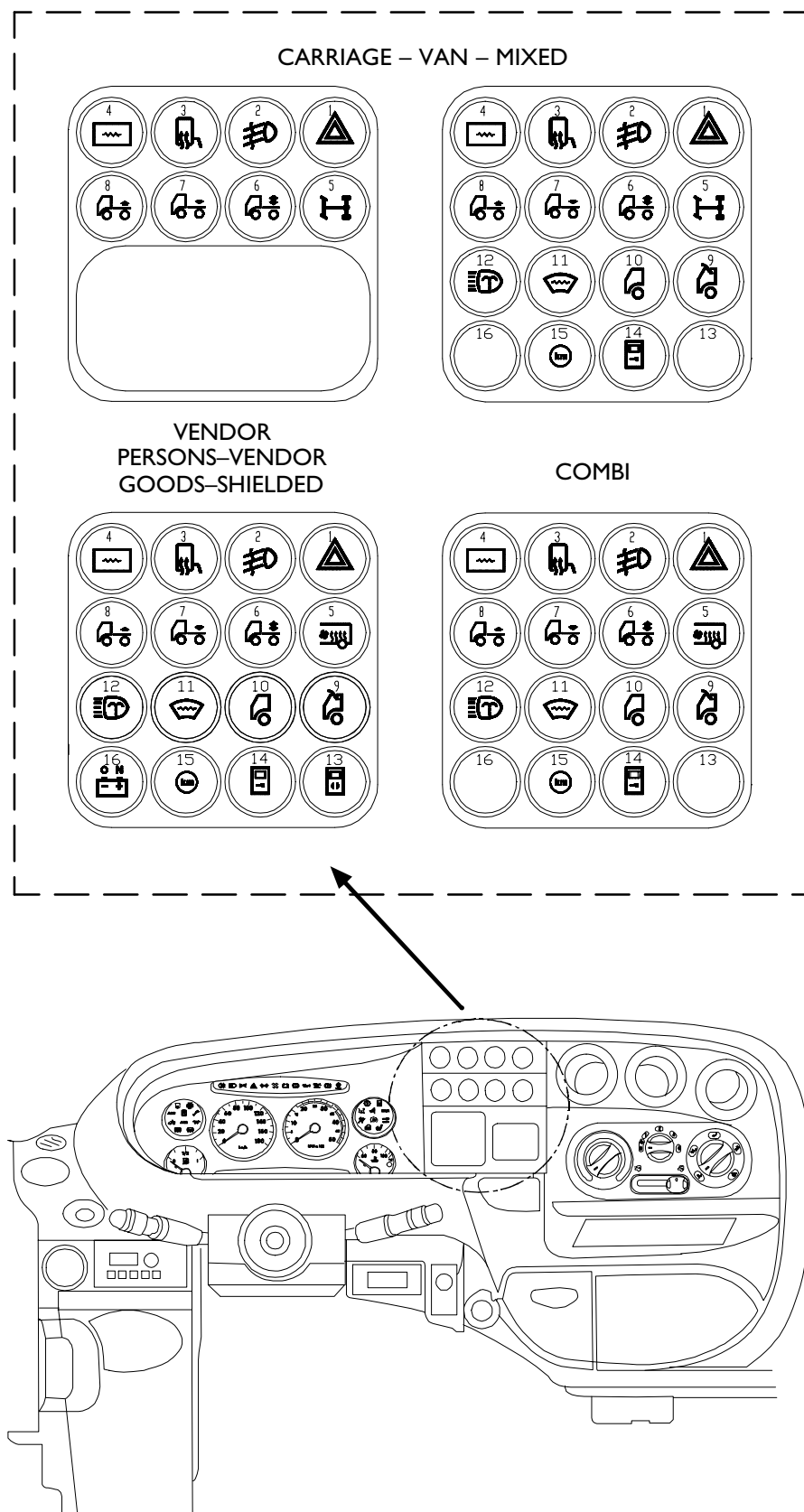
8647

Ref.	Cable colour code	Function
D	5503	Low engine oil pressure warning light
	6662	Handbrake engaged warning light
	5525	Low power steering fluid level warning light
	—	—
	—	Rotary translating door failure warning lamp (bus)
	7778	Battery charge failure warning light
	8016	Brake failure warning light
	1386	Trailer direction indicators on warning light
	—	Handle lock pilot lamp
	—	—
	5583	EGR failure warning light
	6660	Total power takeoff (PTO) on warning light

SWITCH ASSEMBLY

All the switches and push buttons of this module incorporate a warning led.

Figure 98



74271

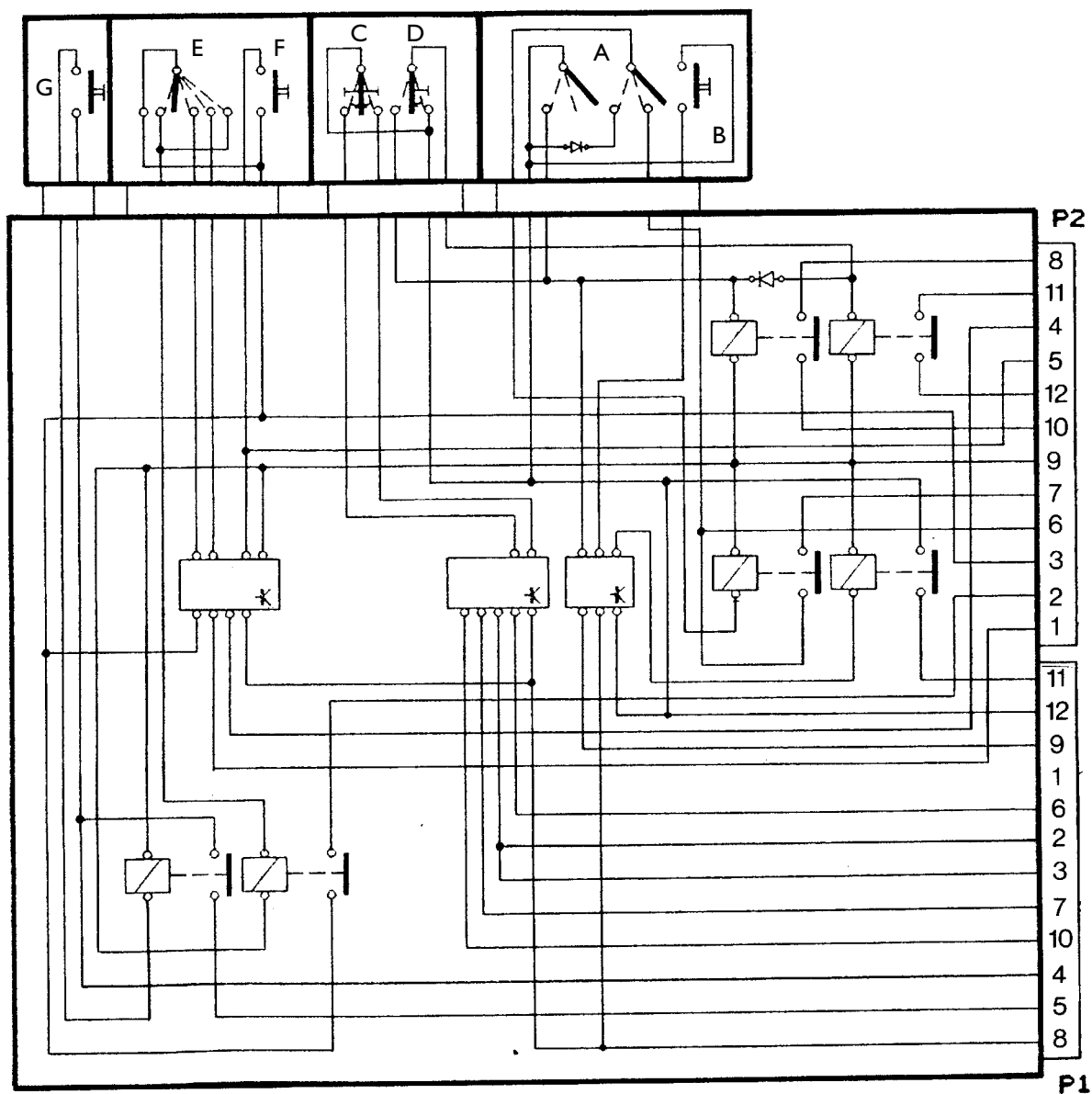
Legend

Position	Function	Vehicles	
		CARRIAGE – VAN – MIXED	
		Standard	Optional
1	Emergency lights	X	
2	Fog lights		X
3	Heated rearview mirrors		X
4	Rear glass		X
5	Rear differential lock		X
6	Level compressed air suspension		X
7	Lower compressed air suspension		X
8	Lift compressed air suspension		X
9	Electrical manhole opens		X
10	Electrical manhole closes		X
11	Heated windshield		X
12	Light washer		X
13	Cap		
14	Rear door lock		X
15	Speed limiter adjustment		X
16	Cap		
		VENDOR PERSONS–VENDOR GOODS–SHIELDED	
1	Emergency lights	X	
2	Fog lights		X
3	Heated rearview mirrors		X
4	Rear glass		X
5	Air heating		X
6	Level compressed air suspension		X
7	Lower compressed air suspension		X
8	Lift compressed air suspension		X
9	Electrical manhole opens		X
10	Electrical manhole closes		X
11	Heated windshield		X
12	Light washer		X
13	Rotary travel holder	X	
14	Rear door lock		X
15	Speed limiter adjustment		X
16	Battery sectioner reset	X	
		COMBI	
1	Emergency lights	X	
2	Fog lights		X
3	Heated rearview mirrors		X
4	Rear glass		X
5	Air heating		X
6	Level compressed air suspension		X
7	Lower compressed air suspension		X
8	Lift compressed air suspension		X
9	Electrical manhole opens		X
10	Electrical manhole closes		X
11	Heated windshield		X
12	Light washer		X
13	Cap		
14	Rear door lock		X
15	Speed limiter adjustment		X
16	Cap		

STALK UNIT

54032

Figure 99

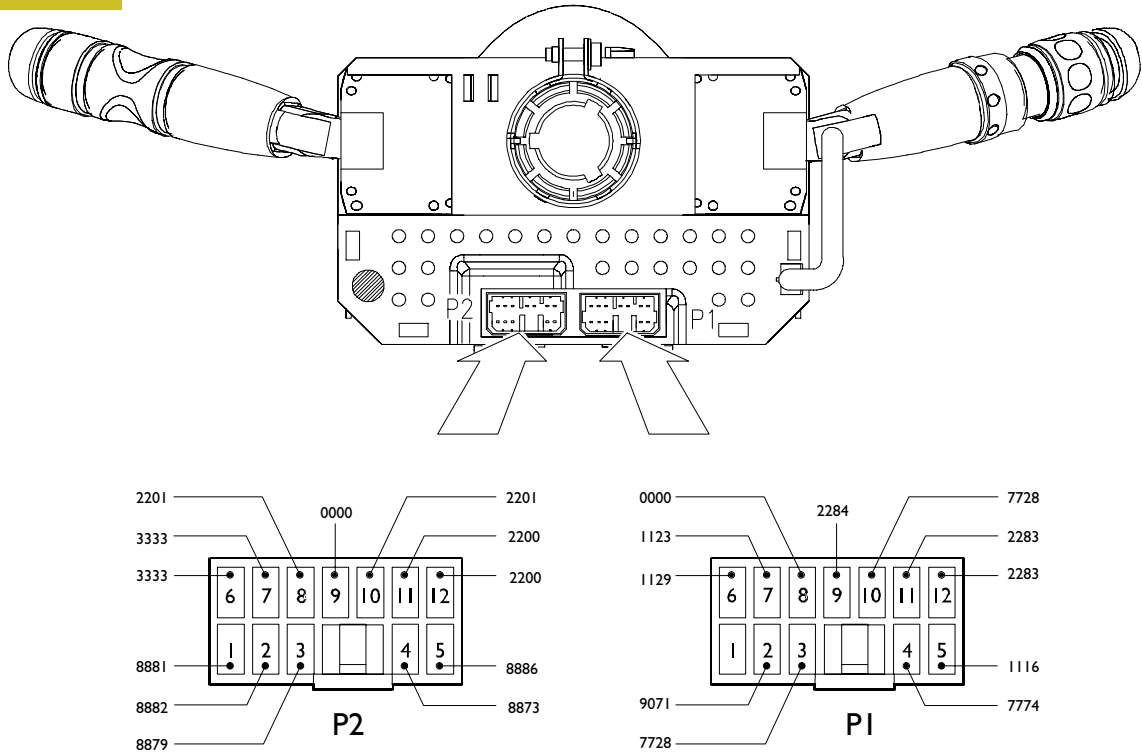


8651

WIRING DIAGRAM

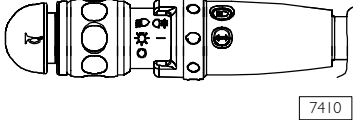
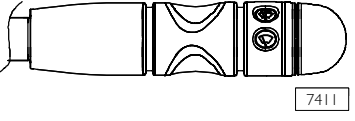
A. Exterior light switch – B. Rear fog guard button – C. Direction indicator switch –
 D. High beam headlamp/flasher switch – E. Windscreen wiper switch – F. Windscreen washer pump button –
 G. Horn button

Figure 100



TECHNICAL VIEW OF COMPONENT AND CONNECTORS OF CAB/BONNET CABLE TO BE CONNECTED TO STALK UNIT

Ref.	Function	Cable colour code
P1 (Black)	1 –	–
	2 Input from alarm	9071
	3 Input from hazard warning lights	7728
	4 Supply (+30) for horn button	7774
	5 Homs	1116
	6 Left-hand direction indicator	1129
	7 Right-hand direction indicator	1123
	8 Signal earth	0000
	9 Positive with fog lights on	2284
	10 Supply (+30) for direction indicators and hazard warning lights	7728
	11 Rear fog guard and rear fog guard on warning	2283
	12 Supply (+15)	2283
P2 (Blue)	1 Windscreen wiper first speed	8881
	2 Windscreen wiper second speed	8882
	3 Supply (+15/A) for windscreen wiper	8879
	4 Windscreen wiper reset input	8873
	5 Electric pump for windscreen washer	8886
	6 Supply (+30) for side lights switch	3333
	7 Side lights	3333
	8 Supply (+15/A) for switching on low beam headlamps	2201
	9 Power earth	0000
	10 Low beam headlamps	2201
	11 Supply (+15/A) for switching on high beam headlamps	2200
	12 High beam headlamps	2200

Technical view	Position	Electrical function
	A	Position "0"
	A1	Right lane change (unstable)
	A2	Right direction
	A3	Left lane change (unstable)
	A4	Left direction
	A5	Light flashing
	A6	High beam headlamps
	A7	Side light switch position "0"
	A8	Side lights
	A9	Low beam headlamps plus high beam and fog guard enable
	A10	Rear fog guard stable position (on or off)
	A11	Rear fog guard unstable position off Rear fog guard unstable position on only with light switch at position A9 of rear fog guards on
	A12	Horn
	B	Windscreen wiper reset
	B1	Intermittent device
	B2	First Speed
	B3	Second speed
	B4	Second speed unstable
	B5	Windscreen washer

Cruise Control

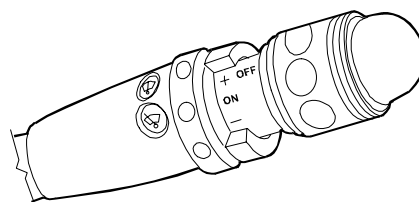
The cruise control is operated by push-buttons (Figure 102) installed on the windscreen wiper control lever with following functions:

- ☐ adjust engine idle speed;
- ☐ read and adjust drive rpm;
- ☐ set and store travel speed.

Cruise control is deactivated by pressing the clutch pedal, the brake pedal and keeping the accelerator pedal pressed for over 10 seconds or by turning the switch to "OFF".

Cf. the Use and Maintenance Handbook on board the vehicle for further information on use.

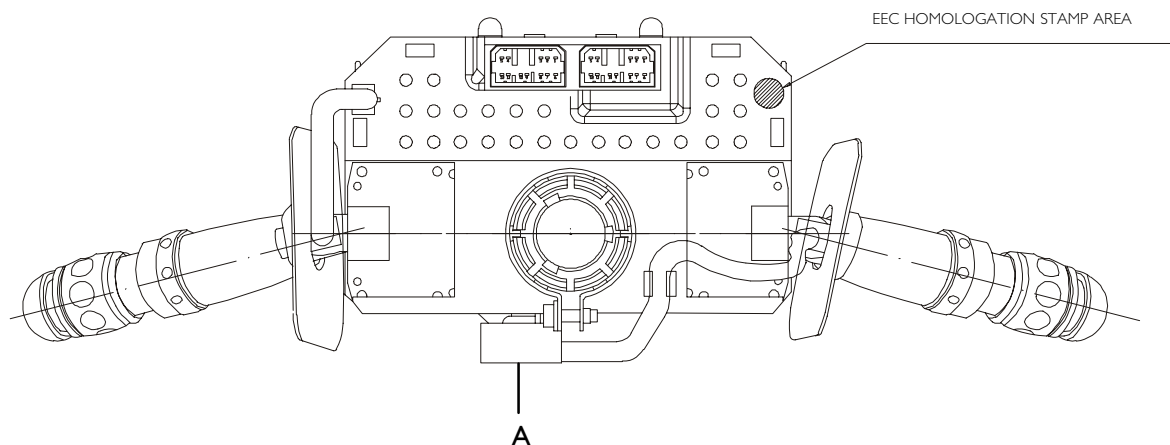
Figure 102



000245t

CRUISE CONTROL

Figure 103



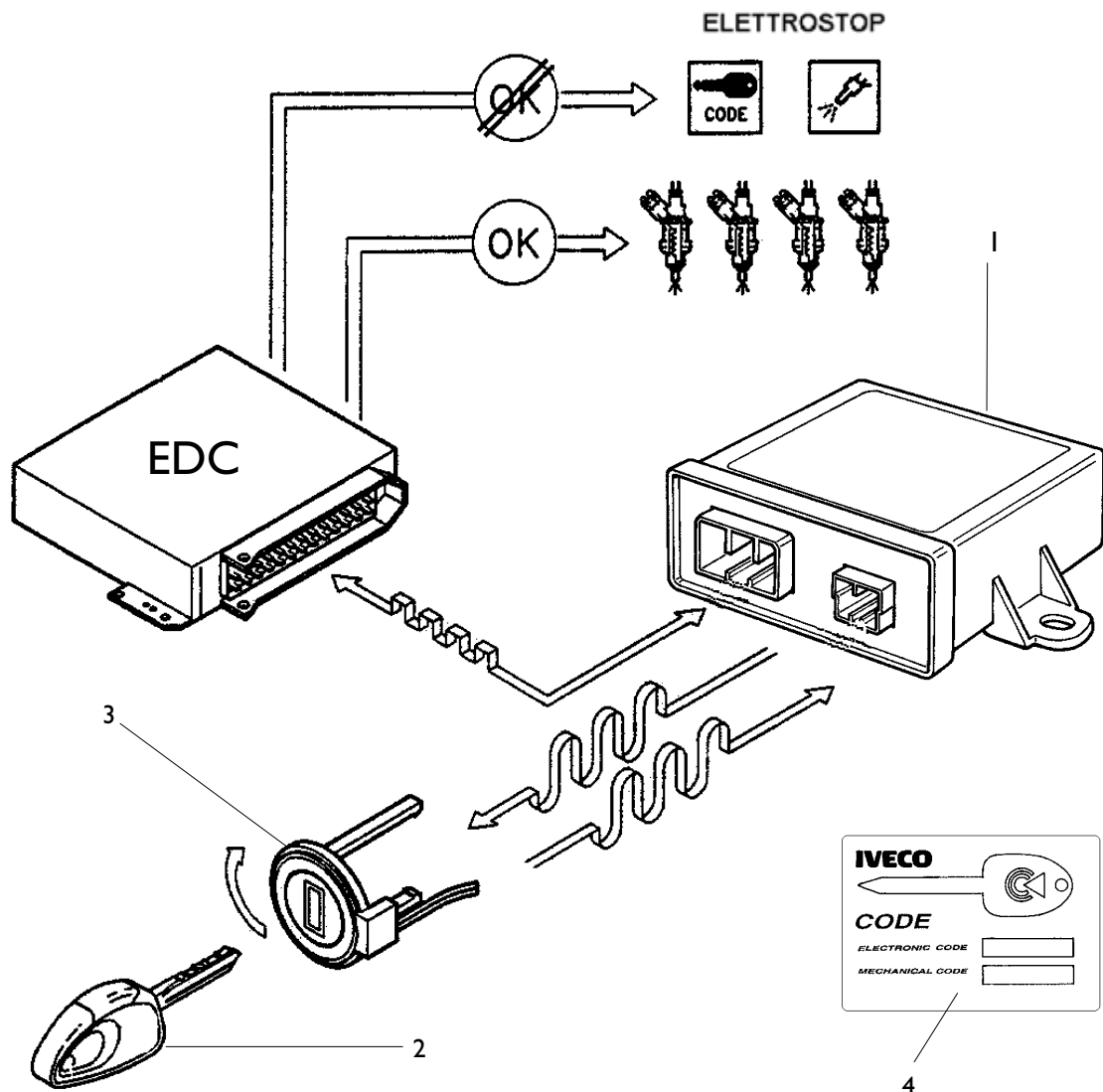
TECHNICAL VIEW OF THE COMPONENT
A. Connector (Cruise control)

ELECTRONIC SYSTEMS

Immobilizer

In order to increase protection against attempted theft, the vehicles have been fitted with an electronic engine block system called "Immobilizer" which is activated automatically removing the ignition key. In fact, the keys are fitted with an electronic "Transponder" that sends a code signal to a special electronic control unit which, only allows the engine to be started if it recognises the code sent. del motore.

Figure 104



WARNING LIGHT MAIN SYSTEM COMPONENTS

1. Immobilizer electronic control unit – 2. Electronic key containing the transponder – 3. Aerial used for sending/reading the key code – 4. Code card containing the code for emergency starting

000245t

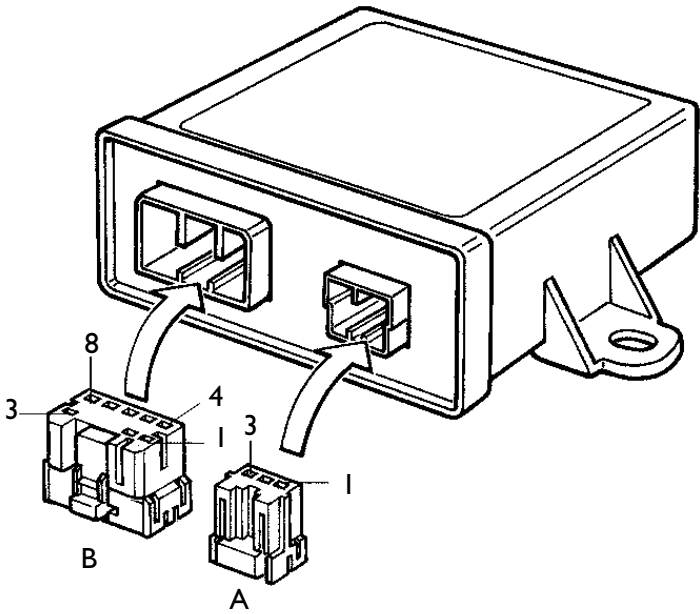
System components

Electronic control unit

This is located behind the steering wheel column.
It is able to communicate with "EDC" central unit by a "CAN line".
The main functions of the control unit are:

- ❑ recognising insertion and rotation of the key in the ignition switch,
- ❑ activating and reading the secret code emitted by the Transponder,
- ❑ managing control and processing the codes,
- ❑ communicating with the "EDC" control unit,
- ❑ memorising any faults,
- ❑ system diagnostics.

Figure 105



00290t

Pin	Function
A1	Aerial
A2	Aerial
A3	—
B1	CAN line L for EDC control unit pin 39 (EDC I6 – pin 61)
B2	Line K for 38-pin diagnostic connector pin 12
B3	Negative for Immobilizer failure warning lamp
B4	CAN H line for EDC control unit pin 8 (EDC I6 – pin 62)
B5	—
B6	Earth
B7	Key-operated positive supply
B8	—

Electronic keys

An electronic device called "Transponder" is fitted in the key grip which is not supplied by any battery, it is not removable and contains and transmits the secret code.

Inserting the key, the "Transponder" is activated, then, when it receives the radio waves emitted by the aerial (fitted on the ignition switch block), it automatically answers transmitting the secret code.

If the two codes correspond, the control unit enables starting, otherwise it shuts off the flow of fuel thereby preventing the engine from starting.



The keys provided are two. Each key contains a "Transponder" with the corresponding secret code. It is highly to carry out the correct key teaching procedure.

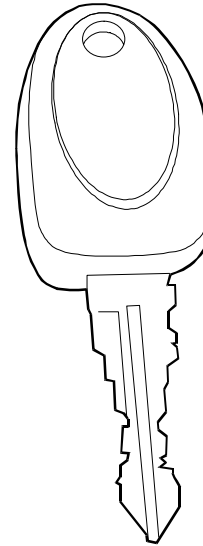
Aerial

This is fitted coaxially with the ignition switch and it has the task of:

- ☐ supplying the energy to the "Transponder" for sending the secret code
- ☐ receiving the signal from the Transponder and sending it to the control unit.

It is connected to the control unit on pins A1/A2.

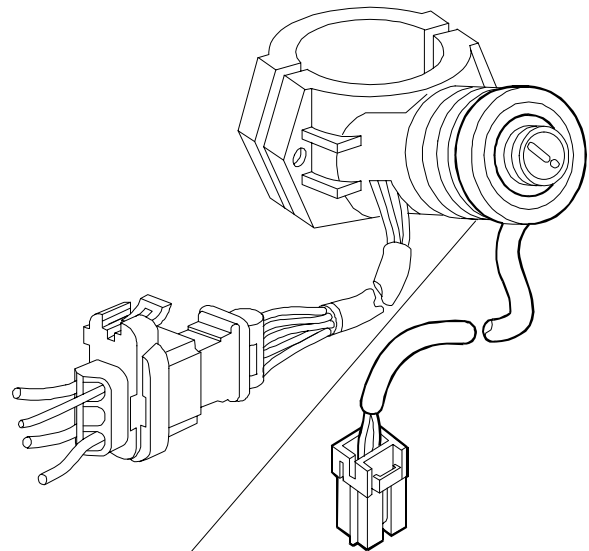
Figure 106



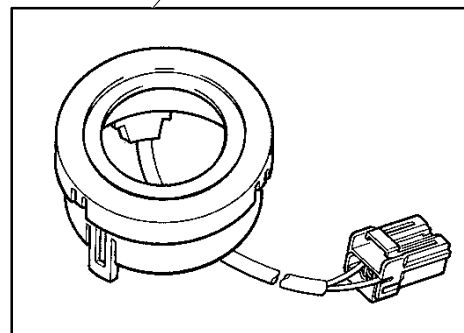
ELECTRONIC KEYS

000291t

Figure 107



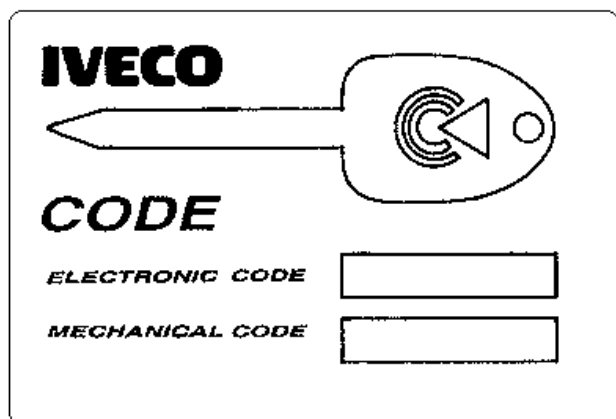
8659



000292t

AERIAL

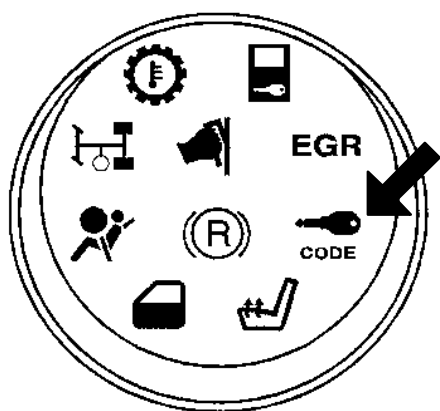
Figure 108



CODE CARD

000293t

Figure 109



000294t

WARNING LIGHT

Code card

Card containing the Electronic code and the Mechanical code.

- ☐ Electronic code or PIN code
Code absolutely necessary for the starting procedure in the event of an emergency.
- ☐ Mechanical code
Code needed for request to duplicate the mechanical part of the key.

Warning light

This is located on the instrument cluster and informs the driver whether the system is working properly or of any possible faults.

Moving the ignition key to drive the control unit tests the system turning on the warning light for about "4 seconds".

If it goes out after this time, this means that the key has been recognised and the system is working, if it does not it indicates the possible system faults.

Key teaching procedure

In the event of loss or replacement of the keys it will be absolutely necessary to follow a precise procedure solely through the diagnostic tools.

- ☐ Connect the diagnostic tool.

Key teaching

- ☐ Engage the first key and turn it to drive.
- ☐ Wait for the "code" warning light on the cluster to go out and turn the key to stop.
- ☐ Engage the new key and turn it to drive, repeat the operations described previously.

If the code warning light does not go out during the teaching procedure, this means that the operation has not been carried out correctly.

Possible causes:

- ☐ The same key has been engaged twice.
- ☐ The key has not been turned to stop in useful time.
- ☐ An attempt has been made to teach more than three keys.
- ☐ Keys already used with another control unit have been used.
- ☐ Aerial reception problems.

NOTE After following this procedure the control unit memorises the keys and will no longer make this operation possible without entering the PIN code.
The keys enabled for starting will never be more than three.
A previously memorised key but not included in the last teaching process is no longer able to start the vehicle.

Emergency procedure

It is absolutely necessary to enter the "PIN code" given on the "Code Card", only using the accelerator pedal as described below:

- ☐ Move the ignition key to Drive.
- ☐ After about ~ 2 seconds the EDC warning light starts to flash quickly
- ☐ Press the accelerator pedal and keep it pressed for about ~ 15 seconds.
- ☐ The EDC warning lamp starts flashing slowly.
- ☐ When the number of flashes corresponds to the first digit of the PIN code press the accelerator completely and then release it again. (During this press the EDC warning light stays off).
- ☐ Continue with the reading and corresponding pressing on the accelerator pedal for the remaining four numbers of the "PIN code".
- ☐ At the end of the sequence if the code entered is correct and there are no system faults, the EDC warning light stops flashing. The operation has been concluded correctly.
- ☐ Start the vehicle.

System self-diagnostics

After the initial test, depending on the behaviour of the "code" warning light, the system is capable of informing the operator of possible system faults, as follows:

- ☐ Warning light "flashing continuously" with a frequency of "0.3 sec. ON" and "3 sec. OFF" means that there is an error or that the emergency starting procedure has not been carried out correctly.
- ☐ Warning light "flashing continuously" with a frequency of "0.3 sec. ON" and "0.3 sec. OFF", means that no key teaching procedure has been carried out.
- ☐ Warning light "always on" means that the key teaching procedure has not been carried out correctly.

For preliminary information, it is possible to display any fault codes on the warning light module of the instrument cluster activating the Blink code.

For complete, more thorough diagnosis it is however indispensable to use the tools available to the service network, such as MODUS.

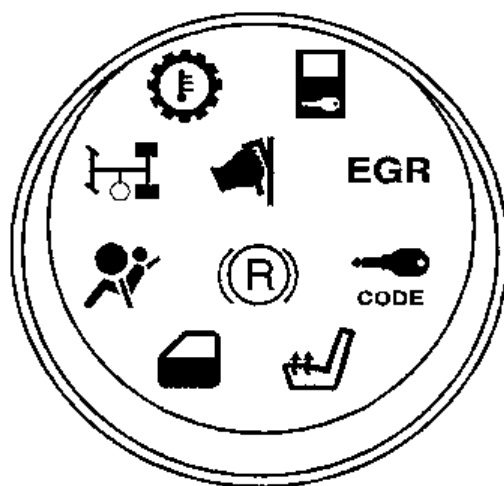
Diagnostics through the Blink Code

To activate the Blink code it is necessary to earth line K for at least 2 seconds with the ignition key at drive to display the first error.

Repeat this operation (line K at earth) to display any other faults.

The table below shows the various error codes that may appear during diagnostics.

Figure 110



000294t

Number of blinks	Type of fault
1	Not connected or not configured central unit
2	Central unit not communicating with EDC
	Central unit not communicating during installation
3	Central unit not installed
4	—
	Short circuit/disturbance on communication line between control unit and "EDC"
5	Key with unknown code
6	Key with code not detected
7	Aerial not connected
8	Internal control unit fault
9	Short circuit on alarm disable line
—	"Code" warning light short circuit
—	"Code" warning light open circuit



If after accurate diagnosis it is necessary to replace one or more components, proceed as described below

PART TO BE REPLACED	PARTS TO BE ORDERED	OPERATIONS AND NOTES
One or two keys (with one still available)	Parts Keys	⇒ Cut the keys according to the mechanical code. ⇒ Perform "Key memorising" in Immobilizer diagnostics (*). NB. In this procedure also enter the remaining working keys otherwise they will NO LONGER be enabled for starting.
Addition of a key		
Steering lock and/or ratchets	KEYS KIT with: 2 cut Parts keys Steering lock + Ratchets	⇒ Change the steering lock and ratchets. ⇒ Perform "Key memorising" in Immobilizer diagnostics (*). ⇒ State the new mechanical code on the Code Card.
ALL the keys		
Ratchets (excluding steering lock)	KIT with: 2 traditional keys + ratchets	⇒ Change the ratchets. NB. Traditional keys means keys without Transponder.
Code card (due to loss)	Code Card	⇒ Complete the "Code Card Duplicate Request Form" printing it from MODUS and send it as mentioned in the ORDER MANAGEMENT form of the IVECO SPARES warehouse concerned.
Aerial	Aerial	⇒ Replace.
ECU Immobilizer	KIT ECU Including: Immobilizer ECU – 2 keys to be cut New Code Card	⇒ Cut the keys according to the mechanical code. ⇒ Perform "Key memorising" in Immobilizer diagnostics (*). NB. At the end of this operation, check that the electronic code shown on the screen corresponds to the one printed on the Code Card, if it is different, put the one on the screen on the Code Card. ⇒ In EDC diagnostics perform "new key recognition". ⇒ Put the old mechanical code on the new Code Card. ⇒ Scrap the old keys as it will be NO LONGER possible to use them. ⇒ Complete the "New Immobilizer Installation Form" printing it from MODUS and send it as mentioned in the ORDER MANAGEMENT form of the IVECO SPARES warehouse concerned.
EDC control unit	EDC control unit	⇒ The system recognises the new actuator automatically the first time the ignition key is moved to Drive. NB. To order the EDC Control Unit complete the specific form printing it from Modus and sending it to the Market Technical help Desk.



(*) With release MODUS 2.0 (Windows), 1.6C (DOS) or IWT 1.4, before "Storing new keys", it is necessary to disconnect EDC central unit. Once the operation has been completed, clear the fault memory to prevent the error from staying memorised on the actuator.

⇒ For subsequent releases keep to any new instructions given on the screen.

ABS/EBD/ABD

General

The braking of a vehicle in motion and the consequent deceleration and stopping space mainly depend on the grip between the tyre surface and the type of road surface.

With a perfectly efficient braking system, further improvement of braking can be obtained only acting on the tyre friction characteristics or on the quality of the road surface.

Even in these optimum conditions, absolute braking safety is not however guaranteed when needing to cope with particular critical situations, such as low grip due to the conditions of the wet or icy road surface: this compels the driver to moderate the braking action to prevent one or more wheels from partially locking, with the possibility of dangerous skidding.

The friction between the tyre and the road surface does not correspond to the friction between rigid bodies but to the

skidding (or slipping) between the tyre and the road in the contact area.

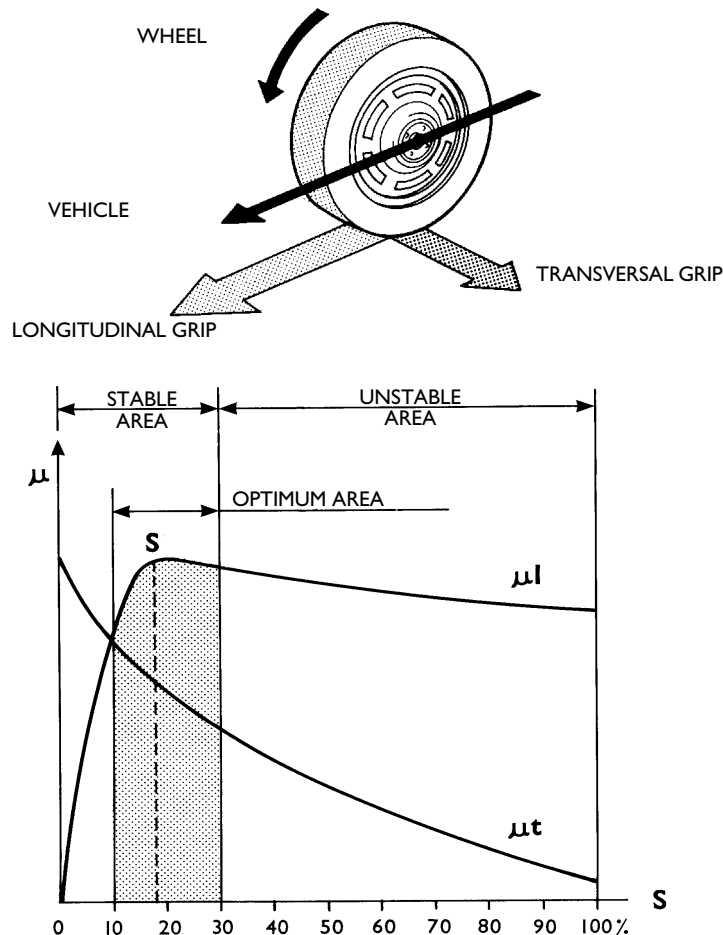
The figure shows the indicative trend of the longitudinal " μ_l " and transversal " μ_t " rip coefficients in relation to the percentage of slipping "S".

The diagram reveals that the maximum value of the longitudinal grip coefficient is not when the wheel is locked, but for a much lower slipping value.

Therefore, the road-tyre grip can be exploited to optimise longitudinal or transversal control of the vehicle. Namely, it is possible to utilise the grip trying to make the braking distance as short as possible or to ensure the best possible handling.

An excellent compromise is obtained using slip rates near the point S^* in which there is a high μ_l value which ensures optimum braking and a μ_t value that offers good lateral roadholding.

Figure 111



TREND OF WHEEL LONGITUDINAL AND TRANSVERSAL GRIP IN RELATION TO SLIPPING

In order to better understand the logic of the system and the parameters that govern it a few basic concepts should be briefly examined.

Due to the effect of the braking action the tyre, that before was rolling freely, undergoes a deformation, called "braking" deformation, in its area of contact with the road and slows down rotation reducing its peripheral speed, to a higher extent than the linear speed of the vehicle.

At the limit, with the wheels completely locked under the braking action, and thus with a wheel peripheral speed of zero, there is the maximum deviation between the wheel rotation speed and the linear speed of the vehicle.

Slipping varies its values within the limits set by the two extreme conditions in which the wheel and vehicle speed may be in.

When the wheel is free, not braked, thus turning at the same speed as the vehicle, the slipping coefficient is 0%.

When the braked wheel is completely locked and the vehicle continues moving forwards due to inertia, the skidding coefficient amounts to 100%.

Experimentation has made it possible to establish that the most effective braking condition is obtained for optimum slipping values contained between about ~ 5% and 20%.

The need to contain the slipping values within precise limits is imposed by the behaviour of the tyres under the braking action, during which the braking friction coefficient comes into play.

The higher this coefficient, the more braking is effective.

If the relation between slipping S and the friction coefficient μ , we see how there are the highest friction coefficient values for slipping values between an average of 5% and 20%.

As the friction coefficient is directly proportionate with the applicable braking force, the result is that the "ABS" device acts in such a way as to apply the maximum braking force exactly in correspondence with the best friction coefficient, and this system tends to bring any type of vehicle within this sector.

Figure 112

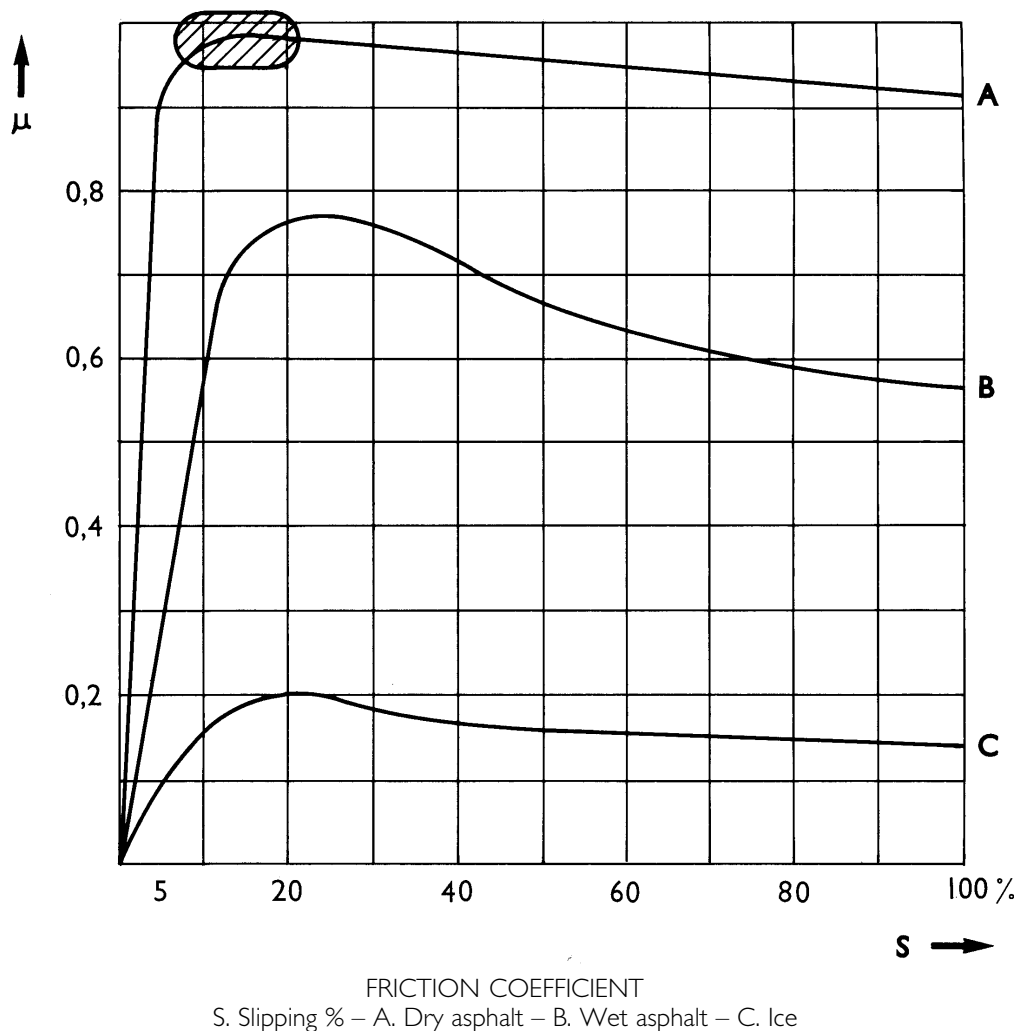
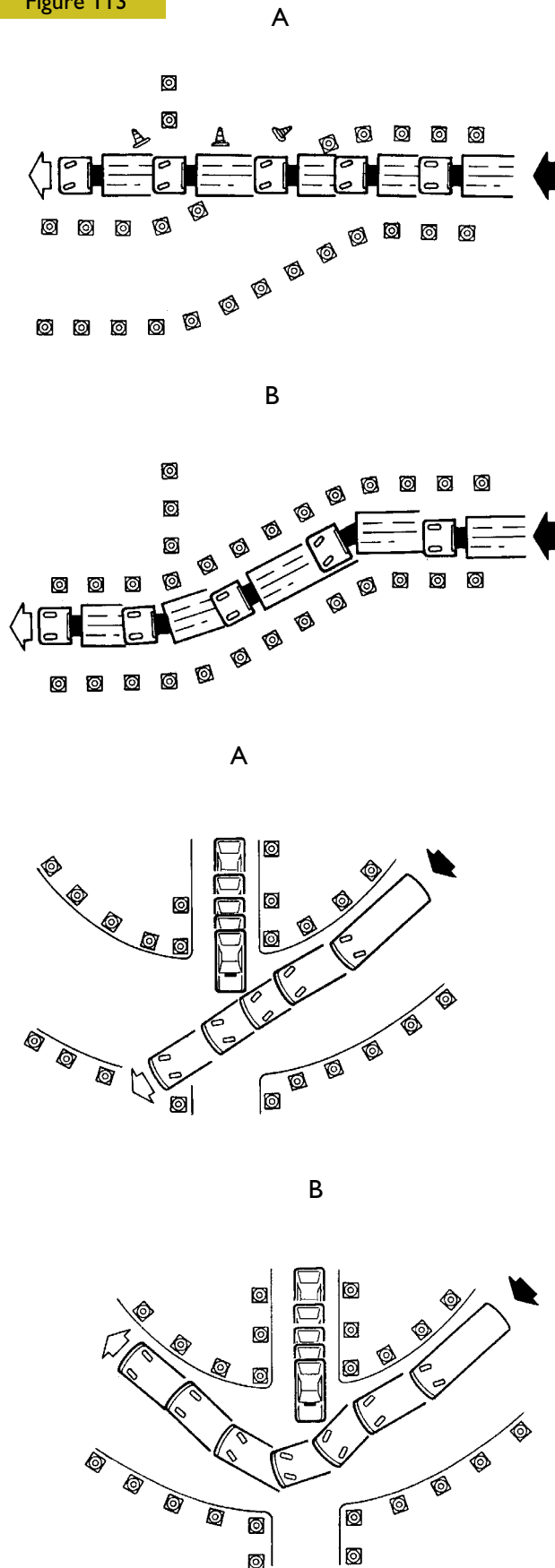


Figure 113



VEHICLE BRAKING TREND
WITHOUT AND WITH ABS

A. Vehicle without ABS – B. Vehicle with ABS

The locking of one or more wheels, due to excessive pressure on the brake pedal, (anyway higher than allowed by the grip) can cause loss of steering control of the vehicle if both front wheels lock, skidding, resulting in about-facing, if both rear wheels lock.

The need for abrupt braking on a bend, compels the driver to immediate action to avoid losing control of the vehicle, resulting in skidding.

Lastly, the so-called panic braking (caused by a sudden obstacle), compels the driver to press the brake pedal as forcibly as possible in the desperate attempt to stop in time..

So what is the solution even for the most expert drivers?

Being able to avail of a braking system capable of fully exploiting all the grip available without locking the wheels, except at a minimum pre-established speed.

The ABS Antibrake Locking System has been developed exactly to obtain this.

A device inserted in the braking system, with the task of preventing wheel locking when the brake operating pressure is too high in relation to the grip of the tyre on the ground.

Therefore, the purpose of the "ABS" device is to ensure vehicle stability (under all braking conditions) preventing locking of the wheels regardless of the conditions of the road surface, thereby ensuring total use of the grip available.

Also in the vent of emergency braking, the system makes it possible to maintain "steerability" of the vehicle, i.e. acting on the steering to avoid obstacles, without the danger of skidding.

Keeping the rotation and grip of the wheels within the optimum parameters, the system makes it possible to obtain those braking distances that only an expert driver would be able to approach; this way even the less expert driver is able to act like one of the best.

The diagram opposite shows some examples of braking without ABS and with ABS.

In order to be able to intervene effectively the system must not only be precise in response, but also very quick.

This is now possible thanks to electronic information which warrants reliability, precision and rapidity, with a minimal number of components and lower system cost.

In addition to the ABS, the system in question also incorporates the EBD and ABD functions.

EBD – Electronic Brake Force distribution

This device replaces and optimises the function of the present hydraulic braking action proportioning valve, better controlling the braking force on the rear wheels.

It is obtained by adding specific software to the ABS and it acts in a determinate interval before the cutting in of the ABS.

It ensures control on any locking of the rear wheels in relation to the front ones, optimising the braking force in the different load, driving and conditions of use of the vehicle.

ABD – Automatic Brake Differential

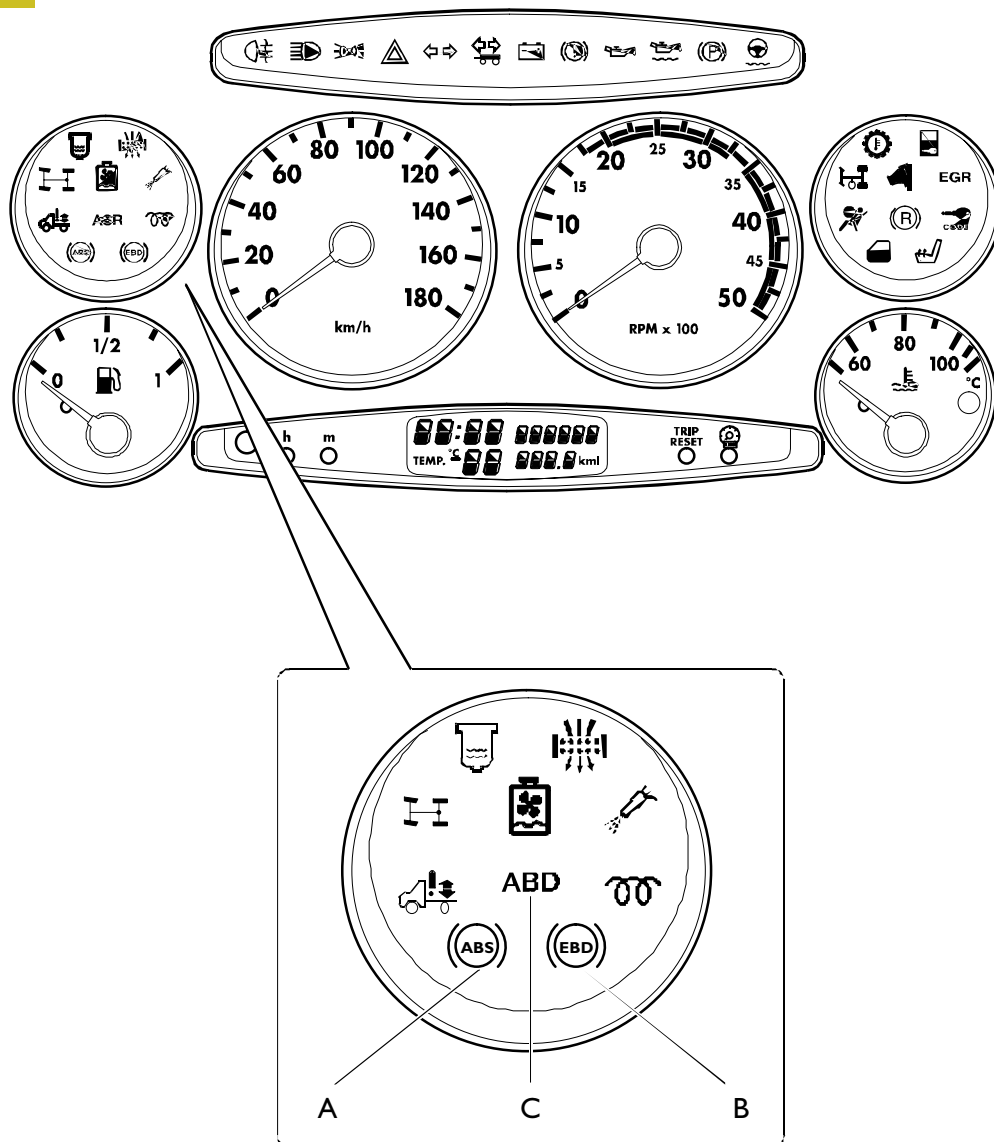
This device comes into action automatically braking the driving wheel that has less grip when moving off or travelling and tends to skid transferring the torque through the differential to the wheel with higher grip.

It cuts in up to a speed of about 40 km/h and does not interfere with the normal braking system.

It cuts in up to a speed of about 40 km/h and does not interfere with the normal braking system.

The driver is informed when this system is engaged by the flashing of the yellow warning light on the dashboard.

Figure 114



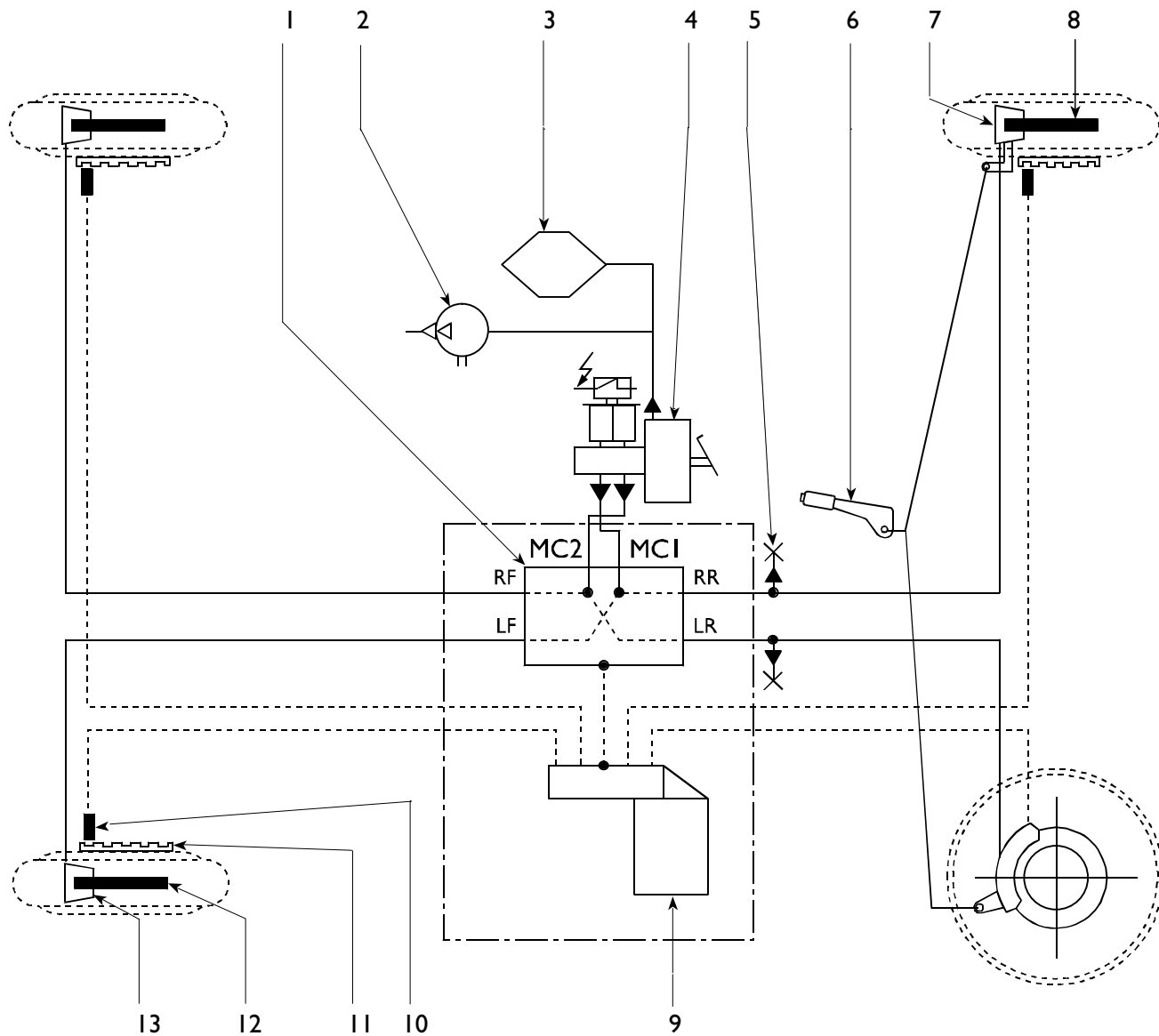
SYSTEM WARNINGS ON DASHBOARD

A. ABS failure warning light 58703 – B. EBD failure warning light 58734 – C. ABD failure warning light 58704

SYSTEM WITH 4 CROSSED CHANNELS (X)

The system shown is used on the following vehicles: 29 L – 35 S

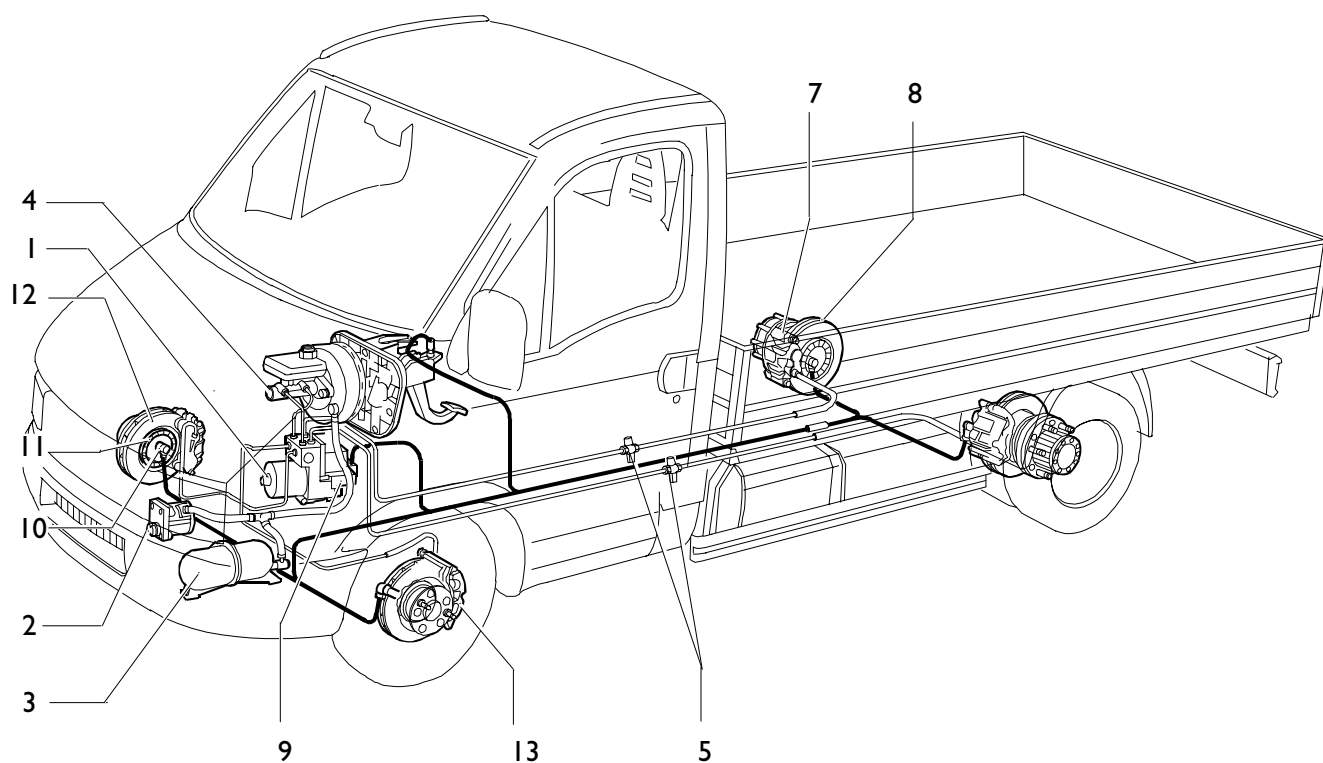
Figure 115



1. Electrohydraulic modulator – 2. Vacuum pump – 3. Vacuum tank – 4. Vacuum servobrake – 5. Hydraulic control sockets
 – 6. Parking brake lever – 7. Rear brake caliper – 8. Rear disk brake – 9. Electronic control unit – 10. Sensor –
 11. Phonic wheel – 12. Front brake disk – 13. Front brake caliper

Location of components

Figure 116



1. Electrohydraulic modulator – 2. Vacuum pump – 3. Vacuum tank – 4. Vacuum servobrake – 5. Hydraulic control sockets – 7. Rear brake caliper – 8. Rear disk brake – 9. Electronic control unit 88000 – 10. Sensor 88001 – 11. Phonic wheel – 12. Front brake disk – 13. Front brake caliper

Sensor on phonic wheel

The sensors supply the electronic control unit all the useful information, with the necessary continuity, for it to be able to control the solenoid valves.

The signals are obtained by magnetic flux lines which are closed through the teeth of an iron toothed wheel facing the sensor and turned with the wheel.

The passage from solid to hollow, due to the presence or absence of the tooth determines a change of the magnetic flux which is enough to create an electromotive force induced at the terminals of the sensor and thus an alternate electric signal, which is sent to the electronic control unit.

The distance between the sensor and the wheel, called gap, must clearly have a pre-established value so that correct signals are supplied ($0,8 \div 1,6 \text{ mm}$).

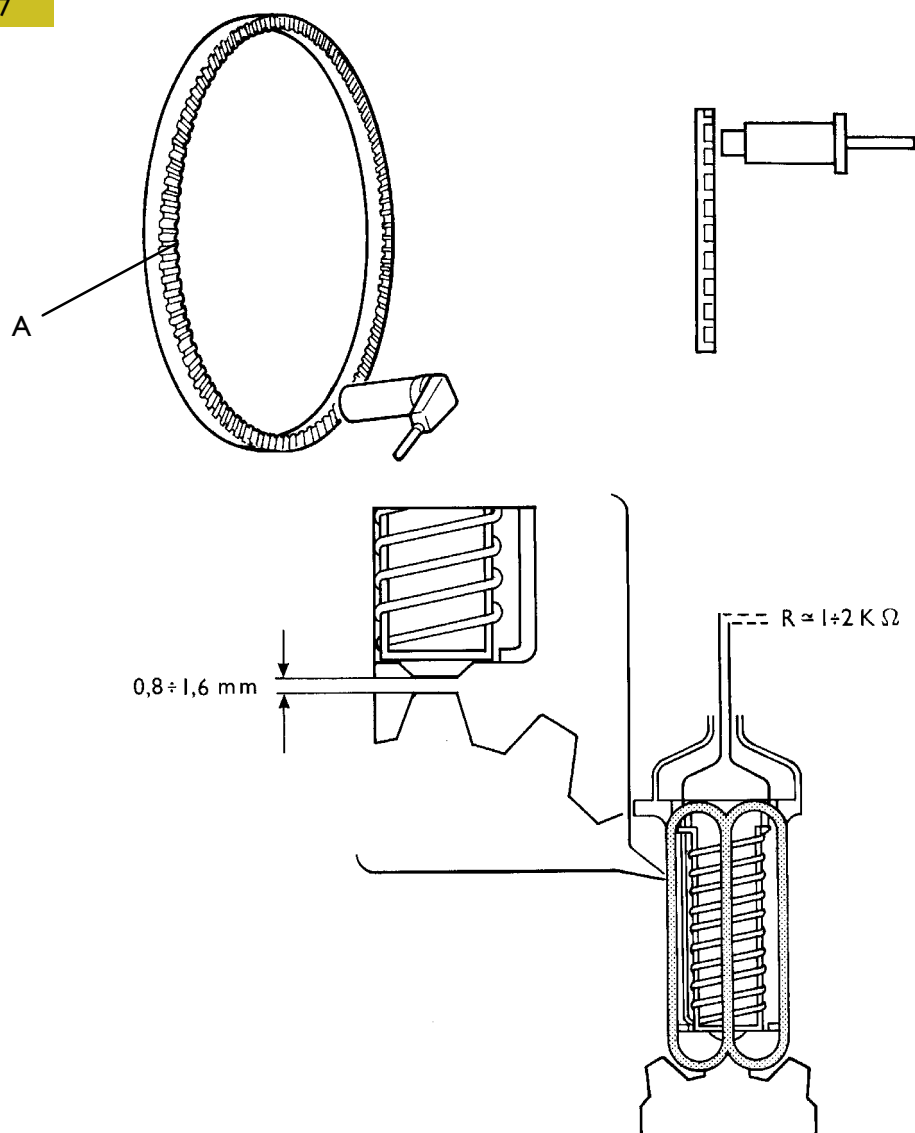
The resistance rating of each single sensor detected at the ends of the connector is between $1 \div 2 \text{ k}\Omega$.

The toothed wheel is called phonic wheel because the signal it generates has the frequency of a sound wave.

The frequency of this signal serves to define the wheel rotation speed.

The changes of the frequency, i.e. the rapidity with which the signals follow one another serve to define the acceleration and deceleration values.

Figure 117



SENSOR ON PHONIC WHEEL

A. Phonic wheel

Electrohydraulic control unit/modulator

The component integrates both the electronic part and the hydraulic modulator for system management.

It comprises the following:

A – Electronic control unit

B – Electrohydraulic modulator

C – Accumulator

It is connected to the hydraulic system by the following connections:

MC1 – Diagonal supply LF/RR

MC2 – Diagonal supply RF/LR

LF – Left front axle output

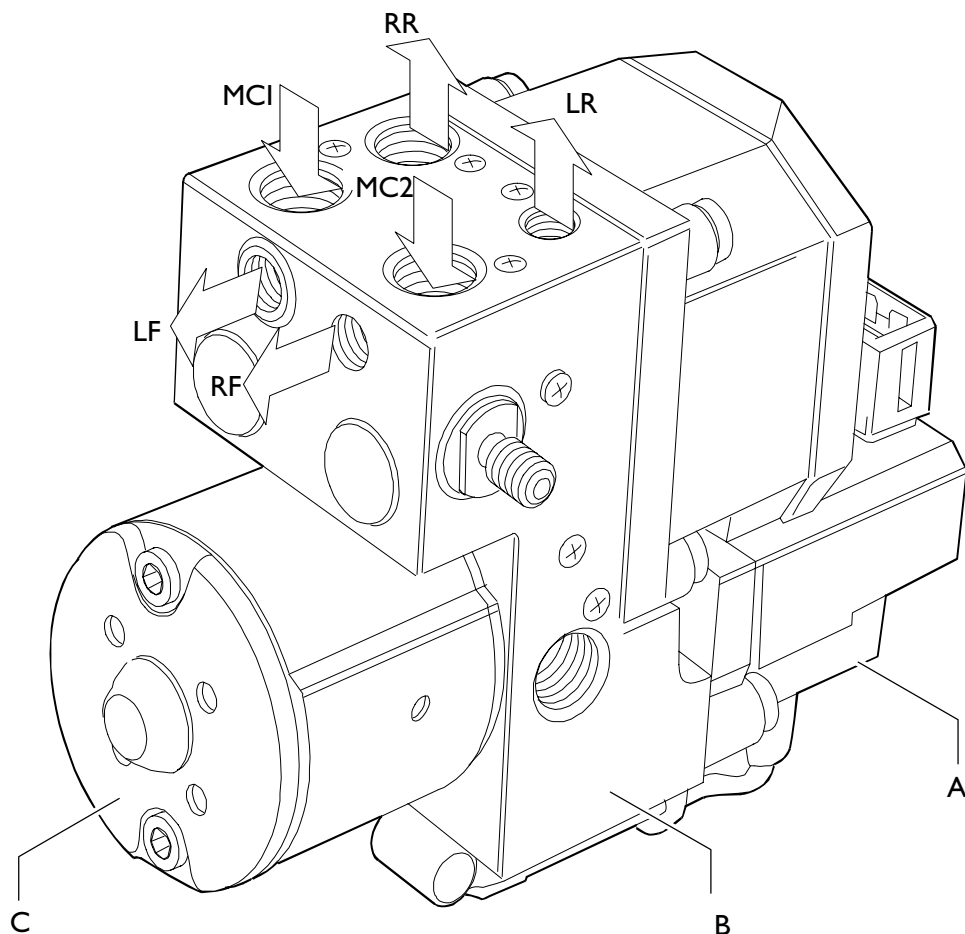
RR – Right rear axle output

RF – Right front axle output

LR – Left rear axle output

It is connected to the electric system through a 31-pin connector.

Figure 118



CONTROL UNIT/ELECTROHYDRAULIC MODULATOR

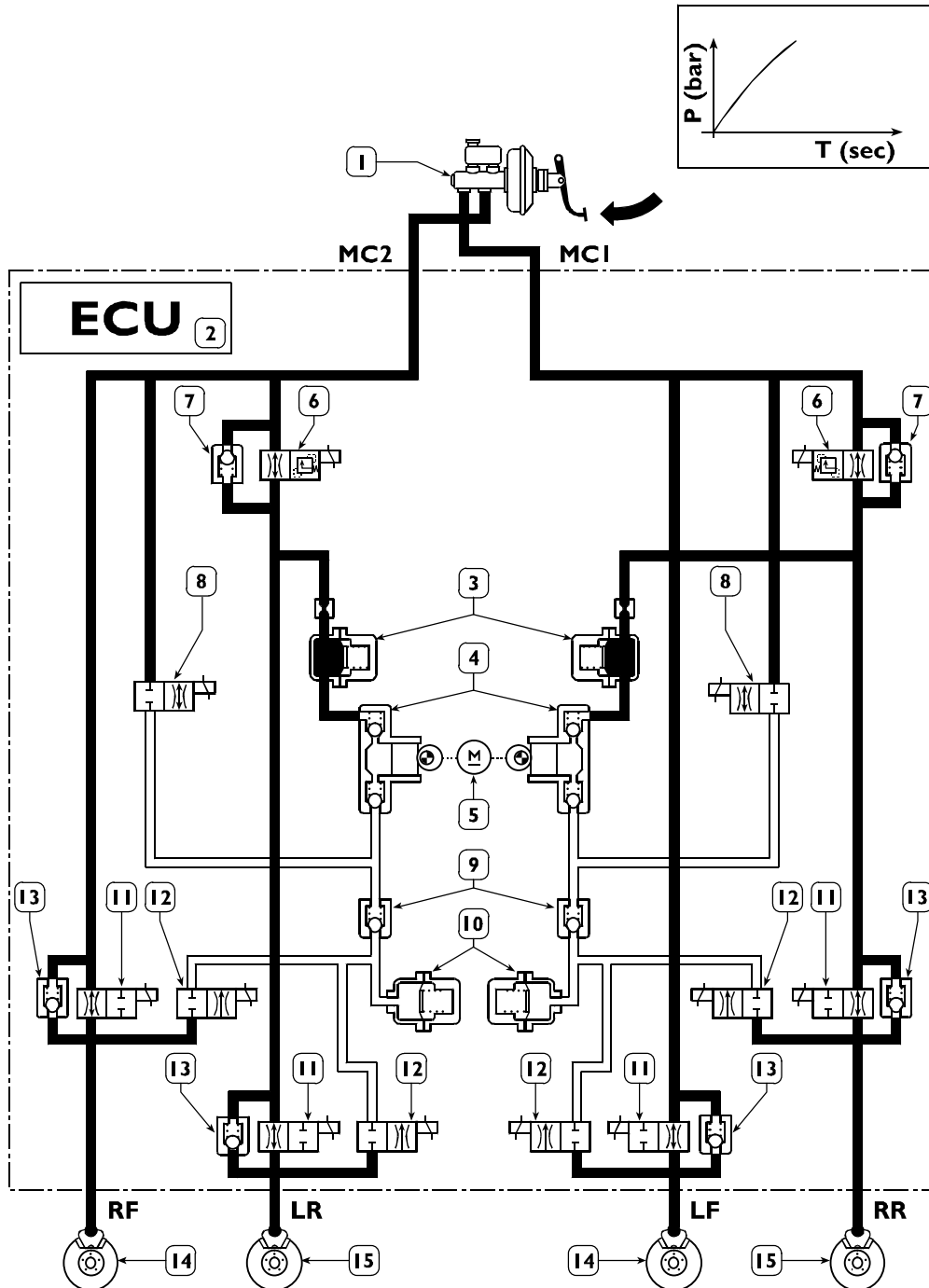
Electrohydraulic modulator

The figure schematically shows the connection of the modulator and its internal components.

Pressure increase

Operating the brake pedal, the pressurised oil can reach the brake calipers as both the drive solenoid valves "6" and the supply solenoid valve "11" are open.

Figure 119



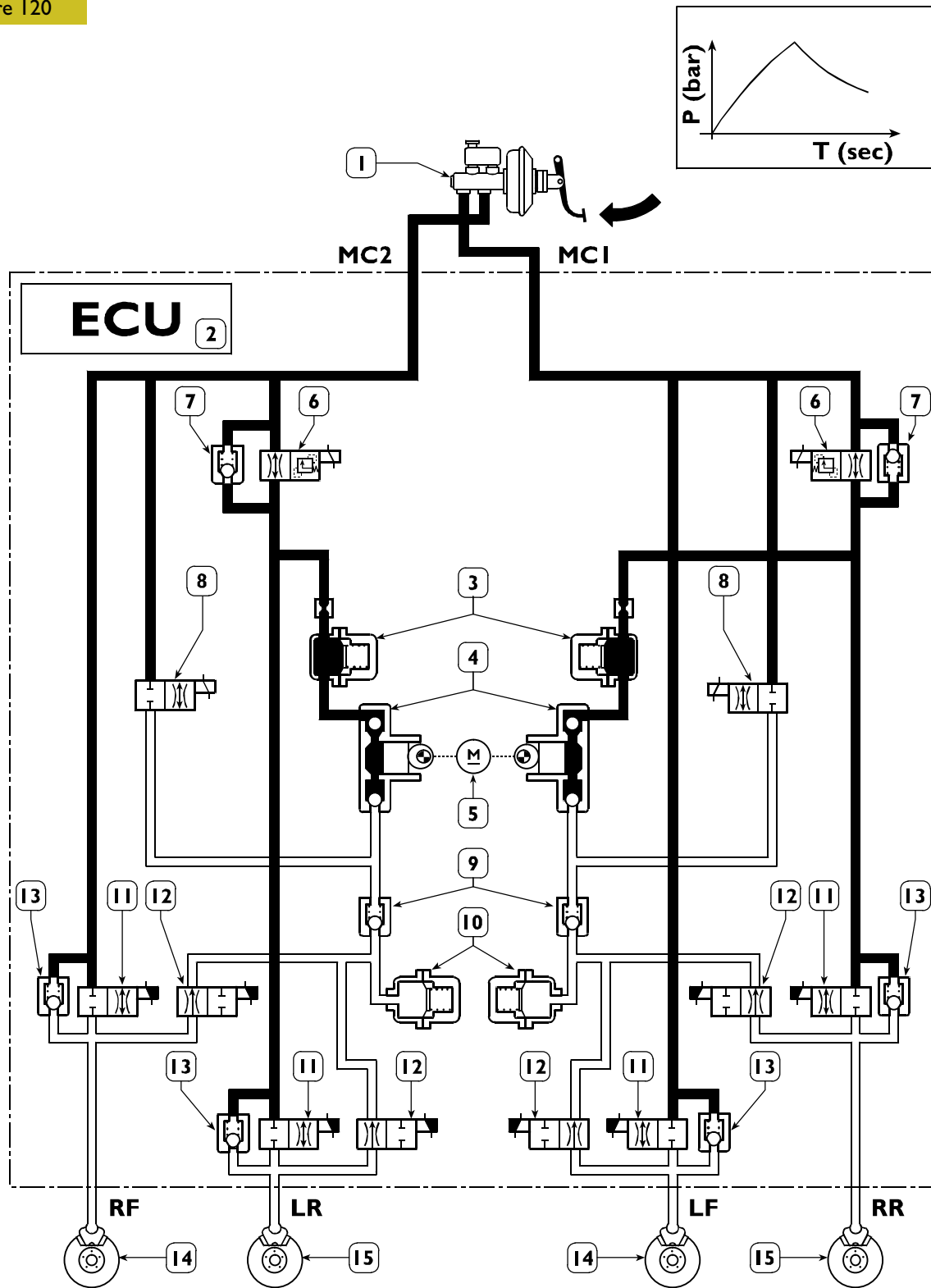
1. Vacuum servobrake – 2. Electronic control unit – 3. High pressure accumulator – 4. Recovery pumps – 5. Recovery pump drive motor – 6. ABD control solenoid valves (setting 90 bar) – 7. One-way safety valves – 8. ABD intake solenoid valves – 9. ABD one-way valves – 10. Low pressure accumulator – 11. Supply solenoid valves – 12. Discharge solenoid valves – 13. One-way quick pressure reduction valves – 14. Front axle disk brakes – 15. Rear axle disk brakes

Pressure lowering

If the sensors detect the tendency of a wheel or wheels to lock, they inform the control unit which reduces the braking force activating the supply solenoid valves "11" and the discharge solenoid valves "12".

At the same time, supplying the motor "5" that drives the pumps "4" it will be possible to recover the excess oil volume in the high pressure accumulators "3".

Figure 120

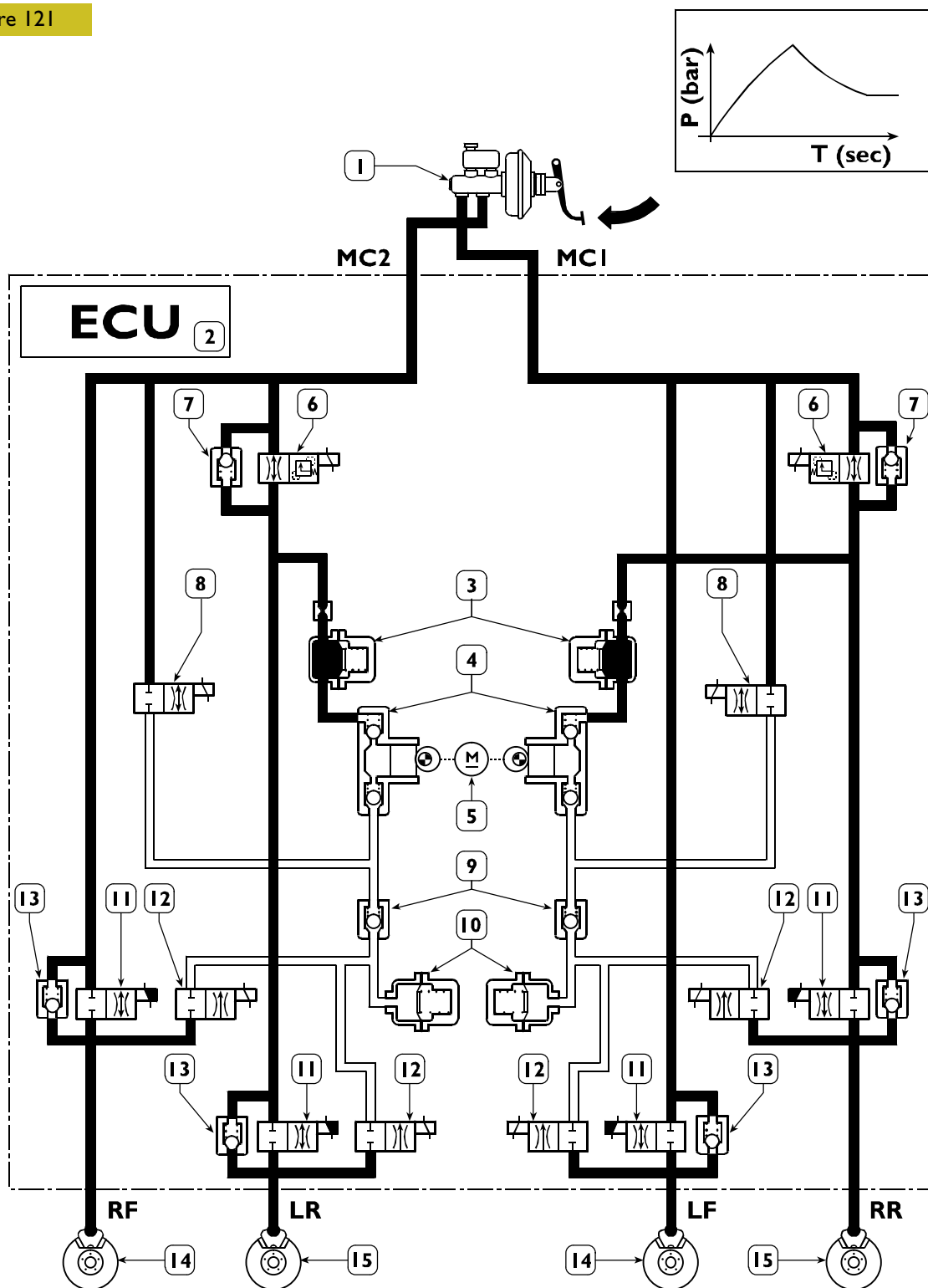


SYSTEM OPERATING LAYOUT SHOWING THE PRESSURE MAINTENANCE PHASE

Pressure maintenance

Once the optimum braking force has been reached, the control unit can keep it constant, no longer energising the discharge solenoid valves "12", the motor "5" and the associated recovery pumps "4", while the supply solenoid valves "11" continue being supplied.

Figure 121



SYSTEM OPERATING LAYOUT SHOWING THE PRESSURE MAINTENANCE PHASE

Cutting in of EBD device

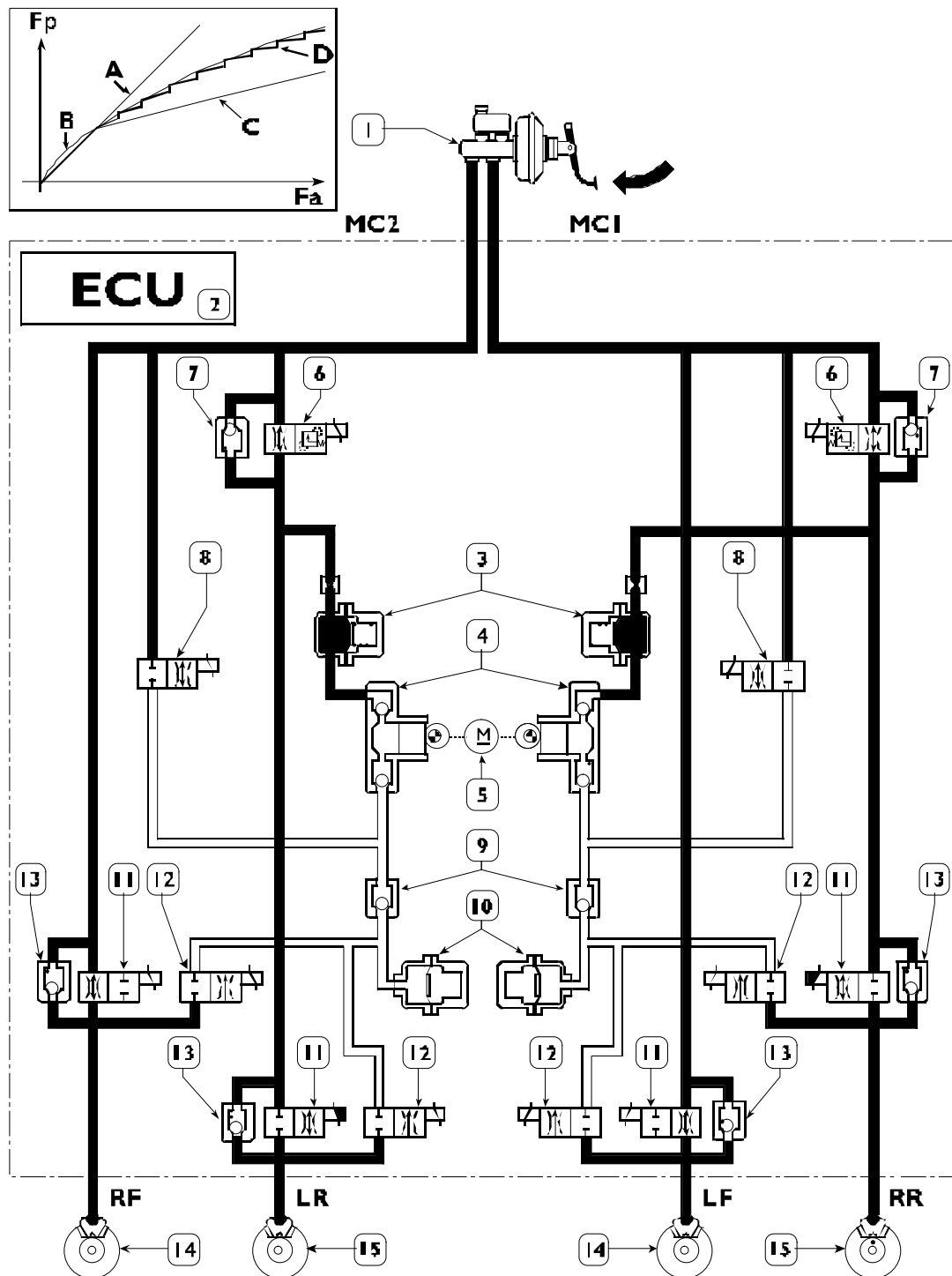
If the sensors detect that one or both wheels tend to lock in relation to the front wheels, they inform the control unit which suitably supplies the supply solenoid valves "II" of the rear axle to optimise the braking force.

A fault on certain system components causes the ABS system to be cut off (displayed by turning on the special warning light), still leaving the EBD system operational.

If both failure warning lights turn on (ABS and EBD) this means an EBD system failure.

Under these conditions the braking force will not be distributed between the axles, therefore rear wheel locking may occur with the possibility of skidding.

Figure 122



FP. Rear axle braking force – FA. Front axle braking force – A. Servobrake distribution curve – B. Ideal distribution – C. Curve – distribution curve of any porportioning valve – D. EBD system distribution curve

Cutting in of ABD device

If the sensors detect that one or both rear axle wheels tend to skid, they inform the control unit which supplies the intake solenoid valves "8" and the drive valves "6".

At the same time, supplying the motor "5" which drives the pumps "4" it will be possible to generate the pressure to be sent to the brake calipers concerned.

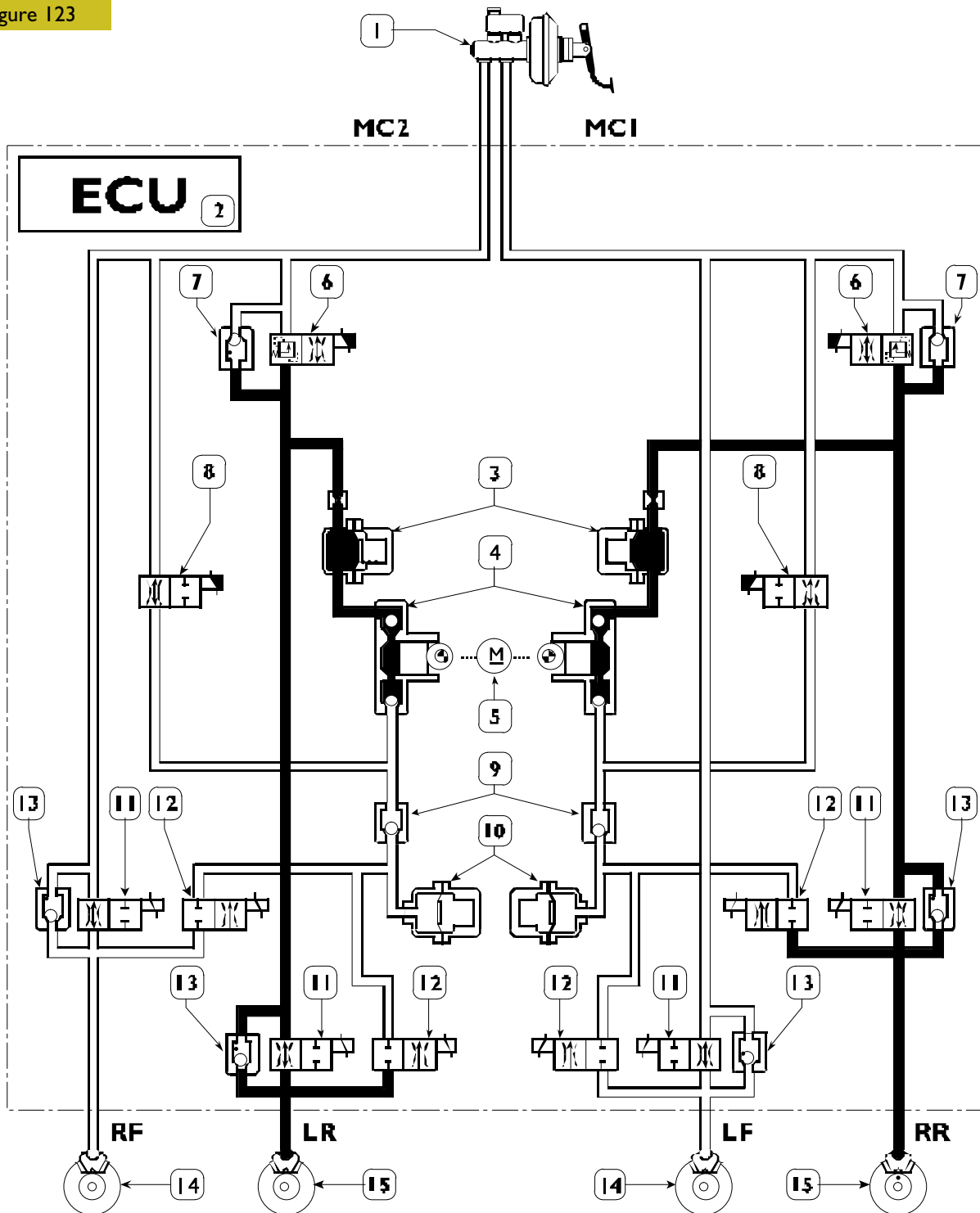
The braking force generated in this manner is modulated by solenoid valves "11" and "12".

During this phase the maximum pressure is limited by the drive solenoid valves "6" and cannot exceed 90 bar.

The driver is informed of the cutting in of this system by the flashing of the corresponding warning light on the dashboard.

If the warning light turns on and glows steadily, there is an ABD system failure.

Figure 123

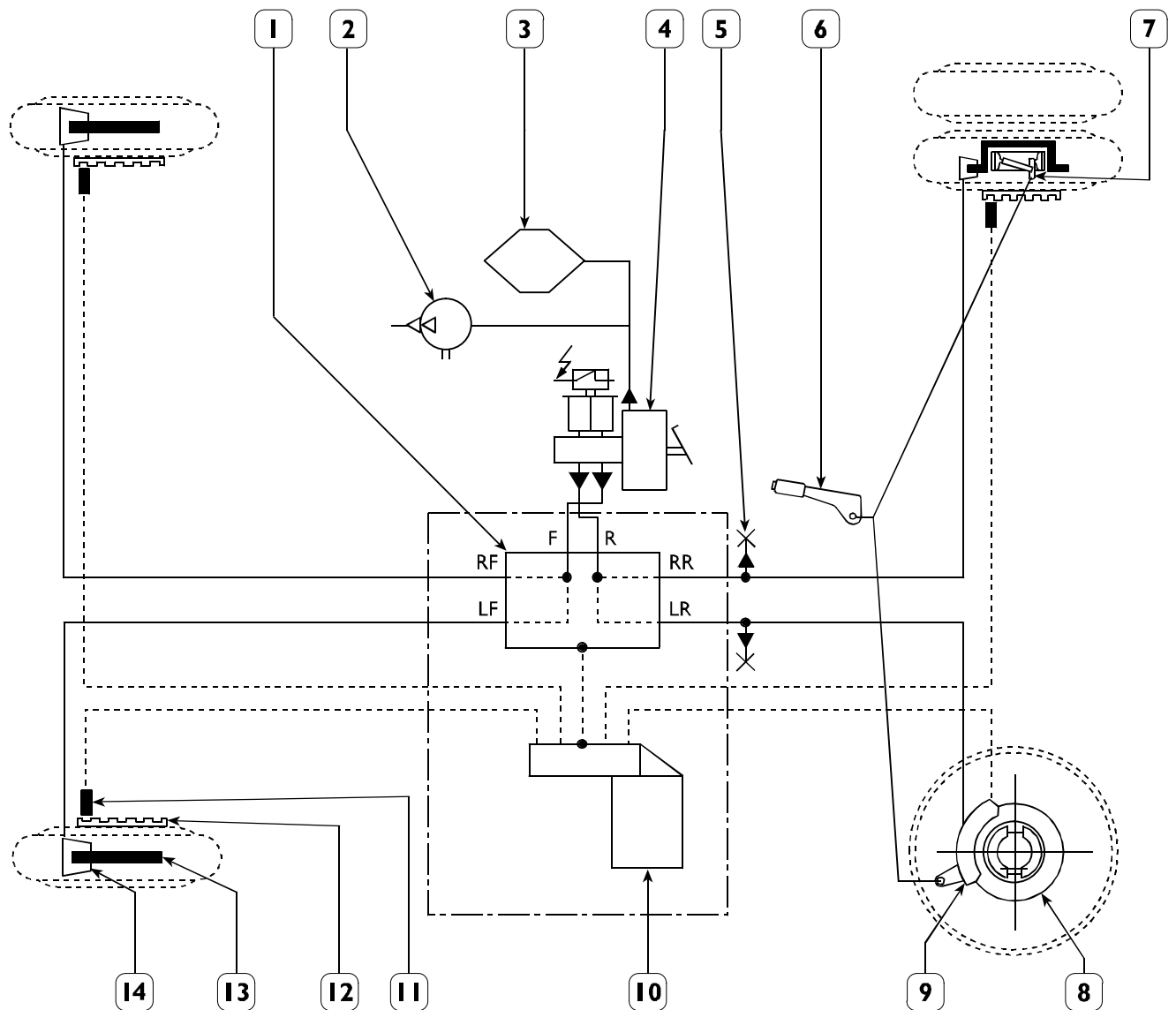


ABD DEVICE OPERATING LAYOUT

SYSTEM WITH 4 PARALLEL CHANNELS (II)

The system shown in Figure 124 is used on the following vehicles: 35 C – 40 C – 45 C – 50 C – 65C.

Figure 124



1. Electrohydraulic modulator – 2. Vacuum pump – 3. Vacuum tank – 4. Vacuum servobrake – 5. Hydraulic control sockets – 6. Parking brake lever – 7. Parking drum brake – 8. Rear brake disk – 9. Rear brake caliper – 10. Electronic control unit – 11. Sensor – 12. Phonic wheel – 13. Front brake disk – 14. Front brake caliper

Electrohydraulic control unit/modulator

This component integrates both the electronic part and the electrohydraulic system control modulator.

It comprises the following:

A – Electronic control unit

B – Electrohydraulic modulator

C – Accumulator

It is connected to the hydraulic system by the following connections:

F – Front axle supply

R – Rear axle supply

LF – Left front axle output

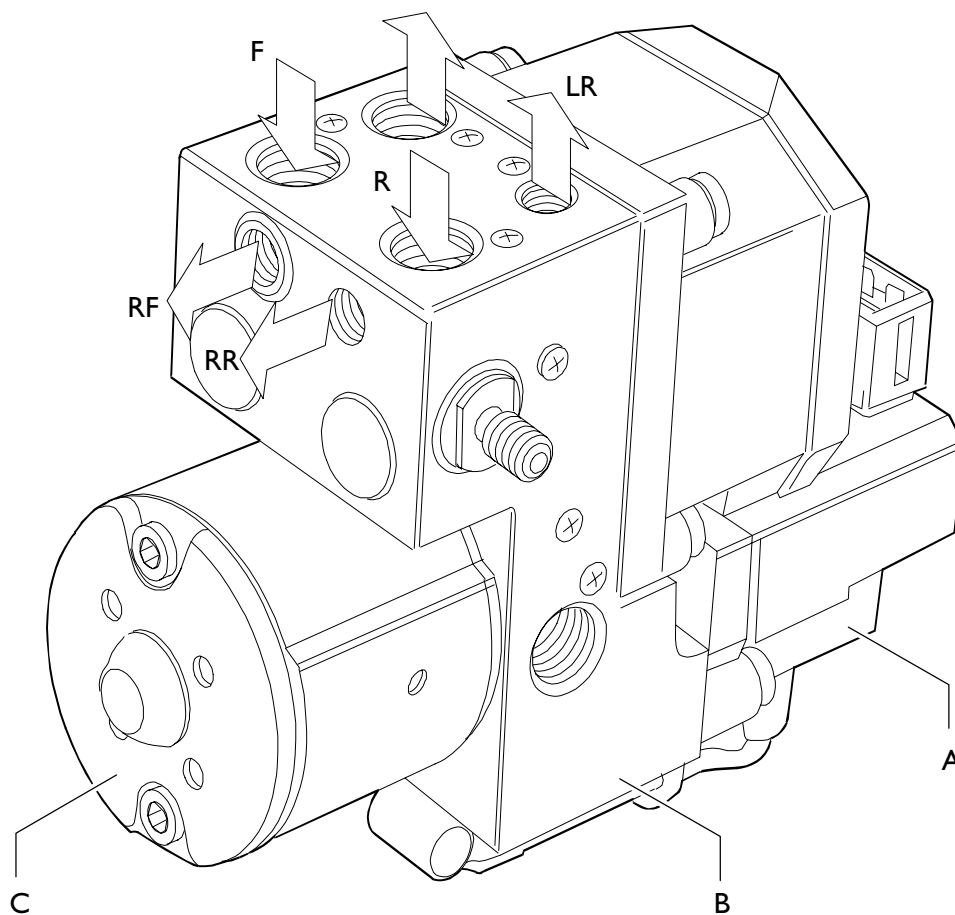
RF – Right front axle output

LR – Left rear axle output

RR – Right rear axle output

It is connected to the electric system through a 31-pin connector.

Figure 125



CONTROL UNIT/ELECTROHYDRAULIC MODULATOR

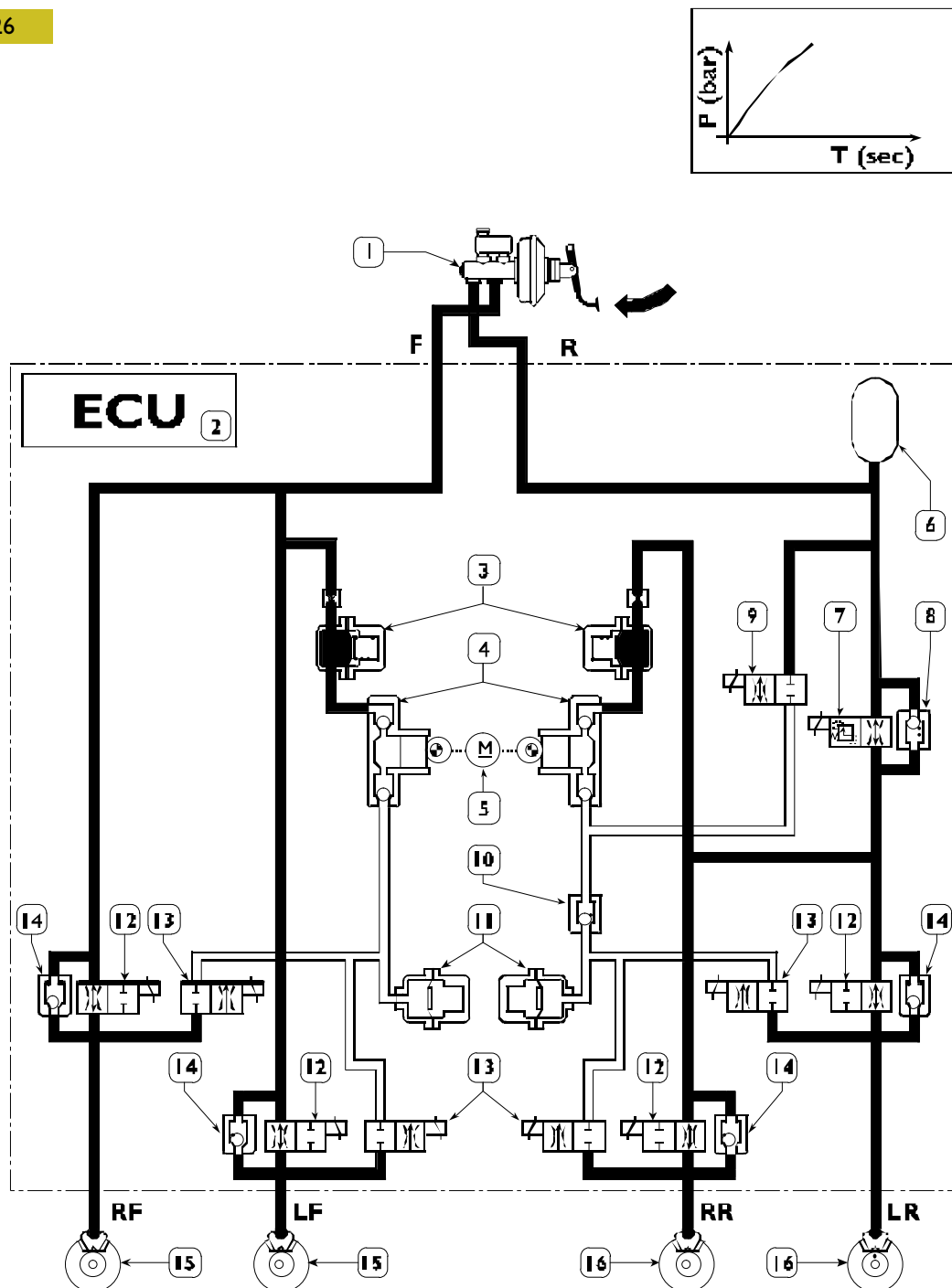
Electrohydraulic modulator

The figure shows the layout of the connection of the modulator and its internal components.

Pressure increase

Operating the brake pedal, the pressurised oil can reach the brake calipers as both the drive solenoid valve "7" and the supply solenoid valves "12" are open.

Figure 126



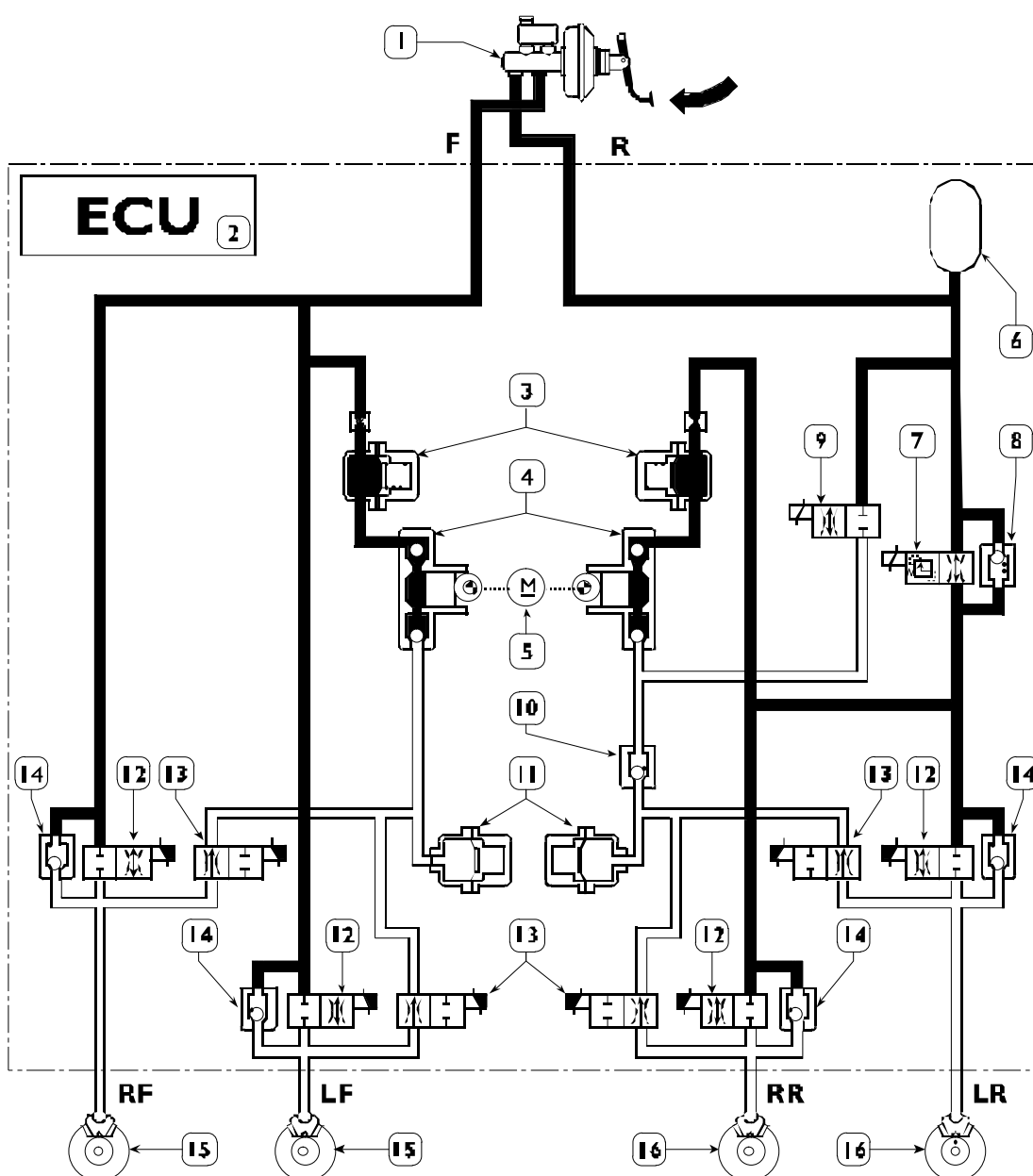
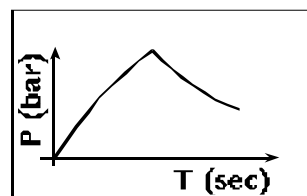
- 1. Vacuum servobrake – 2. Electronic control unit – 3. High pressure accumulators – 4. Recovery pumps –
- 5. Recover pump drive motor – 6. Accumulator for rear axle system – 7. ABD drive solenoid valve (90 bar setting) –
- 8. One-way abd valve – 9. Intake abd solenoid valve – 10. ABD one-way valve – 11. Low pressure accumulators –
- 12. Supply solenoid valves – 13. Discharge solenoid valve – 14. One-way quick pressure relief valves – 15. Front axle disk brake – 16. Rear axle disk brakes

Pressure reduction

If the sensors detect that a wheel or wheels tend to lock, they inform the control unit which reduces the braking force activating the supply solenoid valves "12" and the discharge solenoid valves "13".

At the same time, supplying the motor "5" which drives the pumps "4", it will be possible to recover the excess oil in the high pressure accumulators "3".

Figure 127



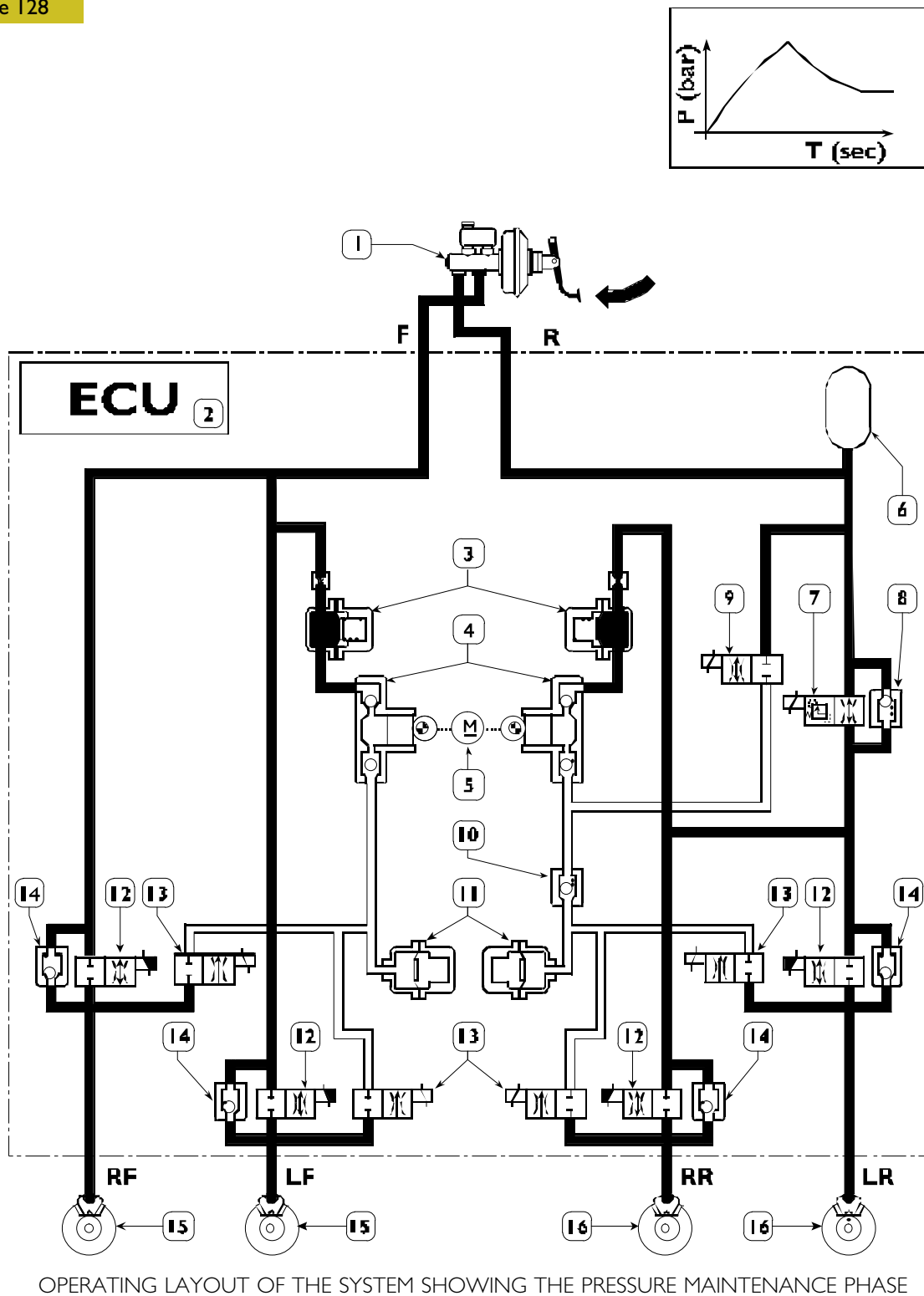
OPERATING LAYOUT OF THE SYSTEM SHOWING THE PRESSURE REDUCTION PHASE

Pressure maintenance

Once the optimum braking force has been reached, the control unit is able to keep it constant by no longer

energising the discharge solenoid valves "13", the motor "5" and the corresponding recovery pumps "4"; while the supply solenoid valves "12" continue being energised.

Figure 128



Cutting in of the EBD device

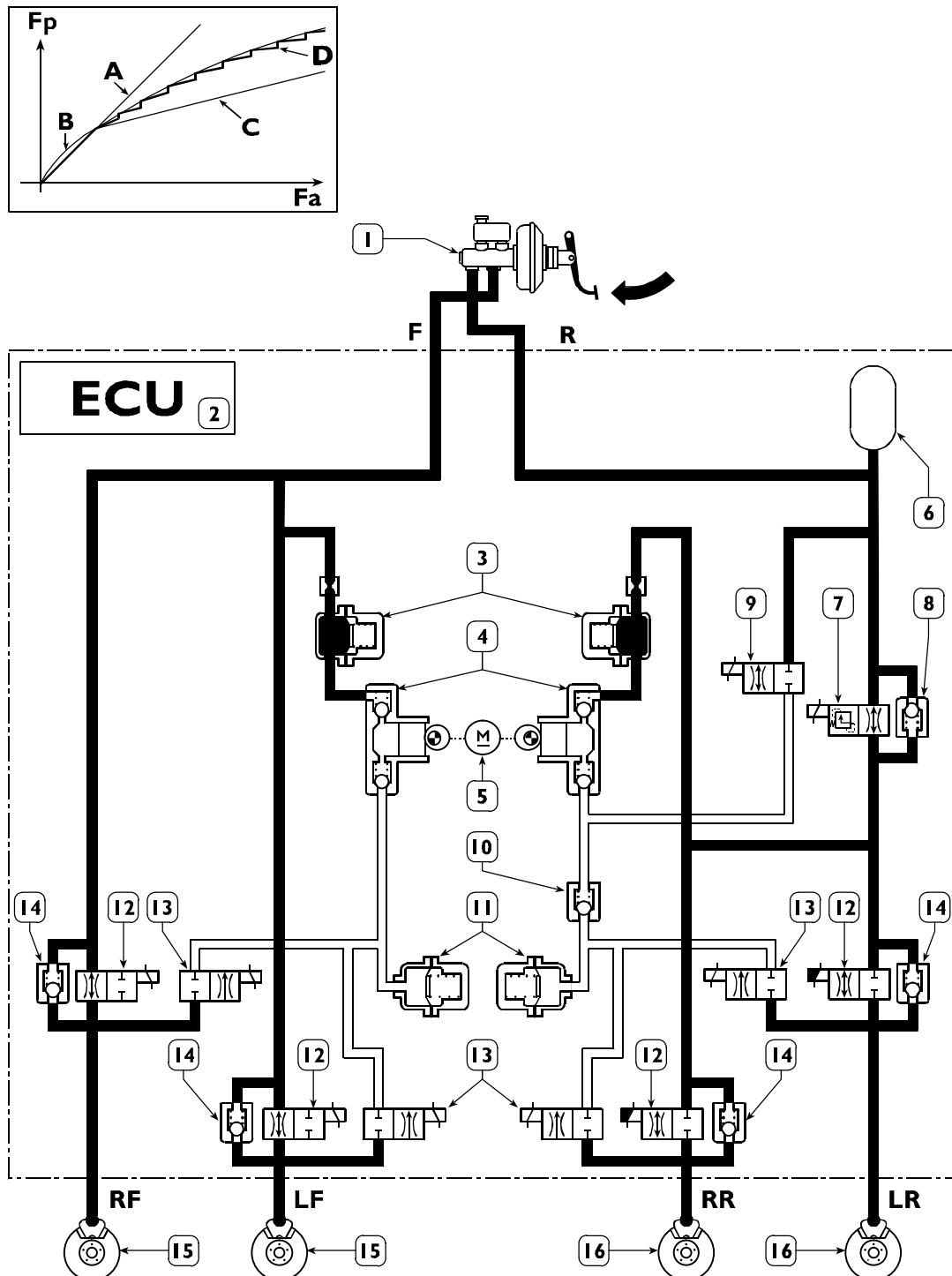
If the sensors detect that one or both rear wheels tend to lock in relation to the front wheels, they inform the control unit which appropriately supplies the supply solenoid valves "12" of the rear axle, in order to optimise the braking force.

A fault on certain system components causes the ABS system to be cut off (displayed by turning on the special warning light), still leaving the EBD system operational.

If both failure warning lights turn on (ABS and EBD) this means an EBD system failure.

Under these conditions the braking force will not be distributed between the axles, therefore rear wheel locking may occur with the possibility of skidding.

Figure 129



FP. Rear axle braking force – FA. Front axle braking force – A. Servobrake distribution curve – B. Ideal distribution curve – C. Any proportioning valve distribution curve – D. EBD system distribution curve

Cutting in of the ABD device

If the sensors detect that one or both rear axle wheels tend to skid, they inform control unit which supplies the intake solenoid valves "9" drive valves "7".

At the same time, supplying the motor "5" which drives the pumps "4" it will be possible to generate the pressure to be sent to the brake calipers concerned.

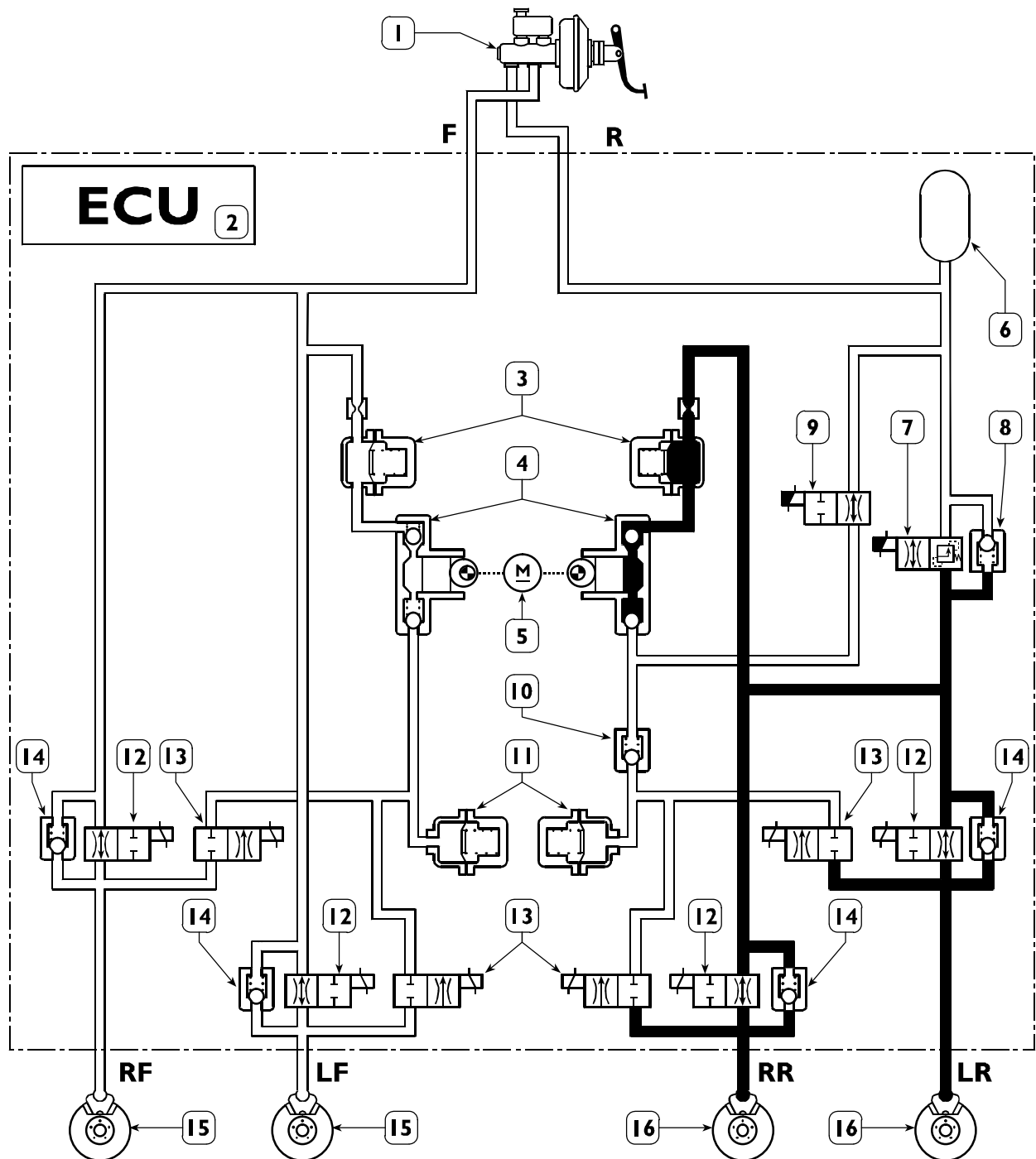
The braking force generated in this manner is modulated by solenoid valves "12" and "13".

During this phase the maximum pressure is limited by the drive solenoid valves "6" and cannot exceed 90 bar.

The driver is informed of the cutting in of this system by the flashing of the corresponding warning light on the dashboard.

If the warning light turns on and glows steadily, there is an ABD system failure.

Figure 130

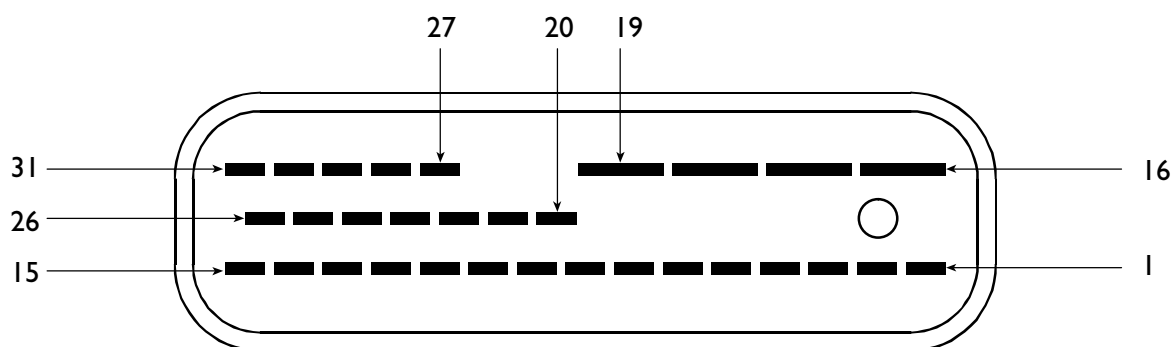


ABD DEVICE OPERATING LAYOUT

Electronic control unit

The following table shows the pin out of the control unit in question.

Figure 131



Pin	Function	Cable colour code
1	Right rear sensor	—
2	Right rear sensor	—
3	Right front sensor (only for 35 C – 40 C – 45 C – 50 C – 65 C vehicles)	—
4	Right front sensor	—
5	Right front sensor (only for 29 L – 35 S vehicles)	—
6	Left front sensor	—
7	Left front sensor	—
8	Left rear sensor	—
9	Left rear sensor	—
10	—	—
11	Line K to pin 4 38-pin connector for diagnostics	8817
12	Line L to pin 3 38-pin connector for diagnostics	8818
13	Negative for third brake relay	0315
14	Positive from stop light switch	1173
15	Key-operated positive	8847
16	Earth	0000
17	Direct positive	7772
18	Direct positive	7772
19	Earth	0000
20	Negative for ABS failure warning light	6670
21	Negative for EBD failure warning light	6673
22	—	—
23	—	—
24	Negative for ABD operating warning light	6674
25	—	—
26	—	—
27	—	—
28	—	—
29	—	—
30	—	—
31	—	—

ABS 8/ESP 8**The ABS 8 system integrates the following functions:**

- ☐ **ABS - Antilock Braking System**
It prevents wheels from being locked during braking, thus making it possible to avoid possible obstacles.
It prevents losing control of the vehicle when braking on a slippery surface (even on one side only → mu-split).
It also reduces the braking distance compared with the one with the wheels locked.
- ☐ **EBD - Electronic Brake Force distribution**
It supersedes and optimizes the function of current hydraulic brake correctors, by better controlling the braking force on rear wheels.
It is implemented by adding a special software to the ABS, and comes into action within a given time interval prior to ABS actuation.
It makes it possible to control any locking condition affecting the rear wheels compared with the front wheels, by optimizing the braking force under different load, running and vehicle utilization conditions.

NOTE The ABD (Automatic Brake Differential) function is available only in systems of the ABS 5.3 type.

The ESP 8 system, in addition to the ESP 8 system, incorporates the following functions:

- ☐ **ESP - Electronic Stability Program**
It monitors the vehicle behaviour continuously (both along straight stretches and bends, when braking or accelerating).
It also monitors the driver's actions: steering the wheel, pressing the brake pedal, accelerator position, and speed.
It is always active in the background, i.e. the ESP system compares the actual vehicle ride with the driver's desired ride 50 times a second. It recognizes dangerous situations before the driver does.
The system considers the different possibilities of coming into operation. It brakes on every single wheel separately.
It operates on the engine control system.
- ☐ **ASR – Acceleration drive control device**
This system prevents driving wheel skid through quick action on the engine and brakes. It allows the vehicle to set off safely and fast even on slippery roads or when one driving wheel is skidding. It also reduces the risk of understeering when you accelerate too much when cornering.

☐ MSR – Engine braking torque control

This system avoids driving wheel drag due to the exhaust brake. It ensures vehicle stability when releasing on slippery roads (e.g. snow, ice), and assists in keeping the path when cornering and shifting down, especially on slippery roads. It requires a slight increase of revs number, through the CAN line.

☐ HHC – Hill holder control

This function allows the vehicle to be kept automatically locked (braked) until the clutch is closed and the driver subsequently presses the accelerator pedal, thus preventing undesired vehicle motion.

The function is actuated automatically: the braking situation is detected by the sensor inside the modulator. When the brake pedal is released, the vehicle will be kept for 2.5 seconds, thus allowing the driver/system to put the gear (and the vehicle to be started). This ensures safe, easy start with any incline, regardless of the weight carried.

☐ HBA (Hydraulic Brake Assistant) – Hydraulic assistant in emergency braking

The main feature of the HBA function is to recognize an emergency braking situation followed by “automatic” increase of vehicle deceleration.

Vehicle deceleration is only restricted to actuation of ABS control, thus taking the greatest advantage of the grip between the tyre and the roadbed currently available. Therefore, ordinary drivers can now achieve braking distances which only experienced drivers could achieve in the past.

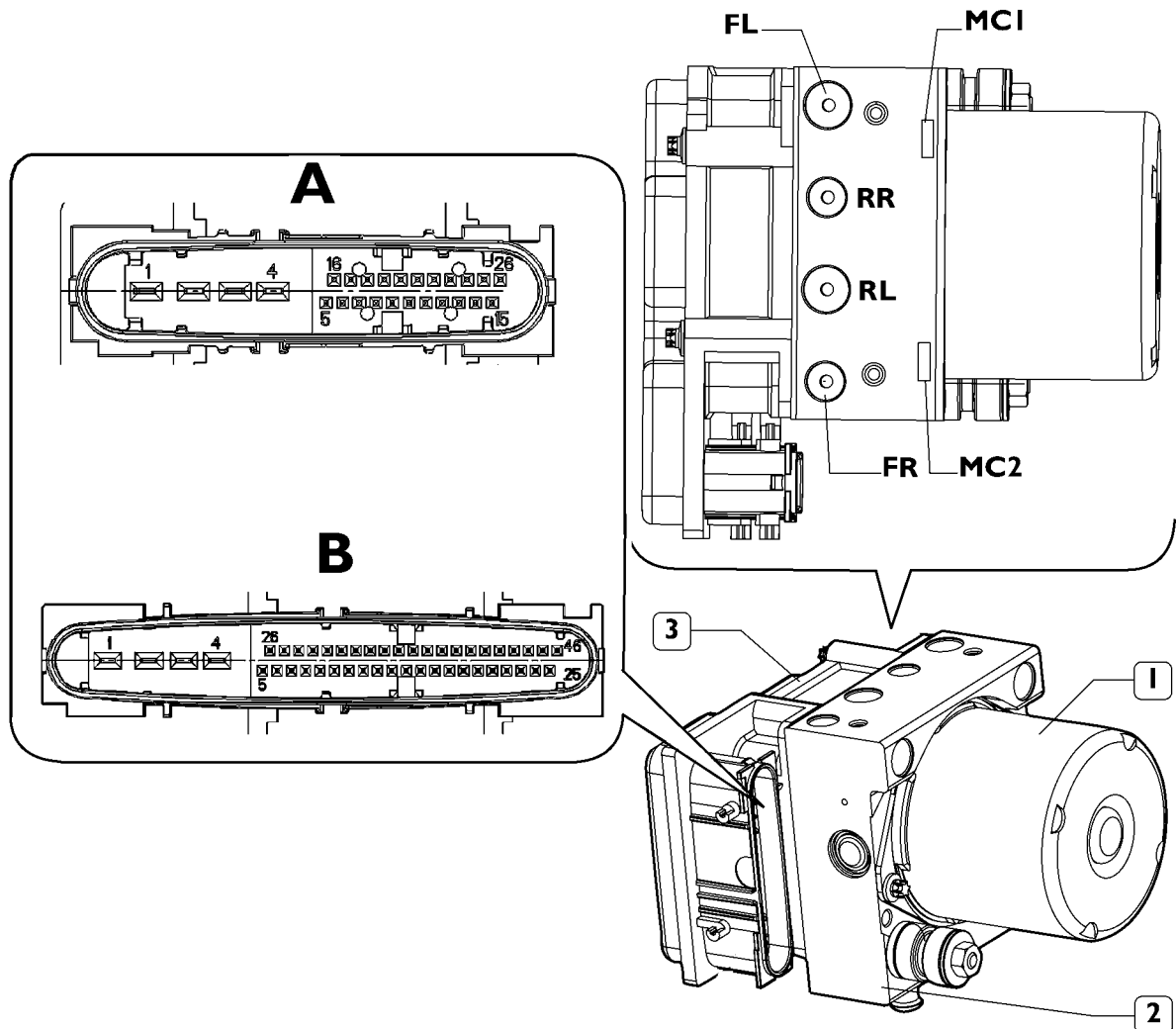
If the driver reduces the braking intensity, vehicle deceleration is reduced depending on the reduction of the force applied onto the pedal.

Therefore, the driver can control deceleration accurately after overcoming the emergency situation.

The extent of the braking request from the driver corresponds to the force applied onto the pedal. Such force is derived from measuring the pressure in the brake pump.

Application of the ABS 8/ESP8 systems

SYSTEM	ABS8		ESP8	
	X (crossed)	II (parallel)	X (crossed)	II (parallel)
VEHICLES	FIA engine with 6 AS 300 VD automatic gearbox		FIA and FIC engines	
	29L – 35S	35C – 65C	29L – 35S	35C

Four crossed channel system (x)**Electro-hydraulic modulator/control unit****Figure 131/I**

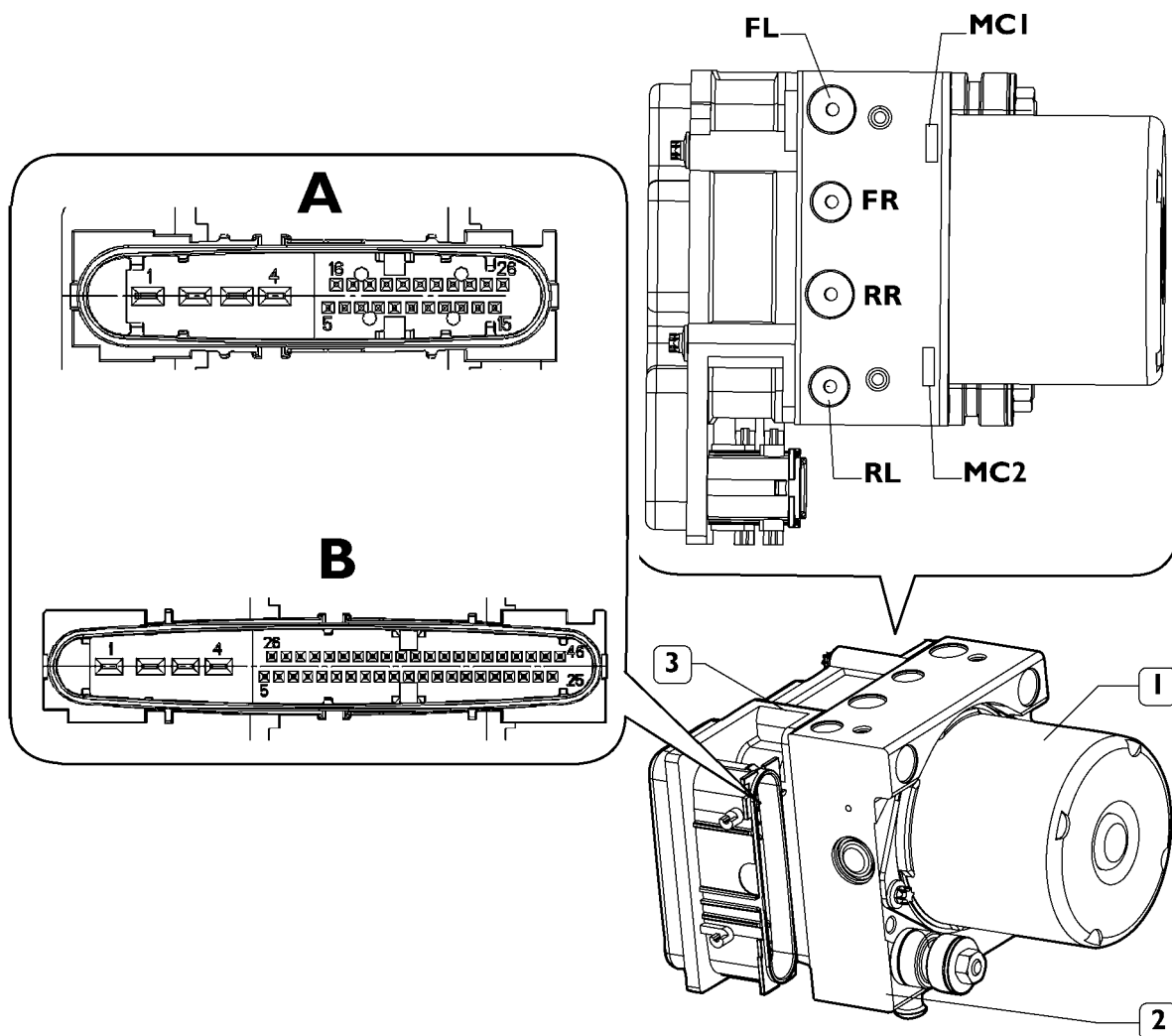
102113

1. Hydraulic accumulator - 2. Electro-hydraulic modulator - 3. Electronic control unit - A. ABS8 connector - B. ESP8 connector - MCI. LF/RR diagonal power supply (or FL/RR with ABS8/ESP8 systems) - MC2. RF/LR diagonal power supply (or FR/RL with ABS8/ESP8 systems) - LF (or FL with ABS8/ESP8 systems). Left front axle output - RR. Right rear axle output - RF (or FR with ABS8/ESP8 systems). Right front axle output - LR (or RL with ABS8/ESP8 systems). Left rear axle output

Four parallel channel system (II)

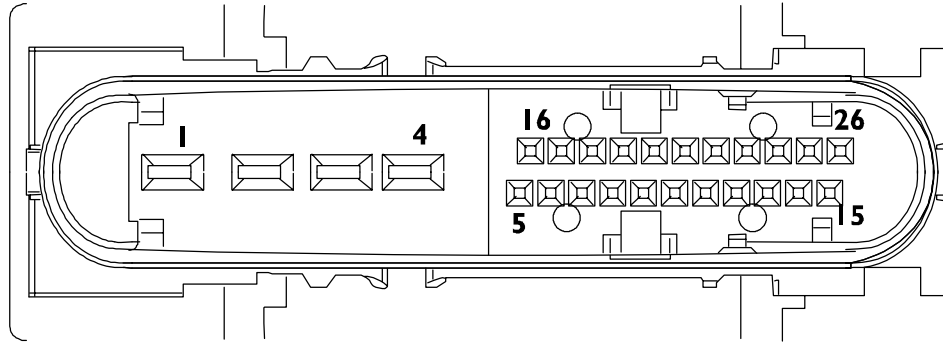
Electro-hydraulic modulator/control unit

Figure 131/2



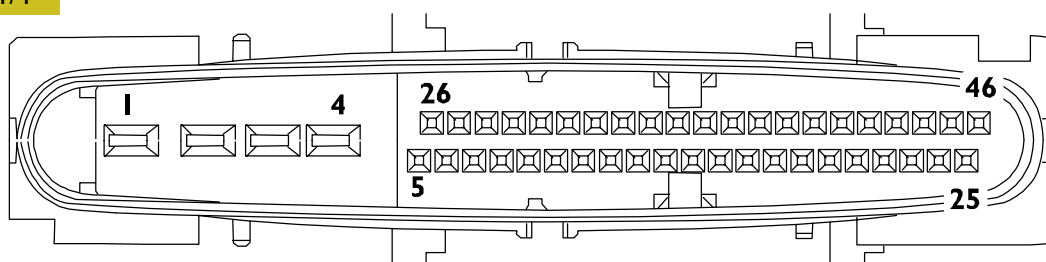
102114

1. Hydraulic accumulator - 2. Electro-hydraulic modulator - 3. Electronic control unit - A. ABS8 connector - B. ESP8 connector - MCI. Front axle power supply - MC2. Rear axle power supply - LF (or FL with ABS8/ESP8 systems). Left front axle output - RR. Right rear axle output - RF (or FR with ABS8/ESP8 systems). Right front axle output - LR (or RL with ABS8/ESP8 systems). Left rear axle output

ABS 8 control unit PIN OUT (X - crossed channels, II - parallel channels)**Figure 131/3**

102245

Pin (X)	Pin (II)	Function	Cable
1	I	Earth (pump control motor)	0000
2	2	Positive after fuse (pump control motor power supply enable relay)	7772
3	3	Positive after fuse (valve lock power supply enable relay)	7772
4	4	Signal earth	0000
5	5	Front left sensor	5570
6	16	-	-
7	7	Rear left sensor	5572
8	8	Rear right sensor earth	5573
9	9	Front right sensor earth	5571
10	10	Front right sensor	5571
11	11	Diagnosis K line	2299
12	12	EBD failure warning light	6673
13	13	Decelerator deactivation with ABS system ON	0315
14	14	-	-
15	15	CAN L line	Green
16	6	Front left sensor earth	5570
17	17	Rear left sensor earth	5572
18	18	Positive after fuse for ABS (KL 15)	8847
19	19	Rear right sensor	5573
20	20	Stop signalling switch – brake lights	1173
21	21	-	-
22	22	ABS failure warning light	6670
23	23	-	-
24	24	-	-
25	25	Diagnosis L line	1199
26	26	CAN H line	White

ESP 8 control unit PIN OUT (X - crossed channels, II - parallel channels)**Figure 131/4**

102246

Pin (X)	Pin (II)	Function	Cable
1	1	Earth (pump control motor)	0000
2	2	Positive after fuse (pump control motor power supply enable relay)	7772
3	3	Positive after fuse (valve lock power supply enable relay)	7772
4	4	Signal earth	0000
5	5	Front left sensor	5570
6	26	-	-
7	7	Rear left sensor	5572
8	8	Rear right sensor earth	5573
9	9	Front right sensor earth	5571
10	10	Front right sensor	5571
11	11	Diagnosis K line	2299
12	12	EBD failure warning light	6673
13	13	-	-
14	14	CAN L line	Green
15	15	Yaw sensor earth	9096
16	16	Yaw sensor signal	9095
17	17	Longitudinal acceleration sensor	9099
18	18	Yaw sensor reference signal	9094
19	19	Voltage stabilization signal to the acceleration sensor	9091
20	20	Side acceleration sensor signal	9092
21	21	Side acceleration sensor signal earth	0050
22	22	Decelerator deactivation with ABS system ON	0315
23	23	-	-
24	24	Diagnosis L line	1199
25	25	-	-
26	6	Front left sensor earth	5570
27	27	Rear left sensor earth	5572
28	28	Positive after fuse for ABS (KL 15)	8847
29	29	Rear right sensor	5573
30	30	Stop signalling switch – brake lights	1173
31	31	Exhaust brake actuation enable switch (ASR/ESP passive switch)	8800
32	32	ABS failure warning light	6670
33/34	33/34	-	-
35	35	CAN H line	White
36	36	Hand brake ON signal	6662
37	37	Yaw sensor test signal	9093
38/39/40	38/39/40	-	-
41	41	Stop signalling switch	1173
42	42	-	-
43	43	Reverse gear signal	2227
44/45	44/45	-	-
46	46	Speed limiter or ASR failure warning light	6672

ESP (Electronic Stability Program) operation

The ESP function controls the vehicle's stability and side dynamics.

The main goals of this function are as follows:

- ☐ to improve stability, especially in understeering and oversteering conditions;
- ☐ to reduce the braking distance in line change conditions and on slippery roads.

The ESP function evaluates the following driver's requests:

- steering-wheel position;
- wheel revs number (speed);
- pressure on the brake pedal or accelerator position.

The ESP control unit microprocessor recognizes the specific manoeuvre and examines the vehicle's behaviour:

- degree of yaw;
- wheel revs number;
- transverse acceleration.

The microprocessor assesses the running behaviour based on the data provided, and the ESP comes into operation by acting on the brakes.

The hydraulic modulator controls brake pressure for every single wheel as quick as possible.

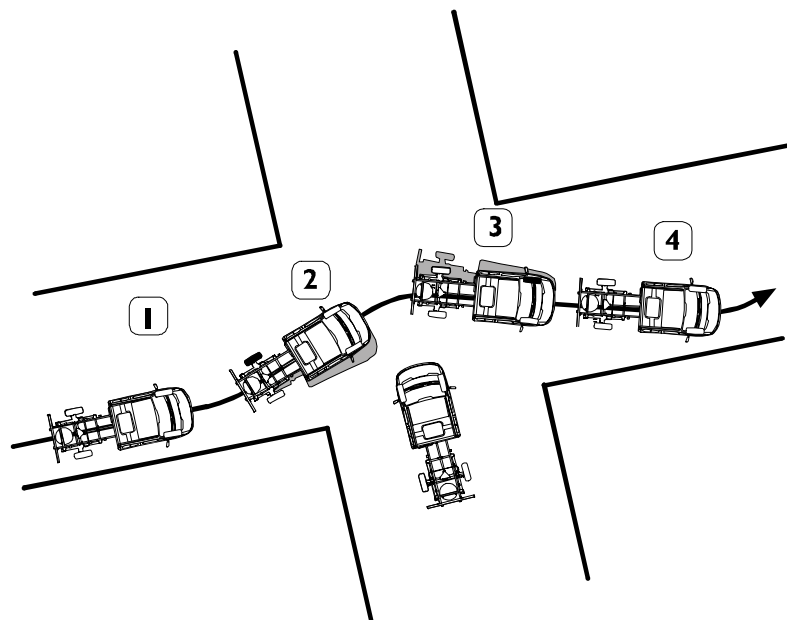
The ESP may, through engine management, reduce the number of revolutions of the engine itself, in order to withstand vehicle deceleration.

The ESP system is always active in the background, i.e. it compares the actual vehicle ride with the driver's desired ride 50 times a second.

Control strategy

Sudden obstacle

Figure 131/5

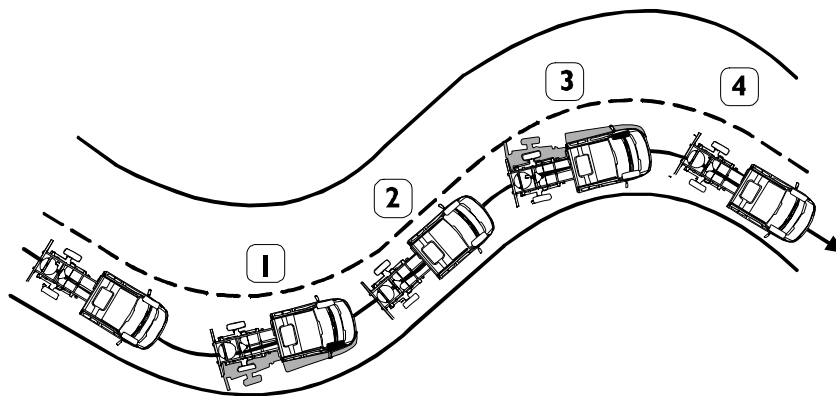


- 1) Acting on the steering-wheel suddenly: danger of understeering.
- 2) The ESP brakes the rear left wheel $\Rightarrow$ the vehicle follows the steering command.
- 3) The driver countersteers: Danger of oversteering $\Rightarrow$ The ESP brakes the front left wheel.
- 4) The vehicle recovers stability.

102246

Sudden steering

Figure 131/6

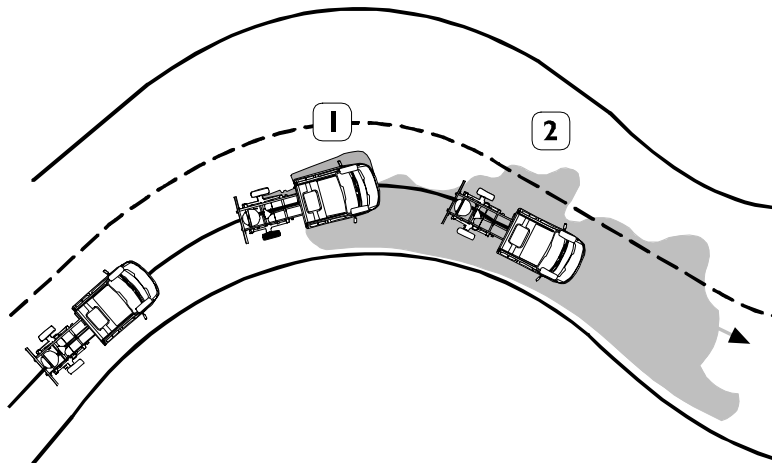


101864

- 1) The vehicle risks skidding (oversteering): the ESP brakes the front right wheel.
- 2) The vehicle recovers stability.
- 3) The vehicle risks skidding (oversteering): the ESP brakes the front left wheel.
- 4) The vehicle recovers stability.

Vehicle running on a slippery road

Figure 131/7



101865

- 1) The vehicle risks skidding (understeering): the ESP brakes the rear right wheel and reduces the engine revs number.
- 2) The vehicle recovers stability.

ASR deactivation strategies

- ☐ Disabling any engine intervention of the ESP and ASR/MSR (torque increase/decrease) over the entire speed range.
- ☐ Traction Control actuation enabled up to the speed of 60 k.p.h. (electronic locking of differential, with no reference to the dragged wheels).
- ☐ Stability intervention (ESP) enabled over the entire speed range.
- ☐ ABS enabled over the entire speed range.
- ☐ EBD enabled over the entire speed range.

NOTE ASR deactivation is recommended when driving with the snow-chains mounted, or when the wheels sink into gravel, sand, etc.

Recovery strategy in case of component failure

System failure (warning light ON)	ESP/ASR	ABS	EBD
Broken component			
Steering angle sensor	X		
Yaw sensor	X		
Brake light switch	X	X	
1 or 2 wheel revs sensors	X	X	
3 ou 4 capteurs tours roue 3 or 4 wheel revs sensors	X	X	X
Electronic control unit	X	X	X
Solenoid valve hydraulic unit	X	X	X
Pressure sensor, ABS pump motor		X	X

NOTE If the warning lights are OFF, all the systems are working.

Warning light legend

Warning light ON: ESP/ASR/MSR not working. No action taken by the ESP/ASR on the engine or the brakes.

N.B. Warning light blinking = ESP/ASR coming into operation.

ABS not working

The front axle may get locked sooner than the rear axle. EBD recovery reduces the rear axle pressure.

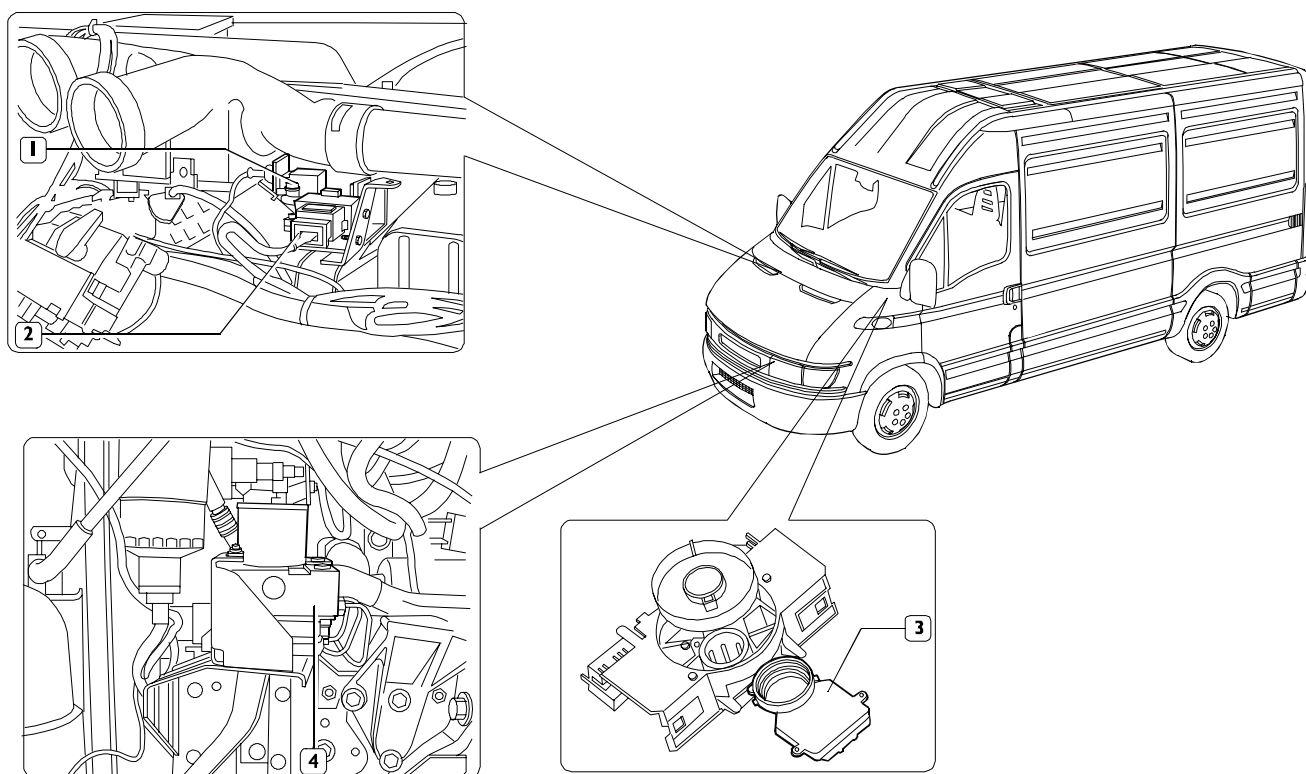
EBD not working

No correction of the rear axle braking pressure: DANGER OF SPINNING THROUGH 180 DEGREES!

The warning light also warns the driver of low brake fluid level, hand brake ON, brake pads worn.

Installing the esp components

Figure 131/8



102108

1. Acceleration sensor - 2. Yaw sensor - 3. Steering angle sensor - 4. Electro-hydraulic modulator/control unit

NOTE Installation of the electro-hydraulic modulator/control unit is similar in ASB 8 systems, too.

ESP system components and calibration

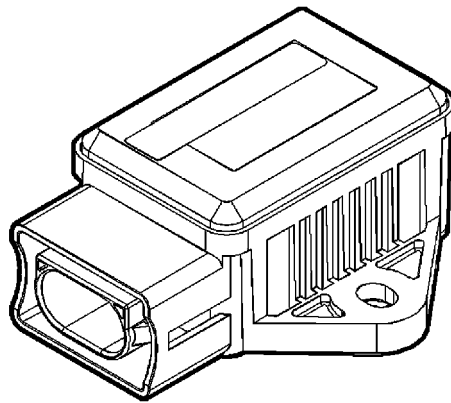
Some modifications or repairs affecting the ESP system components require a calibration procedure.

The repair operations that require the above procedure are detailed as follows:

- ☐ Replacing the system's braking apparatus electronic control unit (incorporated into the electro-hydraulic modulator).
- ☐ Replacing the steering angle sensor fitted into the steering wheel.
- ☐ Replacing the longitudinal acceleration sensor.

Yaw sensor with built-in side acceleration sensor

Figure 131/9



102115

It measures the motion of the vehicle around its own vertical axis (yaw) as well as the vehicle's side acceleration.

These signals continuously inform the control unit about the vehicle's behaviour.

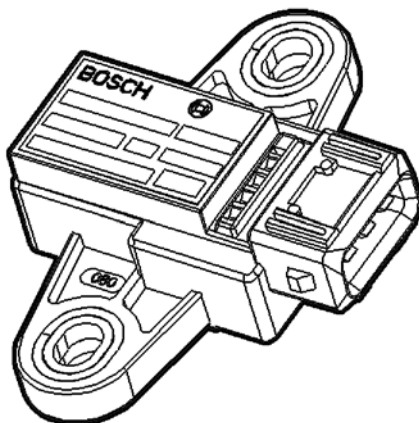
The comparison between these signals and those from the driver (steering-wheel position, wheel spin number/speed and pressure on the brake pedal/accelerator position) allows the ESP control unit to define the actions to be taken. The hydraulic unit controls brake pressure as quickly as possible, separately for every single wheel.

Moreover, the ESP system may decrease the engine revs number by means of the engine control feature.

NOTE Replacing the yaw sensor requires no calibration.

Longitudinal acceleration sensor

Figure 131/10



102116

It measures the vehicle's acceleration and deceleration changes.

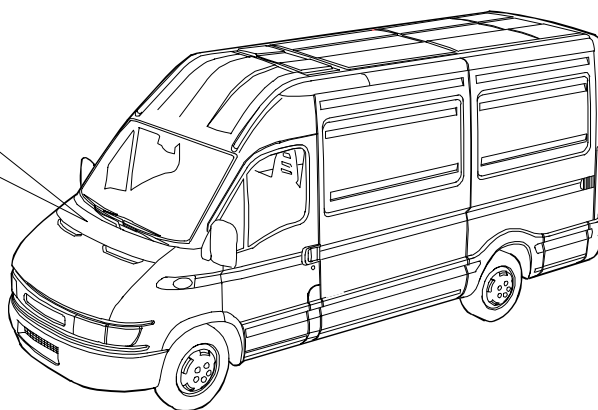
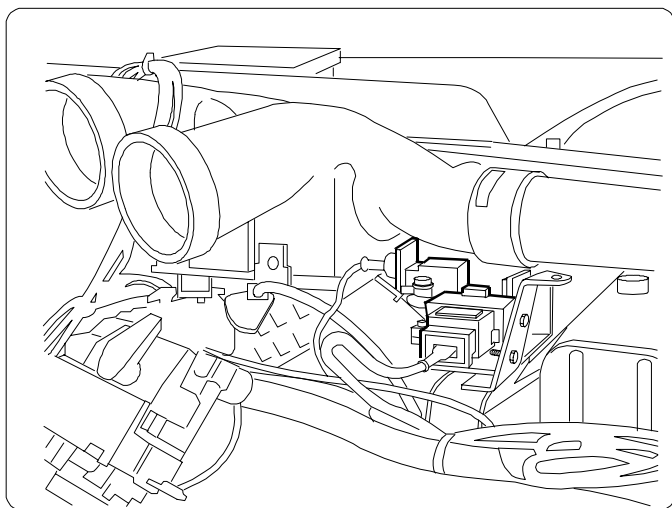
These signals continuously inform the control unit about the vehicle's behaviour.

The comparison between these signals and those from the driver (steering-wheel position, wheel spin number/speed and pressure on the brake pedal/accelerator position) allows the ESP control unit to define the actions to be taken. The hydraulic unit controls brake pressure as quickly as possible, separately for every single wheel.

Moreover, the ESP system may decrease the engine revs number by means of the engine control feature.

Longitudinal acceleration sensor calibration

Figure 131/11



102111

In a horizontal position, you will obtain the sensor "zero" condition through the diagnosis instrument, i.e. you will assign its absolute zero position.

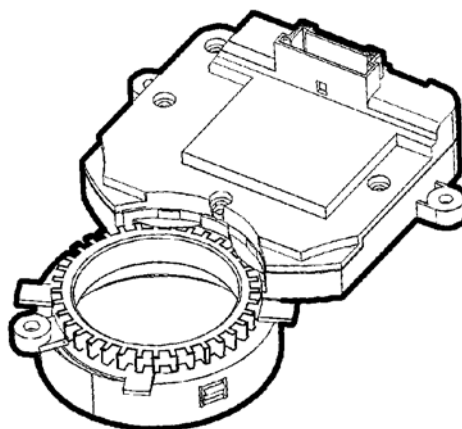
Use a diagnosis instrument to clear the errors.

Carry out a road test, to make the control unit verify whether errors are still found. The vehicle is to be taken to a slight slope and checked if it is kept braked for 2.5 seconds.

Drive back to the service centre, then use a diagnosis instrument to verify that the anomaly is no longer found.

Steering angle sensor

Figure 131/12



102116

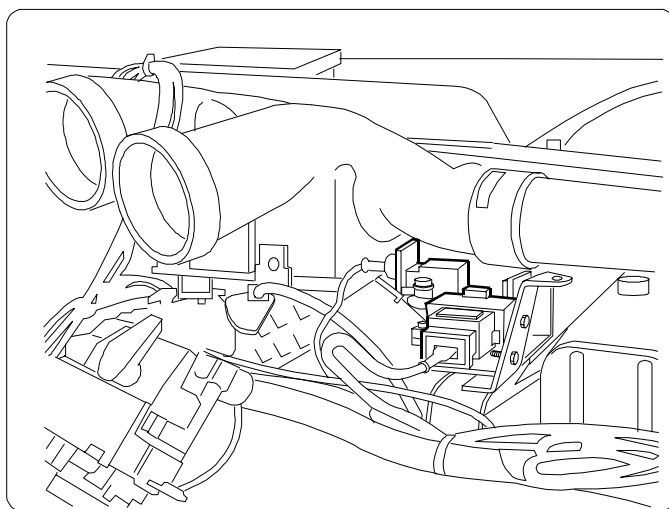
It measures the steering angle required by the driver.

The comparison between this signal and those from all the other sensors allows the ESP control unit to define the actions to be taken. The hydraulic unit controls brake pressure as quickly as possible, separately for every single wheel.

Moreover, the ESP system may decrease the engine revs number by means of the engine control feature.

Steering angle sensor calibration

Figure 131/13



102110

With both the steering-wheel and the wheels in straight position (after checking toe-in), you will obtain the sensor "zero" condition through the diagnosis instrument, i.e. you will assign its absolute zero position.

Use a diagnosis instrument to clear the errors.

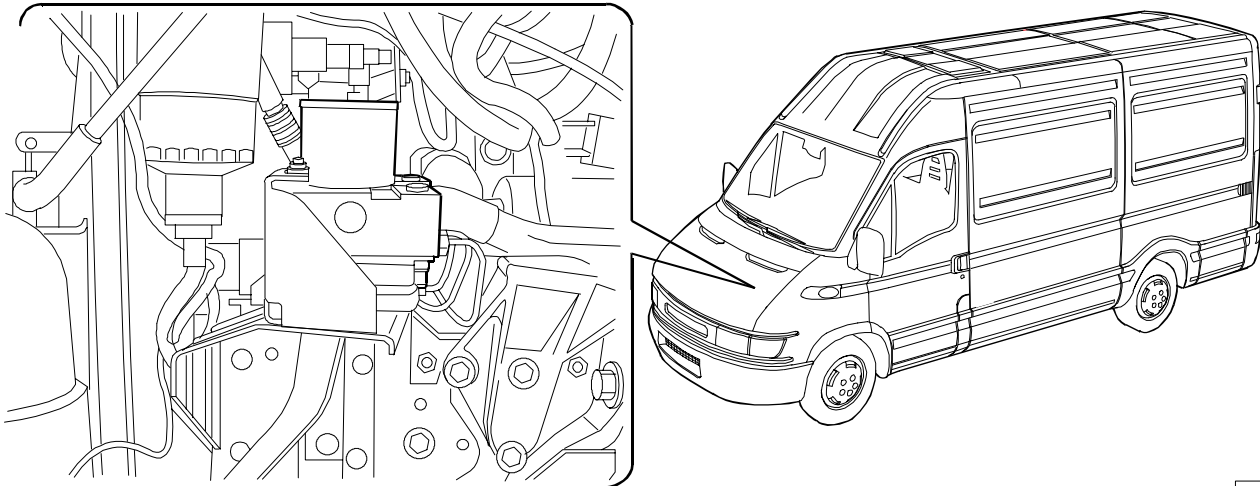
Carry out a road test, to make the control unit verify whether errors are still found. Drive along a straight road at a speed of approximately 50 k.p.h. Steer to the right and then to the left several times, after making sure you do not endanger other drivers.

NOTE You need not oversteer to cause the ESP warning light to come on.

Drive back to the service centre, then use a diagnosis instrument to verify that the anomaly is no longer found.

ESP control unit programming

Figure 131/14



102109

Entering the variant codes: type of drive, engine, MTT, wheelbase, type of front and rear suspensions, height.

Easy compares the type of vehicle (PIC reading) with the control unit code, to avoid installation errors (single wheels instead of dual), and downloads the variant codes into the control unit.

Clear the errors (if any) by means of the diagnosis instrument.

Carry out a road test. Drive along a straight road at a speed of approximately 50 k.p.h. Brake suddenly as if in an emergency, after making sure you do not endanger other drivers: you should feel the system "respond" on the brake pedal. This test makes the control unit verify whether errors are still found.

Drive back to the service centre, then use a diagnosis instrument to verify that the anomaly is no longer found.

Replacing ESP central unit needs the longitudinal acceleration sensor calibration that was already described in previous page.

The steering angle sensor, in this case, is not to be calibrated as it has its own internal memory.

6AS 300 VD AUTOMATIC TRANSMISSION

Description of operation

The gear engaging system of the 6 AS 300 VD gearbox is a combination of a traditional system of the mechanic type and an electric one.

Below are the main components of the system:

- 1) electronic control unit;
- 2) gear selecting/engaging actuator;
- 3) gear shift lever;
- 4) clutch actuator;
- 5) display/buzzer.

In the manual mode, the first gear is used to start the vehicle.

In the automatic mode, you just need to act on the selection lever (A/M): the gear is selected directly by the control unit (1a).

The electronic control unit picks up all the signals required to meet the safety conditions and the parameters programmed in the same. It also drives the electric actuating motor(s) for gear selection/engagement and clutch control.

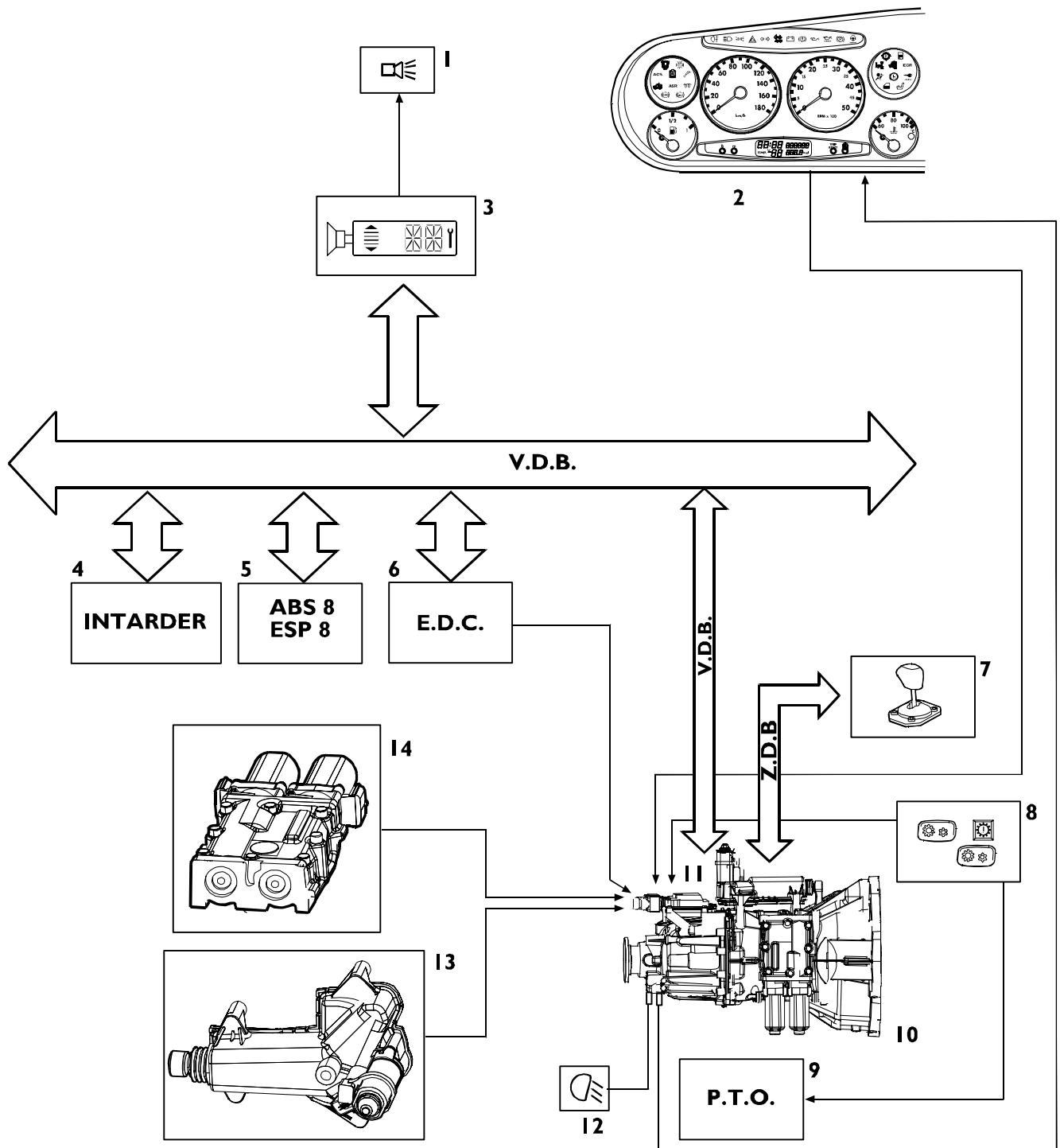
The information required for driving is in any case made available to the driver by means of the display.

NOTE To ensure correct management of the gearbox and the other auxiliary functions provided for by the system, the control unit is interfaced (CAN line) with the other electric and electronic system fitted to the vehicle, such as:

- EDC
- ABS8/ESP8
- IMMOBILIZER
- PTO (POWER TAKE-OFF, if available).

Electronic control

Figure 131/15



101868

1. Buzzer - 2. Tachograph - 3. Display - 4. IMMOBILIZER control unit - 5. ABS8/ESP8 control unit - 6. EDC control unit -
 7. Gear shift lever - 8. PTO icon - 9. P.T.O. - 10. Gearbox - 11. Gearbox control unit - 12. Reversing light actuation -
 13. Clutch actuator - 14. Gearbox actuator.